

ECOTOX Knowledgebase: A Comprehensive Guide for Environmental Researchers and Drug Development Professionals

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ECOTOX Knowledgebase: A Comprehensive Guide for Environmental Researchers and Drug Development Professionals

Abstract

This article provides a thorough overview of the ECOTOX Knowledgebase, the world's largest curated repository of single-chemical ecotoxicity data. Tailored for researchers, scientists, and drug development professionals, it explores the platform's foundational principles, methodological applications for chemical assessment and risk management, strategies for troubleshooting and data optimization, and its validated use in regulatory and research contexts. By synthesizing information on over 13,000 chemicals and 14,000 species from more than 54,000 references, this guide demonstrates how ECOTOX supports ecological risk assessments, chemical safety evaluations, and the development of New Approach Methodologies (NAMs) in environmental toxicology.

Unlocking the ECOTOX Knowledgebase: Core Concepts and Data Landscape

The **Ecotoxicology (ECOTOX) Knowledgebase** is a comprehensive, publicly available application managed by the U.S. Environmental Protection Agency (EPA) that provides information on adverse effects of single chemical stressors to ecologically relevant aquatic and terrestrial species [1]. It stands as the world's largest curated resource for ecotoxicity data, integral to chemical safety assessments and ecological risk evaluations.

The table below summarizes the core quantitative metrics that define the scale of the ECOTOX Knowledgebase.

Table 1: Key Quantitative Metrics of the ECOTOX Knowledgebase (as of September 2025)

Metric	Value
Total Test Records	Over 1 million
Number of Unique Chemicals	Over 12,000
Number of Aquatic and Terrestrial Species	Over 13,000
Number of Source References	Over 53,000

Core Data and Methodology

Data Curation and Scope

Data within ECOTOX are manually curated from the peer-reviewed scientific literature after an exhaustive search and abstraction protocol [1]. For each study, all pertinent information on the species, chemical, test methods, and results presented by the authors is systematically abstracted into the knowledgebase [1]. This process ensures a high level of data quality and reliability. The knowledgebase is proactively maintained, with updates released quarterly to include new data and features [1].

The scope of ECOTOX is specifically focused on ecotoxicology, which is the study of how chemicals interact with organisms in the environment [2]. This field aims to understand the effects of chemicals on populations, communities, and ecosystems, distinguishing it from studies that only measure environmental pollutant levels or focus solely on human health [3].

Experimental Endpoints and Test Organisms

ECOTOX compiles data from standardized toxicity tests, with a historical emphasis on aquatic systems [2]. The most common hazard endpoints are survival (e.g., LC50 - the concentration lethal to 50% of test organisms) and reproduction or growth (e.g., NOEC - No Observed Effect Concentration, or LOEC - Lowest Observed Effect Concentration) [2]. The knowledgebase covers a wide range of ecologically relevant species, including:

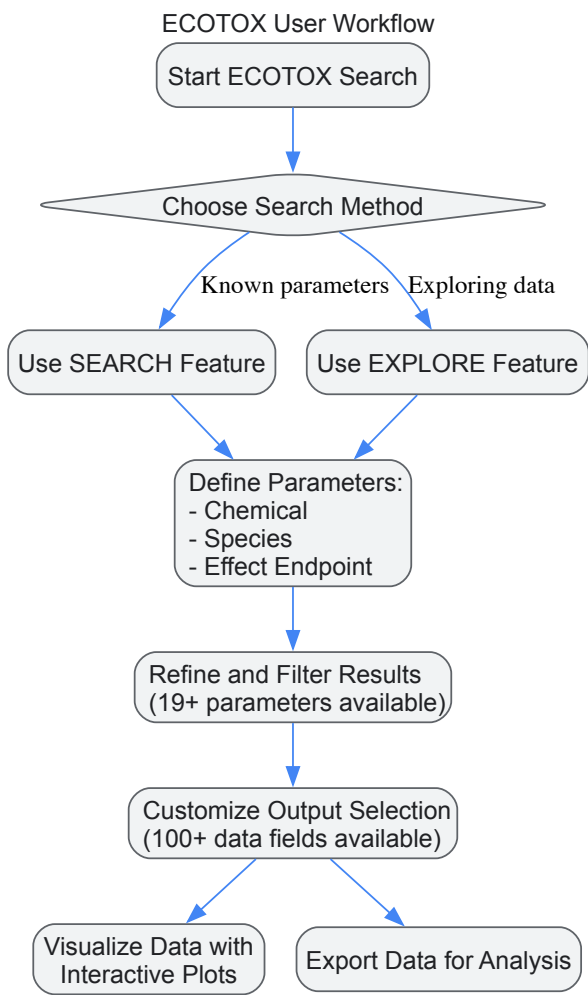
- Primary producers:** plants, fungi, and algae.
- Invertebrates:** worms, insects, beetles, and mollusks.
- Vertebrates:** fish, amphibians, reptiles, birds, and mammals [2].

Table 2: Common Ecotoxicity Test Endpoints in ECOTOX

Endpoint Category	Specific Metric	Description
Acute Toxicity	LC50 / EC50	Concentration causing 50% mortality (Lethal Concentration) or a specified effect (Effective Concentration) in a population over a short-term test.
Chronic Toxicity	NOEC	The highest tested concentration where no significant adverse effect is observed compared to the control group.
	LOEC	The lowest tested concentration that causes a statistically significant adverse effect compared to the control group.

User Workflow and Data Access

The ECOTOX Knowledgebase provides several features to access its vast data holdings. The primary workflow for a researcher involves using one of the search interfaces to define a query, refining the results, and then using visualization tools or exporting data for further analysis.



Search and Exploration Features

ECOTOX offers two main pathways for data retrieval, catering to different user needs:

- **SEARCH Feature:** Used when the user has specific parameters in mind. It allows for targeted searches for data on a particular chemical (with a link to the EPA CompTox Chemicals Dashboard for more chemical information), species, effect, or endpoint [1].
- **EXPLORE Feature:** Useful when the exact search parameters are not fully known, allowing for a more flexible and broad investigation of the data landscape by chemical, species, or effects [1].

Both pathways allow users to refine and filter data searches by 19 different parameters and customize output selections from over 100 available data fields [1].

Data Visualization and Analysis

A key functionality of ECOTOX is its **Data Visualization feature**, which enables users to view results graphically [1]. The generated data plots are interactive; users can hover over data points and zoom in on specific sections to retrieve detailed information of interest [1]. This facilitates rapid data exploration and analysis directly within the platform.

Research Applications and Strategic Importance

The ECOTOX Knowledgebase is a critical tool for various scientific and regulatory applications, serving multiple user groups:

- **Researchers:** The comprehensive nature of the knowledgebase allows for efficient data mining and meta-analyses, reducing the need for new animal tests by maximizing the use of existing data [1].
- **Risk Assessors:** The data collection is uniquely suited for linking traditional biological effects used in regulatory risk assessments with mechanistic responses across different levels of biological organization and species [1].
- **Decision Makers:** Local, State, and Tribal governments use ECOTOX data to develop site-specific environmental quality criteria or to interpret chemical monitoring data [1].

For over two decades, ECOTOX has been a primary source for toxicity data used in developing chemical benchmarks and aquatic life criteria to protect freshwater and saltwater organisms [1]. It also informs ecological risk assessments for chemical registration and re-registration, and aids in the prioritization of chemicals under laws like the Toxic Substances Control Act (TSCA) [1]. Furthermore, the database is used to develop and validate New Approach Methodologies (NAMs), such as Quantitative Structure-Activity Relationship (QSAR) models, which predict toxicity based on a chemical's physical characteristics and structure [1].

The Researcher's Toolkit

The following table outlines key resources and concepts essential for researchers working in ecotoxicology and utilizing a knowledgebase like ECOTOX.

Table 3: Essential Research Reagents and Resources for Ecotoxicology

Item / Concept	Function / Description
ECOTOX Knowledgebase	The central curated database providing single-chemical toxicity data for aquatic and terrestrial species, used for hazard assessment and research [1].
Test Organisms	A diverse set of indicator species (e.g., fathead minnow, water flea (<i>Daphnia</i>), earthworm) used in standardized tests to represent ecosystem responses to chemical stress [2].
GHS Classification	The Globally Harmonized System of classification and labelling of chemicals; provides a standardized framework for categorizing chemical hazards, including aquatic toxicity [2].
Quantitative Structure-Activity Relationship (QSAR)	An in silico method that uses mathematical models to predict a chemical's toxicity based on its structural properties, often trained on data from resources like ECOTOX [1].
Adverse Outcome Pathway (AOP)	A conceptual framework that describes a sequence of events from a molecular initiating event to an adverse outcome at the organism or population level, organizing knowledge for predictive toxicology [2].

The assessment and management of chemical risks to the environment have long required access to reliable, curated ecotoxicity data. In the early 1980s, the **United States Environmental Protection Agency (USEPA)** began addressing this need through the development of several specialized, ecosystem-specific databases to provide regulatory EPA Program Offices with rapid access to ecotoxicity data [4] [5]. These initial systems—AQUIRE (aquatic species), PHYTOTOX (terrestrial plants), and TERRETOX (terrestrial wildlife)—operated as distinct resources, each serving specific assessment needs under various legislative mandates, including the **Federal Insecticide, Fungicide, and Rodenticide Act; the Clean Water Act; and the Toxic Substances Control Act** [4]. However, as the number of chemicals in commerce grew and regulatory requirements expanded, the limitations of these segregated databases became apparent. This fragmentation led to the strategic development of a unified system, the **ECOTOXicology Knowledgebase (ECOTOX)**, which integrated these previously separate resources into a comprehensive, accessible platform supporting environmental research and regulatory decision-making [6].

The evolution from multiple standalone databases to the unified ECOTOX system represents a significant advancement in the field of ecotoxicology, reflecting broader shifts in toxicology testing and regulatory science. This transformation was driven by the need for **cost-effective hazard assessment** methods that could efficiently meet growing regulatory mandates for chemical safety evaluations [4]. The ECOTOX Knowledgebase has since become the **world's largest compilation of curated ecotoxicity data**, embodying decades of systematic literature review and data curation practices that align with contemporary systematic review standards and FAIR data principles (Findable, Accessible, Interoperable, and Reusable) [4] [5].

The Predecessor Systems: Specialized Foundations

The foundational databases that preceded the unified ECOTOX system were each designed with specific ecosystem compartments and taxonomic groups in mind, reflecting the compartmentalized approach to environmental protection prevalent during their development.

AQUIRE (Aquatic Information Retrieval)

The AQUIRE database served as the primary repository for **aquatic toxicity data**, encompassing both freshwater and marine organisms [6]. It was designed to support the development of water quality criteria and assessments under the **Clean Water Act**, compiling test results for fish, invertebrates, and aquatic plants. The database structure focused on capturing critical parameters specific to aquatic testing, including water chemistry variables, exposure pathways, and aquatic-specific endpoints such as swimming performance, larval development, and photosynthetic inhibition.

PHYTOTOX (Terrestrial Plant Toxicology)

The PHYTOTOX database specialized in **terrestrial plant toxicity data**, addressing the effects of chemical stressors on vascular plants, including agricultural species, wildflowers, trees, and other non-crop species [6]. This resource supported pesticide registration and ecological risk assessments particularly for non-target plants, capturing endpoints such as seed germination, root elongation, shoot growth, biomass production, and reproductive effects. The database design accommodated the unique methodological considerations of plant toxicity testing, including soil types, application methods, and growth media characteristics.

TERRETOX (Terrestrial Animal Toxicology)

The TERRETOX database focused on **terrestrial wildlife species**, including birds, mammals, reptiles, amphibians, and terrestrial invertebrates [6]. It supported assessments under statutes such as the **Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA)**, capturing data from both laboratory and field studies. The database structure enabled the recording of complex life history parameters, dietary exposures, and ecological factors relevant to terrestrial ecosystems, with endpoints ranging from acute mortality to reproductive effects and behavioral changes.

Table: Founding Database Systems and Their Specializations

Database	Ecosystem Focus	Taxonomic Coverage	Primary Regulatory Drivers
AQUIRE	Aquatic environments	Fish, aquatic invertebrates, amphibians, aquatic plants	Clean Water Act
PHYTOTOX	Terrestrial systems	Vascular plants, including agricultural species and native flora	FIFRA, Toxic Substances Control Act
TERRETOX	Terrestrial wildlife	Birds, mammals, reptiles, terrestrial invertebrates	FIFRA, Endangered Species Act

The Driving Forces for Unification

Multiple converging factors necessitated the integration of these separate databases into a unified ecotoxicology knowledgebase in the 1990s, with the consolidated ECOTOX system emerging to address critical limitations of the fragmented approach.

The **exponential growth in chemical production** and corresponding regulatory requirements created a pressing need for more efficient data retrieval and synthesis. Risk assessors were increasingly required to evaluate chemical effects across ecosystem boundaries, a process hampered by the need to search multiple independent databases with different query interfaces and data structures [4]. Furthermore, the **advancement of ecological risk assessment frameworks** emphasized the importance of considering cross-media transfers and food-web exposures, which required integrated data spanning aquatic, terrestrial, and plant systems [4].

The **inconsistencies in data curation practices** across the separate databases also presented significant challenges for comprehensive chemical assessments. Each database had developed its own controlled vocabularies, quality assurance criteria, and data extraction protocols, creating barriers to data integration and increasing the potential for transcription errors during manual data transfer between systems [4]. The maintenance of three separate systems also posed substantial **resource and operational inefficiencies**, with duplicated efforts in literature searching, reference management, and software development [5].

The unified ECOTOX Knowledgebase addressed these limitations by establishing **standardized curation protocols** across all ecological domains, implementing a **single controlled vocabulary** for taxonomic, chemical, and methodological descriptors, and creating a **unified user interface** that supported complex queries across ecosystem boundaries [4] [5]. This integration allowed for more efficient literature surveillance, reduced data management costs, and enabled more sophisticated analyses of chemical effects across biological systems and taxonomic groups.

The ECOTOX Unified System: Architecture and Implementation

The unified ECOTOX Knowledgebase represents a significant architectural and conceptual advancement over its predecessor systems, designed to support the evolving needs of chemical risk assessment and ecological research while maintaining backward compatibility with historically curated data.

Data Model and Integration Framework

The ECOTOX data model was designed to accommodate the diverse data elements from all three predecessor databases while establishing consistent relationships between chemicals, species, test methods, and effects. The core integration challenge involved mapping the **legacy data structures** from AQUIRE, PHYTOTOX, and TERRETOX to a unified schema that preserved the critical details specific to each ecosystem compartment while enabling cross-system queries [6]. This required the development of comprehensive **controlled vocabularies** for species taxonomy, chemical identifiers, test methods, and measured endpoints that could span aquatic and terrestrial domains [4].

The implementation maintained the original test data from all three source databases while ensuring that new data curation followed consistent protocols. The system architecture incorporated **EPA's Chemical Data Reporting (CDR)** rules for chemical identification and established linkages to other agency chemical resources, including the **CompTox Chemicals Dashboard** [1]. For species taxonomy, ECOTOX adopted integrated **taxonomic serial numbers (TSN)** from the Integrated Taxonomic Information System (ITIS) to ensure consistent identification and classification across all records [4].

Systematic Review and Data Curation Pipeline

A cornerstone of the unified ECOTOX system is its rigorous, standardized process for literature identification, study evaluation, and data extraction. The curation pipeline follows **systematic review principles** aligned with contemporary evidence-based toxicology practices, implementing transparent and objective procedures for identifying relevant ecotoxicity literature [4].

The literature review process begins with comprehensive searches of the **open and grey literature**, including government reports and regulatory studies that may not appear in peer-reviewed journal indexes [4]. Identified references then undergo a tiered review process, beginning with title and abstract screening followed by full-text evaluation against established criteria for applicability and acceptability. To be included, studies must present **single-chemical toxicity tests** on ecologically relevant species, provide sufficient methodological details, report explicit exposure concentrations and durations, and include documented control measurements [4].

The data abstraction process captures over 100 data fields describing the chemical, species, test conditions, and results, using standardized controlled vocabularies to ensure consistency [4]. This systematic approach to data curation has enabled ECOTOX to maintain its position as a comprehensive and authoritative source of ecotoxicity data while supporting the needs of regulatory applications requiring transparent and reproducible methods.

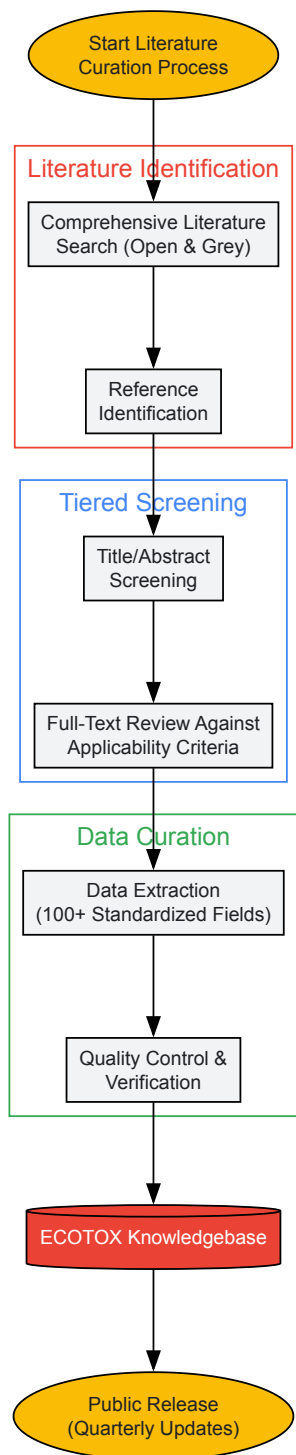


Diagram: ECOTOX Systematic Review and Data Curation Pipeline. The process follows rigorous, standardized procedures for literature identification, screening, and data extraction aligned with systematic review principles.

Quantitative Evolution: From Founding Databases to Modern Knowledgebase

The transition from separate databases to the unified ECOTOX system has enabled exponential growth in both the scope and scale of available ecotoxicity data, supporting more comprehensive chemical assessments and research applications.

As of 2025, ECOTOX has surpassed **1.1 million test records** compiled from over **54,000 references**, covering nearly **14,000 aquatic and terrestrial species** and **13,000 chemicals** [7]. This represents extraordinary growth from the early integrated system, which contained substantially fewer records and chemicals. The knowledgebase continues to expand through quarterly updates that incorporate new scientific literature, with recent additions focusing on emerging contaminants such as **6-PPD quinone**, **cyanotoxins**, and **PFAS** (per- and polyfluoroalkyl substances) [7].

The user base and application of ECOTOX data have similarly expanded, with the system now serving over **16,000 average monthly users** and attracting more than **2,000 new users** in 2023 alone [7]. This widespread adoption reflects the system's critical role in supporting regulatory

decision-making, chemical prioritization, and ecological research across multiple sectors.

Table: Growth Metrics of the ECOTOX Knowledgebase

Metric	Pre-Unification Separate Databases (Circa 1990s)	ECOTOX Unified System (2021)	ECOTOX Current (2025)
Test Records	Not publicly documented	1,018,565 [6]	1.1 million+ [7]
Chemical Substances	Not publicly documented	12,223 [6]	13,000+ [7]
Species Covered	Not publicly documented	13,266 [6]	14,000+ [7]
Source References	Not publicly documented	50,000+ [4]	54,000+ [7]
User Base	Limited to EPA program offices	Publicly accessible	16,000+ monthly users [7]

Functionality and Applications of the Unified System

The modern ECOTOX Knowledgebase provides sophisticated tools for data discovery, analysis, and visualization that far exceed the capabilities of the original standalone systems, supporting diverse applications in regulatory science and ecological research.

Data Access and Exploration Tools

ECOTOX offers multiple pathways for accessing toxicity data tailored to different user needs and levels of specificity. The **Search feature** enables targeted queries for specific chemicals, species, or effects, with results that can be refined and filtered using 19 different parameters [1]. The output can be customized from over 100 available data fields for export and use in external applications. For less structured discovery, the **Explore feature** allows users to investigate data when exact search parameters are unknown, providing flexibility in data exploration [1]. The system also incorporates advanced **Data Visualization features** with interactive plots that enable users to examine data distributions, zoom into specific data regions, and access underlying record details through hover tools [1].

Research and Regulatory Applications

The unified ECOTOX system supports a broad spectrum of applications in environmental science and regulation, serving as a critical resource for researchers, risk assessors, and decision-makers. For **regulatory risk assessors**, ECOTOX provides foundational data for developing water quality criteria, sediment quality guidelines, and chemical-specific benchmarks under various statutory frameworks [1]. The database supports pesticide registration and re-registration processes, chemical prioritization under the Toxic Substances Control Act, and ecological risk assessments for contaminated sites [4] [1].

For **research scientists**, ECOTOX enables meta-analyses, data gap identification, and the development and validation of predictive models, including quantitative structure-activity relationships (QSARs) and species sensitivity distributions (SSDs) [1]. The comprehensive curated data also supports the development and evaluation of **New Approach Methodologies (NAMs)**,- including in vitro and in silico methods, by providing traditional in vivo toxicity data for comparison and validation [4]. Additionally, the system serves **local, state, and tribal governments** in developing site-specific criteria and interpreting environmental monitoring data for chemicals without established regulatory benchmarks [1].

The modern ECOTOX ecosystem encompasses a suite of resources designed to support effective utilization of the knowledgebase across diverse user groups and applications.

Table: Key Resources in the ECOTOX Researcher's Toolkit

Resource	Function	Target Audience
Web Interface (epa.gov/ecotox)	Primary access point for data queries, exploration, and visualization	All users
CompTox Chemicals Dashboard Integration	Provides additional chemical information, properties, and related data	Researchers, risk assessors
Standardized Controlled Vocabularies	Ensures consistency in data curation and retrieval	All users, particularly those conducting systematic reviews
Training Webinars & Archives	Instruction on effective use of search, explore, and visualization features	New users, students

Resource	Function	Target Audience
NAM (New Approach Methodologies) Training	Guidance on using ECOTOX with alternative testing methodologies	Regulatory scientists, researchers
Technical Support (ecotox.support@epa.gov)	Assistance with technical issues and complex data queries	All users

The ECOTOX team provides comprehensive user support through multiple channels, including regularly scheduled training webinars that attract hundreds of participants [7]. These training resources are particularly valuable for leveraging ECOTOX data in the context of New Approach Methodologies, which represent a significant shift in toxicological testing toward non-animal alternatives [1] [7]. The knowledgebase also maintains interoperability with other relevant databases and tools, enhancing its utility as part of an integrated chemical safety assessment framework [4].

The historical evolution from AQUIRE, PHYTOTOX, and TERRETOX to the unified ECOTOX Knowledgebase represents a paradigm shift in how ecotoxicity data is curated, managed, and utilized for environmental protection. This transition from fragmented, specialized databases to an integrated knowledgebase has dramatically enhanced the accessibility, reliability, and utility of ecotoxicity data for regulatory decision-making and scientific research. The unified system has grown to become the **world's largest curated collection of ecotoxicity data**, embodying decades of methodological refinement and technological advancement [4] [5].

Future development of ECOTOX will likely focus on enhancing interoperability with emerging data sources and methodologies, particularly as the field continues to shift toward **New Approach Methodologies (NAMs)** that reduce reliance on animal testing [4]. The knowledgebase's role in validating and contextualizing data from these novel approaches will be increasingly important, bridging traditional apical endpoint data with mechanistic information from high-throughput in vitro assays and computational models [4]. Additionally, the ongoing adoption of **FAIR data principles** (Findable, Accessible, Interoperable, and Reusable) will further refine ECOTOX's capabilities for data exchange and integration within the broader landscape of chemical safety assessment resources [4] [5].

The continued growth and refinement of the ECOTOX Knowledgebase ensures that it will remain a cornerstone of ecological risk assessment and chemical safety evaluation, supporting the protection of ecosystems and public health through rigorous, transparent, and accessible toxicity data. Its evolution from separate databases to a unified system stands as a model for the integration of specialized scientific resources into comprehensive knowledgebases capable of addressing the complex environmental challenges of the 21st century.

The ECOTOXicology Knowledgebase (ECOTOX) is the world's largest compilation of curated ecotoxicity data, providing single-chemical adverse effects data for ecologically relevant species [4]. The following tables summarize the core quantitative data available in the knowledgebase.

Table 1: Core Data Content of ECOTOX

Data Category	Count	Description
Test Records	Over 1.1 million	Individual test results on adverse effects [7]
Chemical Stressors	~13,000	Single chemical substances with toxicity data [7]
Ecological Species	~14,000	Aquatic and terrestrial species [7]
Scientific References	Over 54,000	Peer-reviewed papers and grey literature sources [7]

Table 2: Data Applications and Users

User Group	Primary Applications	Regulatory Context
Researchers	Data mining, meta-analyses, model development (e.g., QSARs), and guiding future research [1].	Reduces reliance on direct animal testing [1].
Risk Assessors	Chemical risk characterizations, linking biological effects to mechanistic responses [1].	Supports mandates under FIFRA, Clean Water Act, CERCLA/SARA, and TSCA [4].
Decision Makers	Developing site-specific water quality criteria and interpreting environmental monitoring data [1].	Informs chemical prioritization and assessment under TSCA [1].

Experimental Protocols: Systematic Literature Review and Data Curation

The data within ECOTOX is populated through a rigorous, systematic literature review and data curation pipeline consistent with contemporary systematic review practices and PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [4]. The methodology ensures all data is FAIR - Findable, Accessible, Interoperable, and Reusable.

Literature Search and Study Selection Protocol

The process for identifying and selecting ecotoxicity studies follows a defined sequence to ensure transparency and objectivity [4]:

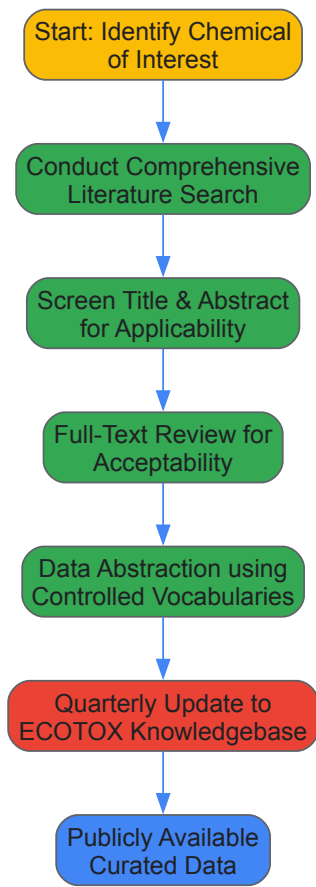
- **Literature Searching:** Comprehensive searches are conducted across the open and grey literature using verified chemicals and developed search terms [4].
- **Citation Identification:** Identified references are screened for applicability based on predefined criteria, including ecologically-relevant species, chemical of interest, and exposure concentration and duration [4].
- **Applicability & Acceptability Review:** References undergo a full-text review against criteria for applicability (e.g., biological species identified) and acceptability (e.g., documented controls and reported endpoints) [4]. Studies must meet these criteria to be included.

From each accepted study, pertinent methodological details and results are extracted into the knowledgebase [4]:

- **Data Abstraction:** Information on chemical(s), species, study design, test conditions, and results is extracted using well-established controlled vocabularies to ensure consistency [4].
- **Data Maintenance:** The knowledgebase is updated quarterly with new toxicity data and features. Standard Operating Procedures (SOPs) govern data maintenance, species verification, and chemical verification [4].

ECOTOX Data Curation Workflow

The diagram below illustrates the systematic pipeline for literature review and data curation within ECOTOX.



The Researcher's Toolkit: ECOTOX Platform Features

The following table details the key functionalities of the ECOTOX Knowledgebase that enable efficient data retrieval and analysis.

Table 3: Essential ECOTOX Research Tools

Tool / Feature	Function
Search Feature	Allows users to search for data on a specific chemical, species, effect, or endpoint. Results can be refined using 19 different parameters [1].
Explore Feature	Useful when exact search parameters are unknown. Enables browsing by Chemical, Species, or Effects with customizable output [1].

Tool / Feature	Function
Data Visualization Feature	Provides interactive data plots. Users can hover over data points and zoom in to retrieve specific information of interest [1].
Customizable Outputs	Allows users to select from over 100 data fields to create customized outputs for export and use in external applications [4].

The ECOTOXicology Knowledgebase (ECOTOX) is the world's largest compilation of curated ecotoxicity data, providing single-chemical environmental toxicity data for use in risk assessment, risk management, and research [5]. Developed by the US Environmental Protection Agency (USEPA) starting in the early 1980s, this authoritative resource has evolved to support regulatory mandates under various acts including the Federal Insecticide, Fungicide, and Rodenticide Act; the Clean Water Act; and the Toxic Substances Control Act [4]. The knowledgebase provides support for chemical safety assessments and ecological research through systematic and transparent literature review procedures that align with contemporary systematic review practices [5]. The recently released version of ECOTOX (Ver 5) provides single-chemical ecotoxicity data for over 12,000 chemicals and ecological species with over one million test results from over 50,000 references [1] [4]. This technical guide details the data curation processes and systematic review protocols that ensure the scientific rigor of the ECOTOX Knowledgebase.

ECOTOX Knowledgebase: Scope and Quantitative Dimensions

The ECOTOX Knowledgebase represents decades of systematic data curation efforts, resulting in an extensive repository of ecologically relevant toxicity information. The quantitative scope and scale of this resource is detailed in Table 1.

Table 1: Quantitative Scope of the ECOTOX Knowledgebase (Version 5)

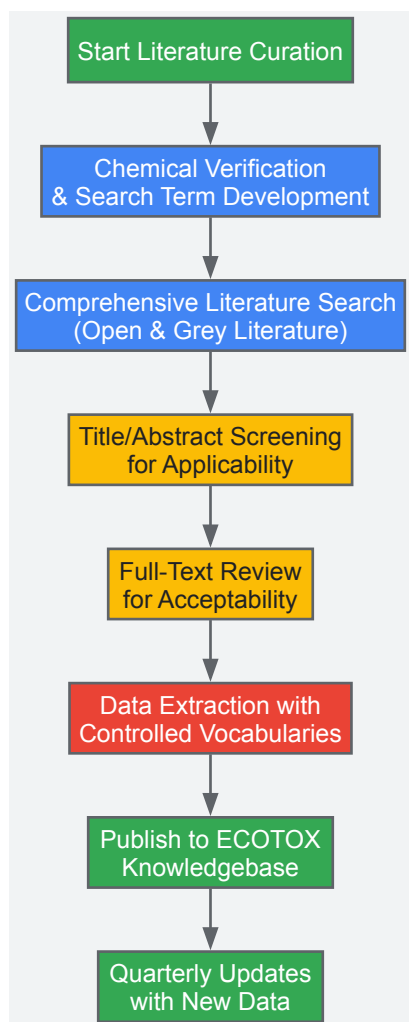
Component	Metric	Reference
Total Test Records	>1,000,000	[1] [5]
Number of Chemicals	>12,000	[1] [4]
Number of Species	>13,000 aquatic and terrestrial species	[1]
Source References	>53,000	[1]
Data Update Frequency	Quarterly	[1]

ECOTOX serves as a reliable source of curated ecological toxicity data that continues to evolve with accessible and transparent state-of-the-art practices in literature data curation [5]. The knowledgebase supports the assessment, management, and research of environmental chemicals through its comprehensive and ever-expanding collection of toxicity test results, with new extracted toxicity data added quarterly to the public website [4].

Systematic Review Framework and Data Curation Pipeline

The ECOTOX team has developed a rigorous literature search, review, and data curation pipeline to identify and provide ecological toxicity data with consistency and transparency. This process follows well-established standard operating procedures (SOPs) that are updated quarterly to reflect improvements and changes in procedures [4]. The systematic methodology ensures both **applicability** (addressing ecologically-relevant species, chemical of interest, proper species identification, and reported exposure concentration and duration) and **acceptability** (documented controls and reported endpoints) of included studies [4].

The workflow diagram below illustrates the comprehensive data curation pipeline:



Literature Search and Screening Protocol

The literature curation process begins with comprehensive searches of both open and grey literature to identify ecologically-relevant toxicity studies for chemicals of interest [4]. The systematic approach includes:

- **Chemical Verification:** Initial verification of chemicals and development of comprehensive search terms to ensure complete literature coverage [4].
- **Comprehensive Searching:** Identification of relevant toxicity studies through exhaustive searching of scientific literature using well-documented systematic and transparent procedures [1] [5].
- **Dual-Level Screening:** Implementation of a two-phase screening process where references are first screened by title and abstract, followed by full-text review to assess applicability and acceptability based on established criteria [4].
- **Inclusion Criteria:** Application of specific criteria for applicability (e.g., ecologically-relevant species, chemical of interest, properly identified biological species, reported exposure concentration and duration) and acceptability (documented controls and reported endpoints) [4].

This systematic approach aligns with contemporary systematic review methodologies and evidence-based toxicology practices, enhancing transparency, objectivity, and consistency in literature evaluation [4].

From each study meeting the applicability and acceptability criteria, relevant details on chemical(s), species, study design, test conditions, and results are extracted following well-established controlled vocabularies [5] [4]. The data abstraction process includes:

- **Structured Data Extraction:** Pertinent methodological details and results are extracted using controlled vocabularies to ensure consistency and interoperability [5].
- **Quality Control:** Implementation of standardized procedures for data verification and validation throughout the abstraction process [4].
- **Interoperability Enhancement:** Formatting data to support interoperability with chemical and toxicity databases and tools, following FAIR principles (Findable, Accessible, Interoperable, and Reusable) [5] [4].

Experimental Protocols and Methodological Framework

The experimental and methodological protocols underlying ECOTOX data curation are designed to ensure scientific rigor and reproducibility. The systematic review process follows protocols consistent with standardized guidelines for systematic reviews and systematic evidence maps [4].

Systematic Review Criteria and Application

The applicability and acceptability criteria used in ECOTOX's systematic review process serve as the foundation for ensuring data quality and relevance. These criteria are detailed in Table 2.

Table 2: Systematic Review Criteria for Study Inclusion in ECOTOX

Criterion Category	Specific Requirements	Purpose	Reference
Applicability	Ecologically relevant species	Ensure ecological relevance	[4]
Applicability	Single chemical stressor	Maintain focus on single-chemical effects	[1] [5]
Applicability	Proper biological species identification	Ensure taxonomic accuracy	[4]
Applicability	Reported exposure concentration & duration	Enable dose-response assessment	[4]
Acceptability	Documented controls	Ensure study quality and reliability	[4]
Acceptability	Clearly reported endpoints	Support quantitative analysis	[4]

Data Management and Interoperability Protocols

ECOTOX implements sophisticated data management protocols to ensure long-term usability and integration with other computational toxicology resources:

- **Controlled Vocabulary Implementation:** Use of standardized terminologies across all data fields to maintain consistency and support data integration [5].
- **Interoperability Framework:** Design of data structures to enable seamless connectivity with related resources including the CompTox Chemicals Dashboard and other computational toxicology tools [1] [5].
- **FAIR Principles Compliance:** Adherence to Findable, Accessible, Interoperable, and Reusable data principles through enhanced data accessibility and reuse capabilities [5] [4].

Research Reagent Solutions: Essential Materials for Ecotoxicology Research

The implementation of rigorous ecotoxicology research and data curation requires specific research tools and materials. Table 3 details key research reagent solutions essential for conducting ecotoxicology studies that could potentially contribute to the ECOTOX Knowledgebase.

Table 3: Essential Research Reagent Solutions for Ecotoxicology Studies

Research Reagent	Function/Purpose	Application in Ecotoxicology
Standardized Test Organisms	Model species with known sensitivity for toxicity benchmarking	Aquatic and terrestrial toxicity testing using species with established response profiles [1] [5]
Reference Toxicants	Chemical controls for quality assurance and comparative assessment	Validation of test organism sensitivity and experimental conditions [4]
Analytical Grade Chemicals	High-purity chemicals for accurate concentration-response characterization	Preparation of precise stock solutions and exposure concentrations for dose-response studies [4]
Water Quality Monitoring Kits	Measurement of critical exposure medium parameters	Monitoring and maintenance of pH, hardness, dissolved oxygen, and temperature during tests [4]
Endpoint-Specific Assays	Specialized reagents for measuring specific biological responses	Quantification of mortality, growth, reproduction, and biochemical endpoints [5] [4]

The ECOTOX Knowledgebase represents a paradigm of scientific rigor in data curation and systematic review protocols within ecotoxicology. Through its comprehensive literature search strategies, transparent study evaluation criteria, systematic data abstraction processes, and commitment to FAIR data principles, ECOTOX provides an authoritative resource that supports both regulatory decision-making and scientific research. The structured methodologies employed ensure the reliability and interoperability of ecotoxicity data while reducing the need for additional animal testing through efficient data mining of existing literature. As chemical innovation continues to advance, the systematic and rigorous approaches exemplified by ECOTOX will remain essential for protecting ecological health through science-based chemical safety assessments.

The ECOTOXicology Knowledgebase (ECOTOX) serves as a critical resource for environmental science and regulatory professionals, offering curated single-chemical toxicity data for ecological species. Developed and maintained by the U.S. Environmental Protection Agency, this comprehensive knowledgebase integrates information from over 54,000 scientific references, encompassing more than 1.1 million test records across nearly 14,000 aquatic and terrestrial species and 13,000 chemicals [7]. This technical guide examines the specialized applications of ECOTOX across three key user groups—researchers, risk assessors, and decision-makers—detailing specific use cases, methodologies, and quantitative data that support informed environmental protection efforts.

ECOTOX Knowledgebase: Scope and Scale

ECOTOX provides a systematically curated collection of ecotoxicity data abstracted from peer-reviewed literature using standardized review procedures that align with contemporary systematic review practices [4]. The knowledgebase is updated quarterly with new data and features, ensuring access to current scientific information [1]. The following table summarizes the quantitative scope of the database.

Table 1: Quantitative Scope of the ECOTOX Knowledgebase

Data Category	Volume	Source
Scientific References	>54,000	[7]
Test Records	>1.1 million	[7]
Aquatic/Terrestrial Species	~14,000	[7]
Chemicals	~13,000	[7]
Data Accessibility	Publicly available, updated quarterly	[1]

Applications for Researchers

Data Mining and Model Development

Researchers utilize ECOTOX's extensive dataset to develop and validate predictive toxicity models and conduct meta-analyses. The database supports:

- **Quantitative Structure-Activity Relationship (QSAR) Modeling:** Physical and chemical properties from ECOTOX help build models that predict toxicity based on molecular structures [1].
- **Species Sensitivity Distributions (SSDs):** Data from multiple species enable statistical determination of chemical concentration thresholds protective of ecological communities [4].
- **New Approach Methodologies (NAMs):** ECOTOX provides empirical in vivo data essential for developing, evaluating, and validating alternative testing methods, including in vitro assays and computational models [4].

Experimental Design and Data Gap Analysis

The knowledgebase aids in identifying research priorities by highlighting areas where toxicity data are lacking. Researchers can efficiently determine:

- Understudied chemicals or species
- Taxonomic groups with limited toxicity information
- Exposure durations or endpoints with insufficient data

Table 2: Key Research Reagent Solutions in ECOTOX

Research Tool	Function	Application Example
Chemical Stressor Data	Single-chemical toxicity information	Mechanistic studies, baseline data for mixture toxicity
Taxonomic Effect Data	Species-specific sensitivity patterns	Cross-species extrapolation, sensitive species identification
Experimental Condition Parameters	Exposure duration, media, endpoints	Test protocol development, methodological standardization
ECOTOX-Integrated Tools (e.g., CompTox Dashboard)	Chemical property data, structure information	Read-across approaches, chemical category development

Applications for Risk Assessors

Ecological Risk Assessment Framework

Risk assessors employ ECOTOX within the established ecological risk assessment (ERA) framework, which comprises three core phases: Problem Formulation, Analysis, and Risk Characterization [8]. The following diagram illustrates how ECOTOX integrates into this workflow.

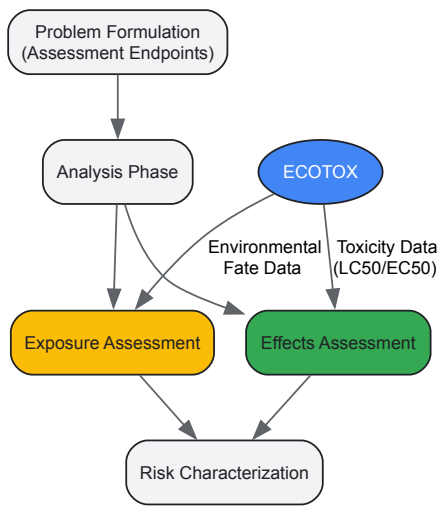


Diagram 1: ECOTOX Integration in Ecological Risk Assessment

Data Evaluation Protocols

Risk assessors follow standardized protocols to evaluate ecotoxicological studies from ECOTOX for regulatory decision-making. The evaluation process includes specific criteria for determining study relevance and reliability [9] [10].

Table 3: ECOTOX Data Evaluation Guidelines for Risk Assessment

Evaluation Phase	Criteria	Regulatory Application
Study Acceptability	- Single chemical exposure- Biological effect on live, whole organisms- Explicit exposure duration- Concurrent control group- Reported chemical concentration	Initial screening for regulatory consideration
Exposure Relevance	- Test substance representativeness- Environmental media realism- Exposure regime appropriateness	Determining environmental applicability
Biological Relevance	- Ecological significance of test organism- Protection goal alignment- Endpoint ecological meaningfulness	Linking effects to assessment endpoints

Supporting Regulatory Mandates

ECOTOX provides critical data for multiple regulatory frameworks:

- **Chemical Registration and Reregistration:** Informs ecological risk assessments for pesticides and industrial chemicals under FIFRA and TSCA [1].
- **Water Quality Criteria Development:** Supports derivation of ambient water quality criteria for protection of aquatic life [1].
- **Superfund Site Remediation:** Provides toxicity data for ecological risk assessments at contaminated sites [8].

Applications for Decision Makers

Environmental Policy and Management

Decision makers at local, state, tribal, and federal levels use ECOTOX to inform environmental protection policies and site-specific management actions. Key applications include:

- **Development of Site-Specific Criteria:** Creating chemical thresholds tailored to local ecosystems when national criteria may not be protective enough [1].

- **Monitoring Data Interpretation:** Contextualizing chemical detection data from environmental monitoring programs with biological effects information [1].
- **Remediation Priority Setting:** Identifying chemicals of greatest ecological concern at contaminated sites to guide cleanup efforts [8].

Risk Management Decision Support

ECOTOX helps decision makers evaluate risk management alternatives by providing:

- Comparative toxicity data for chemical alternatives assessment
- Species-specific sensitivity data for protecting vulnerable populations
- Scientific justification for regulatory standards and cleanup levels

Experimental Protocols and Methodologies

ECOTOX Data Curation Workflow

The ECOTOX team employs systematic literature review and data curation procedures that align with PRISMA guidelines for systematic reviews [4]. The following diagram illustrates this multi-stage process.

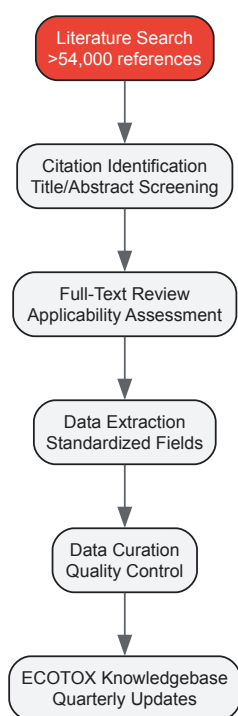


Diagram 2: ECOTOX Systematic Literature Review and Data Curation Pipeline

Key Experimental Data Extraction

For each study meeting acceptability criteria, ECOTOX curators extract detailed information across several domains:

- **Chemical Information:** Identification (CAS RN, DTXSID), properties, verification
- **Species Data:** Taxonomic classification, life stage, source, verification
- **Test Conditions:** Exposure duration, route, media, temperature, endpoints
- **Results:** Quantitative effect values (LC50, EC50, NOEC), statistical measures

Emerging Applications and Future Directions

Integration with New Approach Methodologies (NAMs)

ECOTOX supports the development and validation of NAMs that reduce reliance on animal testing while maintaining environmental protection standards. Specific applications include:

- **Computational Toxicology:** Providing empirical data for training and validating machine learning models and QSAR applications [4] [11].
- **High-Throughput Screening:** Serving as a source of in vivo data for comparing biological pathway activities across species [12].

Behavioral Ecotoxicology

There is growing recognition of behavioral endpoints as sensitive indicators of chemical effects, though standardization for regulatory use remains in development [13]. ECOTOX includes behavioral effect data when reported in the literature, supporting this emerging field.

Access and Technical Support

Users access ECOTOX through multiple functionality options:

- **Search Feature:** Targeted queries by chemical, species, effect, or endpoint with filtering across 19 parameters [1].
- **Explore Feature:** Flexible searching when exact parameters are unknown [1].
- **Data Visualization:** Interactive plots for data exploration and analysis [1].
- **Training Resources:** Available through EPA's NAMs Training Program, including instructional videos and worksheets [1].

Technical assistance is available through ECOTOX Support (ecotox.support@epa.gov), with training resources regularly updated to support user proficiency [1].

Practical Applications: Leveraging ECOTOX for Chemical Assessment and Risk Management

The **Ecotoxicology (ECOTOX) Knowledgebase** is a comprehensive, publicly available application developed by the U.S. Environmental Protection Agency that provides critical information on adverse effects of single chemical stressors to ecologically relevant aquatic and terrestrial species [1]. This sophisticated database serves as a foundational resource for researchers, risk assessors, and decision-makers working in environmental toxicology and chemical risk assessment. The platform's architecture is specifically designed to support complex ecotoxicological queries through three primary functional modalities: **Search**, **Explore**, and **Data Visualization** [1]. For ecotoxicology professionals, understanding how to effectively navigate these core features is essential for leveraging the full potential of this extensive knowledge repository, which contains over one million test records compiled from more than 53,000 references, covering more than 13,000 species and 12,000 chemicals [1].

The integration of these three core functionalities creates a powerful framework for conducting ecotoxicological assessments, chemical prioritization, and ecological risk evaluation. The quantitative data available through these interfaces support various applications including the development of chemical benchmarks for water and sediment quality assessments, design of aquatic life criteria, ecological risk assessments for chemical registration, and prioritization of chemicals under regulatory frameworks like the Toxic Substances Control Act (TSCA) [1].

Search Functionality: Precision Query Capabilities

Core Search Parameters and Methodology

The **Search** feature provides a targeted approach for retrieving specific ecotoxicological data using defined parameters. This functionality enables users to conduct precise queries based on four primary categories, each serving distinct research needs as detailed in the table below [1].

Table 1: Primary Search Categories in ECOTOX Knowledgebase

Search Category	Description	Application in Ecotoxicology Research
Chemical	Search for specific chemicals with linkage to EPA's CompTox Chemicals Dashboard	Identify toxicity data for particular contaminants; includes over 12,000 chemicals
Species	Query by aquatic or terrestrial species	Assess species-specific sensitivity; covers over 13,000 species
Effect	Search based on biological effects or endpoints	Investigate specific adverse outcome pathways
Endpoint	Target particular measurement endpoints	Locate specific experimental results and metrics

The chemical search capability is particularly robust, allowing researchers to identify compounds using various identifiers including Chemical Abstracts Service Registry Numbers (CASRN), chemical names, or DTXSIDs (DSSTox Substance IDs) [1]. This interoperability with the CompTox Chemicals Dashboard provides additional layers of chemical information, enhancing the contextual understanding of ecotoxicological results.

Search Refinement and Output Customization

A critical strength of the ECOTOX Search functionality is the capacity to refine and filter results through **19 distinct parameters**, enabling researchers to narrow results to highly specific experimental conditions or outcomes [1]. Following query execution, users can customize output

selections from **over 100 data fields**, providing exceptional granularity in results reporting. This precision is particularly valuable for meta-analyses and systematic reviews where consistent data reporting across studies is essential.

The search refinement protocol follows a structured methodology:

- **Initial Query Execution:** Conduct broad search using primary categories (Chemical, Species, Effect, or Endpoint)
- **Results Filtering:** Apply relevant filters from the 19 available parameters
- **Output Selection:** Customize data fields for export (up to 100 fields available)
- **Data Export:** Download filtered results for further analysis

This methodological approach ensures reproducible results and enables researchers to tailor extraction to specific research questions or assessment needs.

Explore Functionality: Investigative Data Discovery

Exploratory Search Paradigm and Workflow

The **Explore** feature operates on a fundamentally different paradigm than the standard Search function, designed for situations where researchers lack precisely defined query parameters [1]. This functionality supports investigative research and hypothesis generation by allowing more open-ended navigation through the database's content. The exploratory approach is particularly valuable during preliminary literature reviews, novel chemical assessment, or when investigating emerging contaminants where established research parameters may not yet be defined.

The exploratory methodology follows a flexible workflow:

- **Initial Entry Point:** Begin with broad categories (Chemical, Species, or Effects)
- **Iterative Refinement:** Progressively narrow focus based on initial results
- **Pattern Identification:** Discover relationships and data trends not apparent with structured queries
- **Output Customization:** Configure results for integration with external analytical tools

This exploratory capability makes the ECOTOX Knowledgebase particularly valuable for investigating non-traditional model organisms or novel toxicological endpoints, supporting the expansion of ecotoxicological understanding beyond conventional test species.

Data Integration and Export Protocols

A critical feature of the Explore functionality is its capacity for data export formatted for seamless integration with external analytical tools [1]. Researchers can customize output fields to align with specific analytical requirements, whether for statistical analysis, modeling applications, or visualization in specialized software platforms. This interoperability is essential for modern ecotoxicology research that often involves multi-step analytical workflows across specialized applications.

The export protocol supports various downstream applications including quantitative structure-activity relationship (QSAR) modeling, meta-analyses, data gap analyses, and ecological risk assessment model parameterization. This flexibility addresses the growing need for integrated approaches in chemical safety assessment that incorporate both traditional and new approach methodologies (NAMs) [1].

Data Visualization Features: Analytical Interpretation Tools

Interactive Visualization Capabilities

The **Data Visualization** features represent a sophisticated component of the ECOTOX interface, providing dynamic graphical representation of query results [1]. These interactive plots enable researchers to identify patterns, trends, and outliers within complex ecotoxicological datasets through visual analytics. The platform's implementation of interactive elements includes hover-over data point inspection and zoom functionality, allowing users to investigate specific data regions of interest while maintaining context within the broader dataset.

The visualization methodology employs a structured approach:

- **Automated Plot Generation:** Visual representations created automatically from query results
- **Interactive Exploration:** Hover functionality reveals detailed data point information
- **Focus Investigation:** Zoom capabilities enable examination of specific data regions
- **Pattern Recognition:** Visual identification of trends, clusters, and outliers

These visualization tools are particularly valuable for dose-response analysis, cross-species sensitivity comparisons, and chemical hazard ranking, providing intuitive understanding of complex toxicological relationships.

Application in Ecotoxicological Data Interpretation

The Data Visualization features transform tabular ecotoxicological data into interpretable graphical formats that enhance scientific insight. For example, researchers can visually compare toxicity distributions across multiple species, examine concentration-response relationships across studies, or identify gaps in available data for specific chemical classes. This visual approach to data exploration facilitates more nuanced interpretation of ecotoxicological patterns than traditional tabular reviews.

The interactive nature of these visualizations supports iterative investigation, allowing researchers to quickly test hypotheses about relationships within the data and refine their analytical approach based on visual feedback. This capability aligns with established scientific methodology of iterative hypothesis testing and is particularly valuable for exploring large, complex datasets characteristic of modern ecotoxicology.

Integrated Workflow: From Query to Visualization

Experimental Protocol for Ecotoxicological Data Mining

A comprehensive methodology for leveraging the ECOTOX interface involves sequential application of all three core features, creating an integrated workflow for ecotoxicological data mining and interpretation. The following workflow diagram illustrates this integrated experimental protocol:

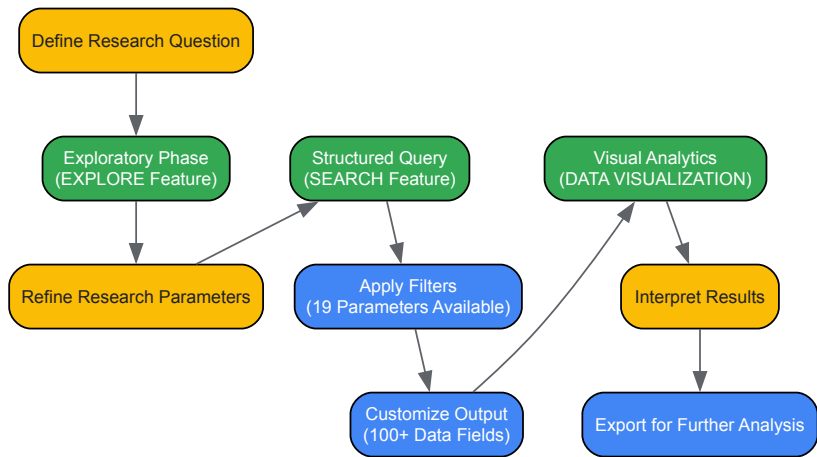


Diagram 1: ECOTOX Integrated Research Workflow

This integrated methodology transforms raw data into actionable ecological insights through a systematic process of discovery, refinement, and interpretation. The protocol emphasizes the complementary nature of the three core features, with each phase building upon the previous to progressively refine understanding of the ecotoxicological question under investigation.

Application in Ecological Risk Assessment

The integrated workflow finds particular application in ecological risk assessment, where researchers must synthesize diverse toxicity data across multiple species and endpoints to characterize chemical hazards. The Search function enables targeted retrieval of toxicity thresholds (e.g., LC50, EC50 values), the Explore function supports identification of relevant test species and sensitive endpoints, and the Data Visualization features facilitate comparison of species sensitivity distributions and derivation of protective concentration values.

This methodological approach aligns with established ecological risk assessment frameworks, including the development of water quality criteria and sediment quality guidelines, where multiple lines of evidence from various aquatic and terrestrial species must be integrated and interpreted [1]. The capacity to efficiently navigate, filter, and visualize these complex datasets significantly enhances the robustness and transparency of ecological risk determinations.

Advanced Technical Applications

Ecotoxicological Modeling and Predictive Toxicology

The ECOTOX interface supports advanced technical applications in ecotoxicological modeling and predictive toxicology through its data export and integration capabilities. The structured data outputs facilitate the development and validation of quantitative structure-activity relationship (QSAR) models, species sensitivity distributions (SSDs), and other predictive approaches that require high-quality, curated toxicity data [1].

The platform's application programming interfaces and data export functionalities enable seamless integration with statistical analysis environments and modeling platforms, creating efficient workflows for high-throughput hazard characterization. This interoperability is particularly valuable for addressing data gaps for chemicals with limited testing data through read-across and quantitative property-property relationship approaches.

Table 2: Research Reagent Solutions for ECOTOX Data Analysis

Tool/Resource	Function	Application Context
CompTox Chemicals Dashboard	Chemical identifier conversion and property data	Contextualizing chemical structures and properties for toxicity interpretation
ToxVal Database	Access to curated toxicity values	Benchmarking and comparison of toxicity results
Quantitative Structure-Activity Relationship (QSAR) Models	Prediction of toxicity based on chemical structure	Addressing data gaps for untested chemicals
Species Sensitivity Distribution (SSD) Models	Statistical analysis of interspecies sensitivity	Deriving protective thresholds for chemical concentrations in the environment
High-Throughput Screening Data	Bioactivity profiles from ToxCast/Tox21	Mechanistic insight into toxicity pathways

The Search, Explore, and Data Visualization features of the ECOTOX Knowledgebase represent a sophisticated trio of interconnected functionalities that support comprehensive ecotoxicological investigation. The methodological approaches outlined in this technical guide provide researchers with structured protocols for leveraging these capabilities across various research contexts, from targeted chemical assessment to exploratory investigation of emerging contaminants. As the field of ecotoxicology continues to evolve toward greater integration of computational approaches and predictive methodologies, the capacity to efficiently navigate, extract, and visualize curated toxicity data will remain fundamental to advancing ecological risk assessment and chemical safety evaluation.

The exponential growth in chemical production has intensified the demand for precise and efficient toxicity assessments. Targeted data retrieval through chemical and species-specific queries addresses this need by enabling researchers to extract highly relevant ecotoxicological information from vast knowledgebases. The U.S. Environmental Protection Agency's Ecotoxicology Knowledgebase (ECOTOX) has emerged as a pivotal resource in this domain, providing curated single-chemical toxicity data for over 12,000 chemicals and 14,000 ecological species compiled from more than 54,000 scientific references [7] [4]. This technical guide examines the methodologies, protocols, and applications of targeted query strategies within the context of systematic ecological risk assessment and drug development workflows.

The evolution of ECOTOX represents a significant advancement in ecotoxicological data management. Originally developed in the 1980s as multiple ecosystem-specific databases, it transformed into a unified web-accessible system in 2000 [14]. The recently released ECOTOX Version 5 incorporates over one million test records and embodies the FAIR principles (Findable, Accessible, Interoperable, and Reusable) through enhanced data queries, retrieval options, and visualization tools [4]. For research professionals, mastering the construction of precise chemical and species-specific queries is fundamental to leveraging this resource for chemical screening, risk assessment, and predictive modeling applications.

ECOTOX Knowledgebase: Architecture and Systematic Review Foundation

Data Curation and Quality Assurance Pipeline

The utility of ECOTOX for targeted queries is underpinned by its rigorous, systematic literature review and data curation pipeline, which shares methodologies with contemporary systematic review practices [4]. The process begins with comprehensive searches of open and grey literature, followed by a multi-tiered screening process. References are initially screened by title and abstract, then progress to full-text review against strict applicability and acceptability criteria, including ecologically relevant species, proper chemical identification, reported exposure concentrations and durations, and documented control conditions [4].

The data abstraction phase extracts detailed information on chemicals, species, study design, test conditions, and results using well-established controlled vocabularies, ensuring consistency and interoperability [4]. This meticulous process is documented through Standard Operating Procedures (SOPs) for literature searches, citation identification, applicability criteria, data abstraction, and data maintenance. The resulting structured data supports complex querying capabilities essential for precise chemical and species-specific investigations. The system's quarterly updates incorporate newly extracted toxicity data, maintaining currentness for research and regulatory applications [4].

Data Scope and Coverage

Table 1: ECOTOX Knowledgebase Content Scope

Category	Scale of Data	Description
Chemical Coverage	Over 13,000 chemicals	Includes pesticides, industrial chemicals, metals, and emerging contaminants like PFAS and 6-PPD quinone [7] [4].
Species Coverage	Nearly 14,000 aquatic and terrestrial species	Represents aquatic and terrestrial organisms, including plants, invertebrates, fish, amphibians, and birds [7] [4].

Category	Scale of Data	Description
Test Records	Over 1.1 million effect results	Individual toxicity measurements from controlled studies [7].
Scientific References	Over 54,000 sources	Peer-reviewed literature and grey literature, with continuous quarterly updates [7] [4].

Designing Chemical-Specific Query Strategies

Chemical Identification and Verification

Chemical-specific queries require precise identifier resolution to ensure comprehensive data retrieval. Successful query formulation begins with chemical verification using standardized nomenclature and identifiers. Researchers should utilize systematic chemical naming conventions (IUPAC), common names, and CAS Registry Numbers (CAS RN) to address synonym variability [4] [15]. The ECOTOX system incorporates chemical verification procedures that resolve these synonyms to ensure accurate data linkage, a critical step given that substances may appear under different names across the scientific literature [4].

For complex chemical categories, specialized approaches are necessary. The database handles metal speciation, chemical categories, and metabolites through defined protocols. For example, chromium toxicity data is speciated between hexavalent and trivalent states, while polycyclic aromatic compounds (PACs) apply a standardized relative potency factor of 18% relative to benzo[a]pyrene [16]. Understanding these category treatments is essential for accurate data interpretation when querying complex chemical groups.

Advanced Chemical Query Methodologies

Beyond basic identifier searches, advanced chemical query strategies incorporate structure-based screening and toxicity prediction. The FDA's Expanded Decision Tree (EDT) tool represents a significant advancement in this area, providing a systematic, science-based approach to evaluate chemical safety based on structural features and estimated toxicity [17]. This methodology is particularly valuable for prioritizing data-poor chemicals, as it uses a refined set of chemical structure-based questions to classify compounds according to their predicted chronic toxic potential [17].

For regulatory applications, integrating chemical-specific queries with toxicity weighting algorithms enhances risk assessment capabilities. The Risk-Screening Environmental Indicators (RSEI) model exemplifies this approach by calculating toxicity weights using standardized factors such as Oral Slope Factors (OSF), Inhalation Unit Risks (IUR), Reference Doses (RfD), and Reference Concentrations (RfC) [16]. These weights are derived from hierarchical data sources including EPA's Integrated Risk Information System (IRIS), Provisional Peer-Reviewed Toxicity Values (PPRTVs), and other consensus sources [16].

Implementing Species-Specific Query Protocols

Taxonomic Resolution and Species Sensitivity Distributions

Species-specific queries require precise taxonomic identification to ensure ecologically relevant data retrieval. The ECOTOX knowledgebase employs controlled vocabularies and taxonomic verification procedures to standardize species nomenclature across records, resolving common synonyms and classification inconsistencies [4]. Query strategies should incorporate hierarchical taxonomic descriptors (phylum, class, order, family, genus, species) to balance specificity with comprehensive retrieval, particularly when investigating related species with potentially similar sensitivity patterns.

The application of species sensitivity distributions (SSDs) represents a powerful analytical framework for extrapolating species-specific toxicity data to ecosystem-level risk assessments [4]. By querying ECOTOX for multiple species within a defined taxonomic group or functional guild, researchers can develop SSD curves that estimate chemical concentrations protective of most species in a community. This methodology transforms discrete species-specific toxicity values into probabilistic risk estimates, supporting regulatory decisions and environmental quality standard derivation [4].

Endpoint-Specific and Life Stage Query Strategies

Targeting specific biological endpoints and life stages refines species-specific query outcomes. The knowledgebase captures extensive endpoint information across multiple levels of biological organization, including biochemical, physiological, behavioral, reproductive, and population-level responses [4] [14]. Effective query construction specifies these endpoints to filter results for particular adverse outcome pathways or regulatory requirements, such as reproductive toxicity for endocrine disruptor screening or neurobehavioral effects for neurotoxicant assessment.

Life stage specificity significantly influences toxicity outcomes, particularly for amphibian, fish, and invertebrate species. Query strategies should incorporate life stage descriptors (e.g., embryo, larval, juvenile, adult) when available, as sensitivity to chemical exposure often varies dramatically throughout development [4]. The ECOTOX data structure supports such refined queries through controlled vocabularies that standardize life stage terminology across studies and species groups.

Experimental Protocols for Ecotoxicity Testing

Standardized Test Guidelines and Methodological Documentation

Retrieving comparable toxicity data requires understanding standardized testing methodologies. Regulatory agencies worldwide have established test guidelines for assessing chemical effects on aquatic and terrestrial species, covering acute toxicity, chronic toxicity, reproductive effects, and developmental toxicity [4] [15]. When querying databases like ECOTOX, knowledge of these methodological frameworks aids in filtering studies based on reliability and relevance criteria similar to those outlined in Klimisch scoring systems [4].

The evaluation of toxicity test quality follows fundamental scientific criteria, including clear description of exposure conditions (dose, route, duration), chemical characterization (purity, stability), use of appropriate controls, and measurement of biologically relevant endpoints [15]. These criteria align with the systematic review methodologies employed in ECOTOX curation, ensuring that extracted data meets minimum quality standards for use in risk assessments and research applications [4].

Data Extraction and Curation Workflows

The process of transforming primary literature into queryable database records follows a defined protocol. The ECOTOX curation pipeline extracts detailed experimental parameters including test duration, exposure route, chemical measurement, endpoint measurement, response values, and statistical significance [4]. This structured extraction methodology enables complex query combinations, such as identifying all freshwater fish species showing significant reproductive effects after 60-day exposures to a specific pesticide.

Table 2: Key Experimental Parameters for Ecotoxicity Data Queries

Parameter Category	Specific Elements	Application in Query Refinement
Test Organism	Species, life stage, sex, source, acclimation	Defines ecological relevance and applicability domain.
Exposure Conditions	Route, duration, medium, temperature, loading rate	Supports cross-study comparability and test condition filtering.
Chemical Verification	Purity, formulation, analytical verification, concentration	Ensures data reliability and appropriate toxicity value assignment.
Endpoint Measurement	Mortality, growth, reproduction, behavior, biochemical	Allows targeting of specific adverse outcome pathways.
Result Metrics	LC50, EC50, NOEC, LOEC, MATC	Facilitates quantitative comparisons and meta-analyses.

Data Analysis, Interpretation, and Visualization

Toxicity Weighting and Comparative Risk Screening

Advanced analysis of queried toxicity data often incorporates toxicity weighting algorithms to support comparative risk assessment. The RSEI model exemplifies this approach by calculating toxicity weights using standardized factors: for carcinogenic effects via inhalation ($IUR/2.8e-7$) and oral exposure ($OSF/1e-6$), and for noncancer effects via inhalation ($3.5/RfC$) and oral exposure ($1/RfD$) [16]. These calculations transform toxicity values into comparable units that can be combined with exposure estimates to generate risk-based screening scores.

The application of these weights varies depending on assessment objectives. The RSEI Score uses the higher of cancer or noncancer weights for each exposure route, while the RSEI Cancer Score and RSEI Noncancer Score use only the relevant cancer or noncancer weights [16]. Understanding these distinctions is crucial for appropriate data interpretation when exporting ECOTOX data for use in risk-screening models and regulatory decision support systems.

Data Visualization and Interoperability

Effective visualization of queried ecotoxicity data enhances pattern recognition and data exploration. The ECOTOX Version 5 interface incorporates visualization tools to aid in data analysis, alongside customizable outputs for export to external applications [4]. For quantitative data summary, histograms effectively display frequency distributions of toxicity values across species or chemicals, while frequency polygons can compare sensitivity distributions between taxonomic groups [18].

Interoperability with other data resources expands the utility of queried ecotoxicity data. ECOTOX supports integration with chemical databases, toxicity prediction tools, and regulatory assessment platforms, enabling integrated workflows that combine empirical data with in silico predictions [4]. This interoperability is particularly valuable for drug development professionals applying New Approach Methodologies (NAMs) that leverage large datasets for faster chemical assessment while reducing animal testing [17] [4].

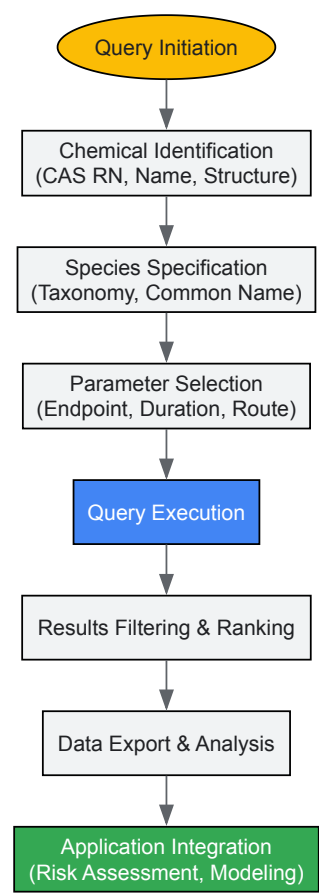


Diagram 1: Chemical and species-specific query workflow for ecotoxicity data retrieval. The process begins with precise identifier resolution and progresses through parameter selection to application integration.

The Scientist's Toolkit: Research Reagent Solutions

Table 3: Essential Research Resources for Ecotoxicology Investigations

Tool/Resource	Function	Application Context
ECOTOX Knowledgebase	Centralized repository of curated single-chemical toxicity data for ecological species [7] [4].	Primary source for chemical- and species-specific toxicity data supporting risk assessments and research.
Expanded Decision Tree (EDT)	Structure-based screening tool for predicting chronic toxic potential of chemicals, especially with limited test data [17].	Chemical prioritization and initial hazard assessment in food chemical safety and drug development.
RSEI Toxicity Weights	Calculated metrics for comparing relative toxicity of chemicals across exposure routes and endpoints [16].	Chemical ranking, comparative risk assessment, and screening-level risk modeling.
IRIS, PPRTVs, HEAST	Hierarchical sources of toxicity values and reference concentrations for human health and ecological risk assessment [16].	Derivation of toxicity criteria and weight-of-evidence evaluations for chemical assessments.
GIS Exposure Tools	Spatial analysis systems incorporating environmental parameters to model chemical fate and exposure [14].	Landscape-scale risk assessment and prediction of chemical impacts on non-target areas and species.

Case Study Application: Integrated Query Strategy

Problem Formulation and Query Design

Implementing an integrated chemical and species-specific query strategy is illustrated through a case study investigating the aquatic toxicity of a hypothetical pesticide. The assessment begins with precise chemical identification using CAS RN and systematic nomenclature, followed by definition of relevant assessment populations (freshwater fish, aquatic invertebrates, algae). Query parameters are then structured to retrieve acute and chronic toxicity data for specific endpoints (mortality, growth, reproduction) across trophic levels.

The query incorporates taxonomic hierarchy to address data gaps through read-across approaches, leveraging toxicity information from taxonomically related species when data for specific assessment species are limited [4]. This strategy employs the systematic search and retrieval capabilities of ECOTOX while applying logical operators to balance dataset comprehensiveness with relevance to the specific assessment context.

Data Analysis and Application

Following data retrieval, the case study applies toxicity weighting algorithms to normalize response values across studies and endpoints [16]. The resulting dataset supports species sensitivity distribution analysis, identifying the most sensitive taxa and calculating protective concentration thresholds [4]. For drug development applications, the data further informs mode-of-action analyses through integration with adverse outcome pathway frameworks, linking molecular initiating events to population-relevant outcomes.

This integrated approach demonstrates how targeted queries transform raw ecotoxicity data into decision-relevant information, supporting chemical prioritization, regulatory risk assessment, and research hypothesis generation. The case study underscores the critical importance of precise query design in extracting meaningful patterns from complex toxicological databases, ultimately enhancing the scientific foundation for chemical management decisions across regulatory and development contexts.

Ecotoxicology, the study of the effects of toxic chemicals on biological organisms in natural environments, generates complex and multi-faceted data. The **ECOTOX Knowledgebase**, maintained by the U.S. Environmental Protection Agency, serves as a comprehensive repository containing over one million test records covering more than 13,000 aquatic and terrestrial species and 12,000 chemicals, compiled from over 53,000 scientific references [1]. For researchers, scientists, and drug development professionals, effectively navigating this vast dataset requires sophisticated filtering strategies. Advanced customization using **duration parameters**, **endpoint selection**, and **test condition refinements** is not merely a convenience but a necessity for extracting biologically relevant and regulatory-ready data. This technical guide provides an in-depth framework for leveraging these advanced parameters to enhance the precision and reliability of ecotoxicological queries, ultimately supporting more accurate ecological risk assessments and research outcomes.

Core Filtering Parameters: A Detailed Technical Examination

Duration Parameters: Defining Temporal Exposure Windows

The duration of chemical exposure is a critical determinant of toxicological outcomes, with standardized test periods established for different taxonomic groups. Filtering by exposure time is essential for comparing data across studies and for extrapolating results to specific regulatory or environmental scenarios.

Table 1: Standardized Test Durations for Major Taxonomic Groups in Ecotoxicology

Taxonomic Group	Standard Acute Test Duration	Common Sub-Acute/Chronic Durations	Primary Regulatory Guideline Examples
Fish	96 hours [11]	7-28 days (early life-stage, growth)	OECD Test Guideline 203 [11]
Crustaceans	48 hours [11]	21 days (reproduction, life-cycle)	OECD Test Guideline 202 [11]
Algae	72-96 hours [11]	Not typically defined	OECD Test Guideline 201 [11]

Implementation Strategy: When querying a knowledgebase, researchers should filter for data within these standardized windows to ensure comparability. For instance, when assessing acute toxicity in fish, applying a filter of `Duration = 96 hours` will retrieve the most relevant and directly comparable LC50 (Lethal Concentration for 50% of the population) values. It is also critical to exclude data from early life stages (e.g., eggs, embryos) if the goal is to assess toxicity to juvenile or adult organisms, as the toxicokinetics and sensitivity can differ significantly [11].

Endpoint Selection: Differentiating Measures of Effect

The biological endpoint is the specific measurable outcome used to quantify a chemical's effect. Different endpoints provide information on distinct types of toxicity, from mortality to sub-lethal impairments.

Table 2: Key Ecotoxicological Endpoints for Advanced Filtering

Endpoint	Full Name	Definition & Measurement	Commonly Applied To
LC50	Lethal Concentration 50	Concentration causing 50% mortality in a test population.	Fish, Crustaceans [11]
EC50	Effective Concentration 50	Concentration causing a 50% effect in a non-lethal endpoint (e.g., immobilization).	Crustaceans (Immobilization) [11]

Endpoint	Full Name	Definition & Measurement	Commonly Applied To
NOEC	No Observed Effect Concentration	Highest tested concentration where no statistically significant effect is observed.	All taxa (chronic studies) [2]
LOEC	Lowest Observed Effect Concentration	Lowest tested concentration where a statistically significant effect is observed.	All taxa (chronic studies) [2]
Growth Inhibition	-	Measurement of reduced biomass or population growth.	Algae, Plants [11]

Implementation Strategy: A precise query must isolate the intended effect. For example, in crustaceans, the endpoint "intoxication (ITX)" often includes immobilization, which is used interchangeably with mortality in standardized guidelines [\[11\]](#). Therefore, a comprehensive search for acute crustacean toxicity should include both `Mortality (MOR)` and `Intoxication (ITX)` endpoints. For algae, where individual mortality is less relevant, endpoints related to `Population (POP)`, `Growth (GRO)`, and `Physiology (PHY)` are more appropriate for assessing toxicity [\[11\]](#).

Test Condition Parameters: Contextualizing the Bioassay

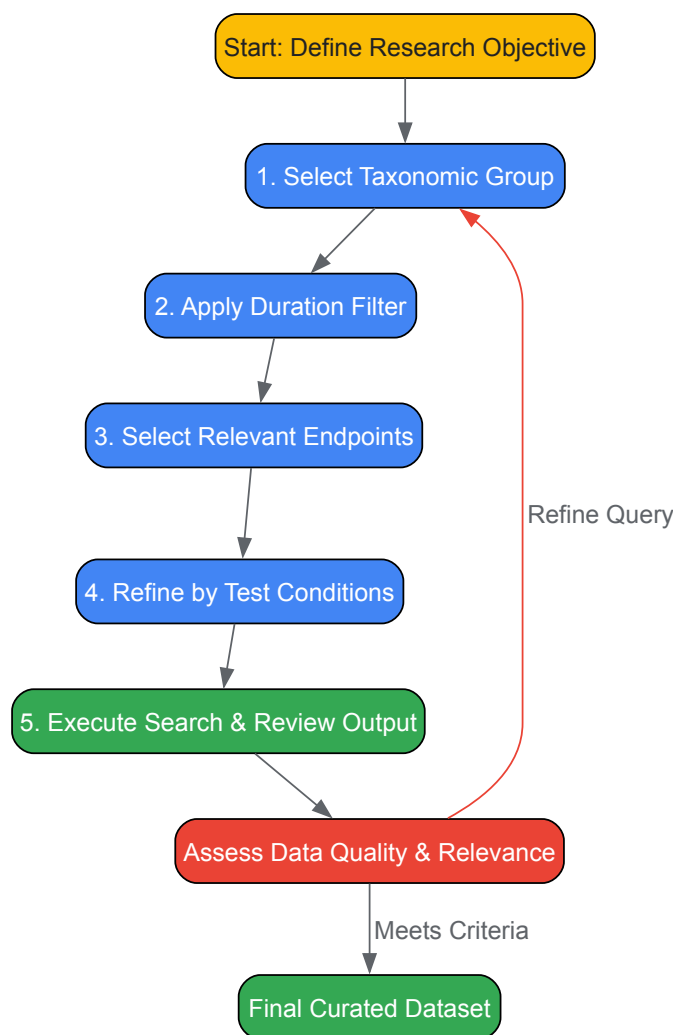
The environmental and methodological context of a test can profoundly influence its outcome. Filtering by test conditions ensures that the retrieved data is relevant to the specific environment or scenario under investigation.

Key test condition parameters include:

- **Test Medium:** The medium in which the test was conducted (e.g., fresh water, salt water, sediment) is a primary filter. Toxicity values can vary dramatically between freshwater and marine environments due to differences in water chemistry and species sensitivity.
- **Laboratory vs. Field Studies:** Laboratory studies offer controlled conditions, while field studies provide greater environmental realism but with more confounding variables. A robust assessment may require data from both, but they should be analyzed with an understanding of their respective limitations.
- **Temperature & pH:** These abiotic factors can influence chemical bioavailability and organism metabolism. Filtering for studies within an environmentally relevant range can improve the accuracy of an ecological risk assessment.

Integrated Workflow for Advanced Data Retrieval

The following diagram illustrates the logical decision process for applying advanced filters to build a precise and relevant ecotoxicology dataset.



Experimental Protocol for Data Extraction and Validation

The methodology for extracting and curating data from a knowledgebase like ECOTOX is critical for ensuring the reliability of subsequent analysis. The following protocol is adapted from established practices in the field [\[10\]](#) [\[11\]](#).

Step 1: Problem Formulation and Scope Definition Clearly define the chemical(s), species of interest, and the environmental compartment (e.g., freshwater aquatic). This determines the core search strategy.

Step 2: Initial Data Retrieval and Harmonization

- Query the knowledgebase (e.g., ECOTOX) for the target chemical and species.
- Download and harmonize data from relevant tables (e.g., species, tests, results).
- Retain critical identifiers like CAS numbers, DTXIDs, and InChIKeys to ensure traceability and compatibility with other chemical databases [\[11\]](#).

Step 3: Application of Core Filters

- **Taxonomy Filter:** Retain only entries for the relevant taxonomic groups (e.g., "Fish", "Crustacea", "Algae").
- **Duration Filter:** Filter the `observation_time_hr` column to include only entries within the standardized durations (e.g., 96h for fish, 48h for crustaceans).
- **Endpoint Filter:** Filter the `effect` and `endpoint` columns based on the required measurements (e.g., MOR, ITX for acute crustacean toxicity).

Step 4: Data Quality Screening This phase involves reviewing the filtered data against established acceptance criteria to ensure verifiability and quality [\[10\]](#). A study should be accepted if it meets the following minimum criteria:

- The toxic effects are related to single chemical exposure.
- A biological effect is reported on live, whole organisms.
- A concurrent chemical concentration/dose and an explicit duration of exposure are reported.
- The study includes a calculated endpoint (e.g., LC50) and an acceptable control for comparison.
- The test location (lab/field) and species are unambiguously reported.

Step 5: Data Curation and Feature Expansion

- Convert effect concentrations to a standardized unit (e.g., molarity) for better comparison across chemicals.
- Expand the dataset with additional features, such as chemical properties from the CompTox Chemicals Dashboard or species-specific phylogenetic data, which can be invaluable for modeling [\[11\]](#).

The Scientist's Toolkit: Essential Reagents and Research Materials

The following table details key reagents, organisms, and tools essential for conducting standardized ecotoxicological tests and for leveraging computational tools for data analysis.

Table 3: Research Reagent Solutions for Ecotoxicology

Item Name	Type	Function & Application in Ecotoxicology
Aliivibrio fischeri	Bacterial Reagent	Used in bioluminescence inhibition assays (e.g., Microtox) for rapid assessment of basal toxicity in water and sediment samples [19] .
Daphnia magna	Live Organism	A model crustacean used in standard acute (48-hour) immobilization and lethality tests, representing primary consumers in freshwater food webs [19] [11] .
Sinapis alba	Plant Organism	Used in phytotoxicity tests (e.g., root and shoot elongation inhibition) to assess chemical impacts on terrestrial plants [19] .
OECD Standardized Test Media	Chemical Solution	Reconstituted water (e.g., OECD 202/203) or soil with defined hardness, pH, and ionic composition to ensure test reproducibility across laboratories.
CompTox Chemicals Dashboard	Digital Tool	A web application providing access to chemical properties, toxicity data, and predictive models; used to annotate and contextualize ecotoxicity data [11] .
Quantitative Structure-Activity Relationship (QSAR) Models	Computational Tool	In silico models that predict ecotoxicity based on a chemical's structural features; used for priority setting and filling data gaps [2] .

Mastering the advanced filtering capabilities of ecotoxicology knowledgebases is fundamental to modern environmental science and chemical regulation. By systematically applying filters based on **duration parameters**, **biological endpoints**, and **test conditions**, researchers can transform a vast, heterogeneous repository of data into a targeted, high-quality dataset fit for purpose. The integrated workflow and quality control protocols outlined in this guide provide a robust framework for supporting reliable ecological risk assessments, informing the development of safer chemicals, and ultimately contributing to the protection of aquatic and terrestrial ecosystems. As the field evolves with increasing data volumes and the integration of machine learning techniques [\[20\]](#) [\[11\]](#), these foundational data curation skills will only grow in importance.

The **ECOTOXicology Knowledgebase (ECOTOX)** is the world's largest compilation of curated ecotoxicity data, providing single-chemical ecological effects data for over **12,000 chemicals** and **13,000 aquatic and terrestrial species** with more than **one million test results** from over **53,000 references** [\[4\]](#) [\[1\]](#). As the need for chemical safety assessments continues to grow, ECOTOX has evolved into an authoritative source supporting research, risk assessment, and regulatory decision-making. A critical advancement in the recently released **ECOTOX Version 5** is its enhanced capability for data export and interoperability with other chemical databases and tools, designed following the **FAIR principles** (Findable, Accessible, Interoperable, and Reusable) to maximize data utility and integration [\[4\]](#).

The interoperability framework of ECOTOX enables seamless data exchange and integration with complementary resources, particularly the **CompTox Chemicals Dashboard** (hereafter "Dashboard"), which provides access to chemistry, toxicity, and exposure data for over one million chemicals [\[21\]](#) [\[22\]](#). This integration creates a powerful ecosystem for chemical safety assessment by connecting detailed ecotoxicity data from ECOTOX with extensive computational toxicology data from the Dashboard, facilitating more comprehensive chemical evaluations and supporting the development of predictive models [\[4\]](#) [\[23\]](#).

Data Export Capabilities and Methodologies

ECOTOX Data Export Features

The ECOTOX Knowledgebase provides robust data export functionalities through its web interface, allowing users to extract curated data for use in external applications. The system offers **customizable outputs** with selection from over **100 data fields**, enabling researchers to tailor exports to specific analytical needs [1]. Data can be refined and filtered using **19 different parameters** before export, including chemical properties, species characteristics, test methods, and effects measurements [1].

The **EXPLORE feature** is particularly valuable when exact search parameters are not known in advance, allowing iterative query refinement before final data export [1]. For programmatic access and enhanced reproducibility, the **ECOTOXr R package** provides direct retrieval of data from the ECOTOX database, enabling seamless integration of ecotoxicity data into statistical computing environments and automated analysis workflows [24].

Table: ECOTOX Knowledgebase Data Statistics and Export Capabilities

Aspect	Specification	Utility for Data Export
Total Test Records	>1,000,000 [1]	Comprehensive data for analysis
Chemical Coverage	>12,000 substances [4] [1]	Broad applicability across chemistries
Species Coverage	>13,000 aquatic & terrestrial species [1]	Diverse ecological representation
Reference Sources	>53,000 references [1]	Extensive literature foundation
Customizable Fields	>100 data fields [1]	Flexible output configuration
Filter Parameters	19 refinement parameters [1]	Precise data subset selection
Programmatic Access	ECOTOXr R package [24]	Automated data retrieval & analysis

Data Curation and Quality Assurance Protocols

The interoperability of ECOTOX data is fundamentally enabled by its rigorous **data curation pipeline**, which follows systematic review practices with well-established **standard operating procedures (SOPs)** [4]. The curation process includes:

- **Literature Identification:** Comprehensive searches of open and grey literature using systematic review protocols [4]
- **Citation Screening:** Multi-stage review of titles, abstracts, and full texts against established applicability criteria [4]
- **Data Abstraction:** Pertinent methodological details and results extraction using controlled vocabularies [4]
- **Quality Control:** Documented controls and endpoint verification to ensure data reliability [4]

This meticulous curation process ensures that exported data maintains consistent structure, terminology, and quality, making it suitable for integration with other databases and tools. The use of **controlled vocabularies** throughout the curation process is particularly important for semantic interoperability, enabling accurate mapping of concepts and terminology across different systems [4] [25].

Integration with Chemical Dashboards and Tools

CompTox Chemicals Dashboard Integration

The ECOTOX Knowledgebase features **deep integration** with the EPA's **CompTox Chemicals Dashboard**, creating a unified chemical safety assessment environment. This integration is implemented through **direct linking** between the systems, where chemical searches in ECOTOX provide connections to the Dashboard for additional chemistry, hazard, and exposure information [1]. The Dashboard serves as a complementary resource by providing:

- **Chemical identification** and structure information for over one million substances [21] [22]
- **Physicochemical properties** and environmental fate parameters [26]
- **Exposure data** including product use categories and functional uses [26]
- **Bioactivity data** from high-throughput screening programs like ToxCast [26]
- **Predictive models** for various toxicity endpoints and properties [21]

This bidirectional interoperability enables users to begin investigations in either platform and seamlessly access complementary data from the other system, significantly enhancing the efficiency of chemical safety evaluations.

Table: ECOTOX and CompTox Chemicals Dashboard Feature Comparison

Feature	ECOTOX Knowledgebase	CompTox Chemicals Dashboard
Primary Focus	Empirical ecotoxicity data [4]	Computational toxicology & chemistry [21]
Data Type	Curated in vivo test results [4]	Experimental & predicted data [21]

Feature	ECOTOX Knowledgebase	CompTox Chemicals Dashboard
Chemical Coverage	~12,000 chemicals [4]	>1,000,000 chemicals [21]
Key Data Sources	Peer-reviewed literature [1]	EPA databases & public sources [21]
Taxonomic Scope	Ecological species [4]	Human health & ecological [21]
Export Capabilities	Customizable data fields [1]	Multiple download formats [26]
Update Frequency	Quarterly [1]	Regular releases [26]

Interoperability with Analysis and Visualization Tools

ECOTOX provides several pathways for interoperability with external analysis and visualization tools. The **DATA VISUALIZATION feature** within ECOTOX enables interactive exploration of results, with functionality to hover over data points and zoom into specific sections of data [1]. These visualization capabilities support data interpretation before export to external tools.

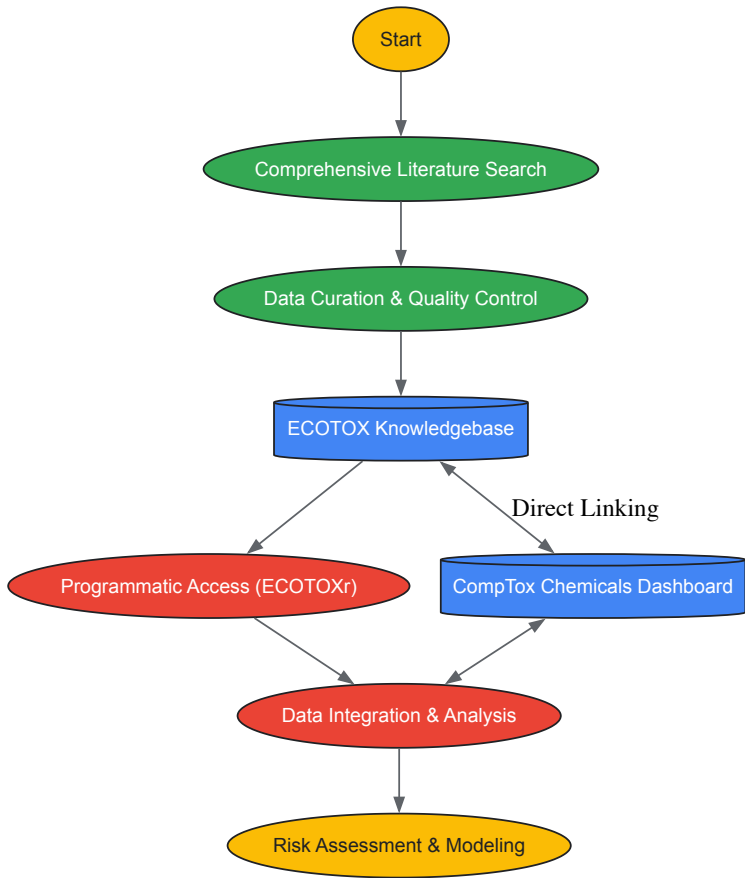
For advanced statistical analysis and modeling, researchers can utilize the **ECOTOXr R package**, which allows direct programmatic access to ECOTOX data within the R environment [24]. This enables seamless integration of ecotoxicity data with specialized statistical analysis, species sensitivity distribution (SSD) modeling, and quantitative structure-activity relationship (QSAR) development workflows.

The interoperability extends to supporting **read-across approaches** and **weight-of-evidence assessments** by providing consistent, well-annotated data that can be combined with information from other sources such as the **Integrated Chemical Environment (ICE)** [23], which provides tools for chemical safety testing including in vitro to in vivo extrapolation (IVIVE) [23].

Implementation Workflows and Experimental Protocols

Data Retrieval and Integration Workflow

The interoperability between ECOTOX and external tools enables sophisticated workflows for chemical safety assessment. The following diagram illustrates a typical workflow for data retrieval and integration:



This workflow demonstrates how data flows from primary literature through curation into ECOTOX, then becomes accessible through both web interfaces and programmatic methods, ultimately supporting integrated analysis and risk assessment.

Protocol for Automated Data Retrieval and Analysis

For researchers requiring automated data access, the following protocol utilizing the ECOTOXr R package enables reproducible retrieval of ecotoxicity data:

Materials and Software Requirements:

- R programming environment (v4.0 or higher)
- ECOTOXr package [24]
- CompTox Chemicals Dashboard access [21]
- Additional R packages for analysis (e.g., ggplot2, dplyr)

Step-by-Step Methodology:

- **Installation and Setup**
 - Install the ECOTOXr package in R environment
 - Load required libraries and initialize connection parameters
- **Query Formulation**
 - Define search parameters using controlled vocabularies
 - Specify chemical identifiers (preferably using DSSTox identifiers for interoperability)
 - Set ecological species filters if required
 - Define endpoint types and effect measurements of interest
- **Data Retrieval**
 - Execute query through ECOTOXr API interface
 - Retrieve results in tabular format with complete metadata
 - Validate data completeness and quality flags
- **Data Integration**
 - Cross-reference chemical identifiers with CompTox Chemicals Dashboard
 - Retrieve additional chemical properties and descriptors from Dashboard
 - Merge datasets using standardized chemical identifiers
- **Analysis and Visualization**
 - Perform statistical analysis and modeling
 - Generate visualizations for data exploration
 - Export results for reporting or further analysis

This protocol ensures transparent, reproducible access to ECOTOX data while leveraging the interoperability with complementary data sources to enhance analytical capabilities.

Table: Essential Tools and Resources for ECOTOX Data Interoperability

Tool/Resource	Type	Primary Function	Access Method
ECOTOXr	R Package	Programmatic data retrieval from ECOTOX [24]	R environment
CompTox Dashboard	Web Application	Chemical information & predictive data [21]	Web browser
DSSTox IDs	Identifier System	Standardized chemical identifiers [26]	Cross-platform
ICE (Integrated Chemical Environment)	Analysis Platform	IVIVE & chemical safety testing tools [23]	Web browser
ToxValDB	Database	Human health toxicity values [26]	CompTox Dashboard
invitroDB	Database	High-throughput screening data [26]	CompTox Dashboard

Advanced Interoperability Applications

Supporting New Approach Methodologies (NAMs)

The interoperability features of ECOTOX play a crucial role in supporting the development and validation of **New Approach Methodologies (NAMs)** in toxicology. By providing curated in vivo toxicity data, ECOTOX enables researchers to:

- **Benchmark in vitro bioactivity** against traditional apical endpoints for assay validation [4]
- **Develop computational models** including quantitative structure-activity relationships (QSARs) [4]
- **Perform cross-species extrapolation** to evaluate the safety of chemicals [1]
- **Build species sensitivity distributions (SSDs)** for ecological risk assessment [4]

These applications are enhanced through interoperability with tools like the CompTox Chemicals Dashboard, which provides access to high-throughput screening data and predictive models [26] [23].

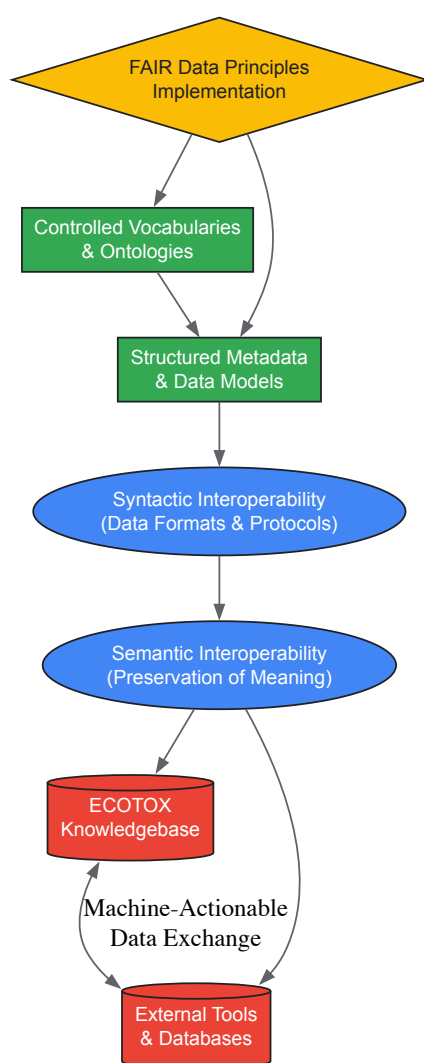
Semantic Interoperability and FAIR Data Principles

The ECOTOX Knowledgebase implements sophisticated approaches to **semantic interoperability** to enhance data reuse and integration. Key implementation strategies include:

- **Controlled vocabularies** for consistent terminology across biological effects, test methods, and chemical identifiers [4]
- **Structured metadata** following FAIR principles to ensure machine-actionability [4] [27]
- **Ontology development** to standardize vocabulary and relationships between entities [27] [25]

These semantic interoperability features enable more accurate data integration and knowledge discovery by preserving meaning across different systems and contexts. The implementation addresses both **syntactic interoperability** (common data formats and exchange protocols) and **semantic interoperability** (preservation of meaning during data exchange) as defined in established data interoperability frameworks [25].

The following diagram illustrates the semantic interoperability framework that enables meaningful data exchange between ECOTOX and external tools:



This semantic interoperability framework ensures that data exchanged between ECOTOX and external tools retains its meaning and context, enabling more reliable integration and analysis across systems.

The data export and interoperability capabilities of the ECOTOX Knowledgebase represent a significant advancement in environmental toxicology data resources. Through its integration with the CompTox Chemicals Dashboard, programmatic access via ECOTOXr, and implementation of FAIR data principles, ECOTOX provides researchers with powerful tools for chemical safety assessment and ecological research. The interoperability framework enables more efficient data analysis, supports the development and validation of new approach methodologies, and facilitates the integration of ecotoxicity data into broader chemical safety assessment contexts. As ECOTOX continues to evolve, its interoperability features will play an increasingly important role in addressing the growing need for rapid, cost-effective chemical evaluation methods while reducing reliance on animal testing.

The ECOTOXicology (ECOTOX) Knowledgebase, maintained by the U.S. Environmental Protection Agency (EPA), is a comprehensive, publicly accessible resource that provides single-chemical ecotoxicity data for ecologically relevant aquatic and terrestrial species. As the world's largest curated database of its kind, it contains over one million test results compiled from more than 53,000 references, covering more than 13,000 species and 12,000 chemicals [1] [5]. The knowledgebase serves as a critical resource for researchers, risk assessors, and decision-makers who require high-quality, curated ecotoxicity data to support environmental protection goals. The systematic and transparent literature review and data curation processes employed by ECOTOX align with contemporary systematic review practices, ensuring the data's reliability and relevance for various applications in environmental research and regulation [5]. This technical guide explores the three primary real-world use cases of the ECOTOX Knowledgebase, detailing methodologies, applications, and implementation frameworks for each.

Supporting Ecological Risk Assessments

Framework and Regulatory Context

Ecological risk assessment (ERA) forms the scientific foundation for evaluating the potential adverse effects of chemical exposures on ecosystems. The ECOTOX Knowledgebase supports the entire ERA process by providing curated toxicity data that inform chemical effects characterization. This application is particularly vital for regulatory programs operating under statutes such as the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), the Toxic Substances Control Act (TSCA), and the Clean Water Act [1] [5]. Risk assessors leverage ECOTOX data to develop toxicity benchmarks, derive species sensitivity distributions, and characterize potential hazards to ecological receptors under various exposure scenarios.

The use of ECOTOX in risk assessment represents a shift toward greater efficiency in chemical evaluations. By providing pre-curated toxicity data, ECOTOX reduces the need for redundant literature reviews and minimizes unnecessary animal testing while ensuring environmental protection [1]. Furthermore, the database supports the development and validation of New Approach Methodologies (NAMs), including in vitro assays and computational models, by providing high-quality in vivo data for comparison and validation [28] [5].

Experimental Protocols and Methodologies

Data Collection and Study Evaluation

The ECOTOX team employs systematic review procedures to identify, evaluate, and extract ecotoxicity data from the scientific literature. The process begins with comprehensive literature searches across multiple scientific databases using standardized search strategies. Identified studies then undergo rigorous evaluation based on predefined criteria for scientific quality and relevance [5].

Trained data curators extract pertinent information on test organisms, chemical exposure conditions, experimental methods, and measured effects using controlled vocabularies and standardized data fields. This meticulous curation process ensures consistency and reliability across the database. The curated data includes detailed information on species taxonomy, chemical identity, test duration, exposure pathway, measured endpoints (e.g., survival, growth, reproduction), and statistical results (e.g., LC50, EC10, NOEC) [5].

Case Study: Aromatase Inhibition and Reproductive Impairment in Fish

A specific example demonstrating the application of ECOTOX data in risk assessment comes from research on aromatase inhibition in fish. The adverse outcome pathway (AOP) linking aromatase inhibition to reproductive impairment provides a mechanistic framework for predicting chemical effects based on in vitro bioactivity data [28].

Experimental Protocol:

- **In Vitro Screening:** Initial identification of imazalil as an aromatase inhibitor using fathead minnow (*Pimephales promelas*) ovarian post-mitochondrial supernatant assays.
- **Computational Modeling:** Use of a mechanistic model of the female fathead minnow hypothalamic-pituitary-gonad-liver (HPG) axis to simulate response to aromatase inhibition and inform in vivo experimental design.
- **In Vivo Validation:** Conductation of 60-hour and 21-day exposures of adult fathead minnows to imazalil concentrations ranging from 0.12–260 µg/L.
- **Endpoint Measurement:** Assessment of multiple key events along the AOP, including:
 - Ex vivo production of 17β-estradiol (E2) by ovary tissue
 - Plasma E2 concentrations

- Vitellogenin (Vtg) mRNA expression
- Vtg plasma concentrations
- Vtg uptake into oocytes
- Cumulative fecundity and spawning metrics [28]

This case study demonstrated that model predictions based on in vitro aromatase inhibition data were consistent with observed in vivo effects, supporting the use of such integrated approaches in ecological risk assessment [28].

Data Analysis and Application

The quantitative data extracted from ECOTOX facilitate various risk assessment applications. For deterministic assessments, the most sensitive endpoint for relevant species is often used to derive protective benchmarks. For probabilistic assessments, species sensitivity distributions can be constructed using multiple toxicity values across taxonomic groups to determine concentrations protective of specified percentiles of species [5].

The interoperability of ECOTOX with other databases and tools, such as the EPA CompTox Chemicals Dashboard, enhances its utility in risk assessment by providing additional chemical information and computational toxicology resources [1]. This integration supports more comprehensive chemical evaluations and facilitates the integration of traditional toxicity data with emerging data from high-throughput screening approaches.

Informing Water Quality Criteria

Development of Aquatic Life Benchmarks

The ECOTOX Knowledgebase plays a fundamental role in establishing water quality criteria designed to protect aquatic ecosystems. The EPA's Office of Pesticide Programs (OPP) derives Aquatic Life Benchmarks (ALBs) from toxicity data compiled in ECOTOX and other sources. These benchmarks represent concentrations below which pesticides are not expected to pose significant risks to aquatic organisms [29]. As of September 2025, the ALB table includes values for hundreds of pesticides and their degradates, with benchmarks provided for freshwater and estuarine/marine vertebrates, invertebrates, and plants [29].

These benchmarks enable state, tribal, and local governments to interpret water quality monitoring data by comparing measured environmental concentrations with effect levels. This comparison helps identify sites and chemicals that may require further investigation or management action [29]. While ALBs are not regulatory standards, they provide critical technical support for developing water quality standards under the Clean Water Act and for implementing various water protection programs.

Methodology for Benchmark Derivation

Data Requirements and Evaluation

The development of ALBs follows rigorous evaluation guidelines for ecological toxicity data. The toxicity values are extracted from the most recent publicly available OPP risk assessments for individual pesticides, typically based on the most sensitive value for each taxon [29]. The data requirements follow the U.S. Code of Federal Regulations (40 CFR 158), with studies evaluated using EPA's Harmonized Test Guidelines and the Evaluation Guidelines for Ecological Toxicity Data in the Open Literature [29].

The ALB table includes multiple types of toxicity values to address different exposure scenarios:

- **Acute values** for vertebrates (LC50/EC50) and invertebrates (LC50/EC50)
- **Chronic values** for vertebrates (NOAEC/EC10) and invertebrates (NOAEC/EC10)
- **Plant effects** measurements (EC50 and NOAEC) for nonvascular and vascular plants [29]

Water Quality Criteria Development Process

The process for developing water quality criteria involves several standardized steps:

- **Data Collection:** Comprehensive literature review and data extraction from ECOTOX and other sources
- **Study Evaluation:** Assessment of data quality and relevance using established criteria
- **Species Selection:** Identification of tested species representing relevant ecological receptors
- **Dose-Response Analysis:** Statistical analysis of concentration-effect relationships
- **Extrapolation:** Application of assessment factors or species sensitivity distributions to derive protective concentrations
- **Peer Review:** Independent scientific review of proposed criteria
- **Implementation:** Adoption by regulatory agencies as water quality standards [29]

Table 1: Example Aquatic Life Benchmarks for Selected Pesticides (as of September 2025) [29].

Pesticide	CAS Number	Freshwater Vertebrates Acute (µg/L)	Freshwater Vertebrates Chronic (µg/L)	Freshwater Invertebrates Acute (µg/L)	Freshwater Invertebrates Chronic (µg/L)	Nonvascular Plants IC50 (µg/L)
3-iodo-2-propynyl butyl carbamate (IPBC)	55406-53-6	33.5	3	209	80	< 3
Abamectin	71751-41-2	1.6	0.52	7.5	0.17	0.01
Acetochlor	34256-82-1	190	130	1050	740	1.43
Acrolein	107-02-8	3.5	11.4	120	16	< 15.5

Application in Monitoring and Assessment

Water quality managers use ECOTOX-derived benchmarks in multiple applications beyond criteria development. The benchmarks help prioritize contaminants of concern in watershed management, interpret biological monitoring data, and design targeted toxicity testing programs. The continuous updating of ECOTOX ensures that these applications incorporate the most recent scientific information, with new benchmarks added and existing benchmarks revised as new toxicity data become available [29].

The integration of ECOTOX data with large-scale monitoring efforts, such as the U.S. Geological Survey's National Water-Quality Assessment Program, enhances the ability to identify emerging contaminants and assess spatial and temporal trends in water quality [30]. This integration supports evidence-based decision-making for water resource management and protection.

Enabling Chemical Prioritization

Framework for Prioritization Approaches

Chemical prioritization represents a critical application of the ECOTOX Knowledgebase, particularly given the vast number of chemicals in commerce and the limited resources available for comprehensive risk assessment. Prioritization frameworks use ECOTOX data to identify chemicals requiring further testing, regulatory attention, or management action based on their potential hazard to ecological receptors [30]. This approach is especially valuable for programs operating under TSCA, which mandates evaluation of existing chemicals in commerce [1] [31].

The stepwise prioritization framework typically incorporates multiple lines of evidence, including the availability and quality of toxicity data, the potency of identified effects, and the potential for environmental exposure. ECOTOX data provide the foundational hazard information necessary to screen chemicals based on their intrinsic potential to cause adverse ecological effects [30].

Methodology for Retrospective Prioritization

Case Study: Great Lakes Tributaries Chemical Prioritization

A recent study demonstrated the application of ECOTOX in a stepwise prioritization of chemicals detected in Great Lakes tributaries between 2008-2018 [30]. The methodology involved:

- **Chemical Monitoring:** 586 chemicals monitored with 334 detected in grab/composite water samples
- **Tiered Prioritization:**
 - **Tier 1:** Identification of chemicals with existing water quality guidelines or criteria
 - **Tier 2:** Evaluation using apical toxicity data from ECOTOX and other databases
 - **Tier 3:** Assessment using alternative data (in vitro bioactivity, non-apical effects, QSAR predictions)
- **Action Categorization:** Classification of chemicals into specific action categories based on data type and quality [30]

This framework accommodated varying levels of confidence in the available data by categorizing chemicals for different management or research actions rather than relying on a single prioritization metric.

Data Integration and Analysis

The chemical prioritization process integrates ECOTOX data with other information sources to support decision-making. Key steps include:

- **Data Extraction:** Retrieval of relevant ecotoxicity endpoints from ECOTOX for target chemicals
- **Potency Evaluation:** Comparison of effect concentrations across taxonomic groups and endpoints
- **Data Gap Identification:** Recognition of chemicals with insufficient toxicity data

- **Risk-Based Screening:** Integration of toxicity and exposure information where available
- **Stakeholder Engagement:** Collaboration with regulatory agencies and environmental managers to refine priorities [30]

The study identified 11 chemicals as high priority across different action categories, with four targeted for environmental management or risk assessment, three for effects-based monitoring, one for apical effects assessment, and three for non-apical effects evaluation. Additionally, 164 chemicals were identified as low priority based on existing water quality guidelines or apical effect concentrations [30].

Advanced Approaches: Integrating New Approach Methodologies

Chemical prioritization is increasingly incorporating New Approach Methodologies (NAMs) to enhance efficiency and address data gaps. The ECOTOX Knowledgebase supports this evolution by providing traditional in vivo toxicity data for validating and contextualizing NAMs [28] [5]. Key integrated approaches include:

- **In Vitro to In Vivo Extrapolation:** Using high-throughput screening data to predict potential in vivo effects, with ECOTOX data serving as a reference for validation
- **Adverse Outcome Pathways:** Placing chemical-specific data within mechanistic frameworks to support extrapolation across chemicals and species
- **Computational Toxicology:** Developing quantitative structure-activity relationship (QSAR) models using ECOTOX data as training sets
- **Weight of Evidence:** Combining traditional toxicity data with NAMs to strengthen prioritization decisions [28]

Table 2: Chemical Prioritization Action Categories and Data Requirements [30]

Action Category	Priority Level	Data Requirements	Potential Follow-up Activities
Environmental Management/Targeted Risk Assessment	High	Established water quality guidelines or consistent apical toxicity data	Regulatory action, site-specific risk assessment
Effects-Based Monitoring	High	In vitro bioactivity or mechanistic data indicative of potential hazard	Targeted field monitoring using bioassays or specific biomarkers
Apical Effects Assessment	Medium-High	Limited apical toxicity data requiring confirmation	Standardized toxicity testing focused on sensitive endpoints
Non-Apical Effects Evaluation	Medium	Only non-apical (subcellular, biochemical) effects data available	Investigation of ecological relevance of effects
Low Priority	Low	Adequate data showing low toxicity potential or high effect concentrations	No immediate action; periodic re-evaluation

The Scientist's Toolkit: Research Reagent Solutions

The implementation of ecotoxicology studies and the effective use of databases like ECOTOX require specific research tools and reagents. The following table details key resources used in the field, with examples drawn from the case studies and applications discussed in this guide.

Table 3: Essential Research Reagent Solutions for Ecotoxicology Studies

Research Reagent	Function and Application	Example Use Cases
Fathead Minnow (<i>Pimephales promelas</i>) Post-mitochondrial Supernatants	In vitro assessment of enzyme inhibition (e.g., aromatase activity)	Screening for endocrine-disrupting chemicals [28]
Imazalil and other Fungicide Reference Standards	Chemical stressors for controlled exposure studies	Investigating aromatase inhibition and reproductive effects in fish [28]
ELISA Kits for 17β-estradiol (E2) and Vitellogenin (Vtg)	Quantification of protein and hormone biomarkers in plasma	Measuring endocrine disruption responses in aquatic organisms [28]
Species-Specific cDNA Probes and Primers	Measurement of gene expression changes via qPCR	Assessing Vtg mRNA expression as a key event in AOPs [28]
Laboratory Cultured Test Organisms (e.g., <i>Daphnia magna</i> , <i>Ceriodaphnia dubia</i>)	Standardized toxicity testing for regulatory applications	Deriving acute and chronic toxicity values for chemical assessments [29] [5]
High-Throughput Screening Assays (e.g., ToxCast)	Rapid in vitro bioactivity profiling	Prioritizing chemicals for further testing [28] [30]

Research Reagent	Function and Application	Example Use Cases
AOP-Wiki Framework	Organizing mechanistic knowledge for chemical assessment	Placing chemical-specific data within biological pathways [28]
Quantitative Structure-Activity Relationship (QSAR) Models	Predicting toxicity based on chemical structure	Screening chemicals with limited empirical data [1] [39]

Experimental Workflow Visualization

The following diagram illustrates a generalized experimental workflow for ecotoxicity studies that generate data suitable for inclusion in the ECOTOX Knowledgebase, integrating both traditional and NAMs approaches:

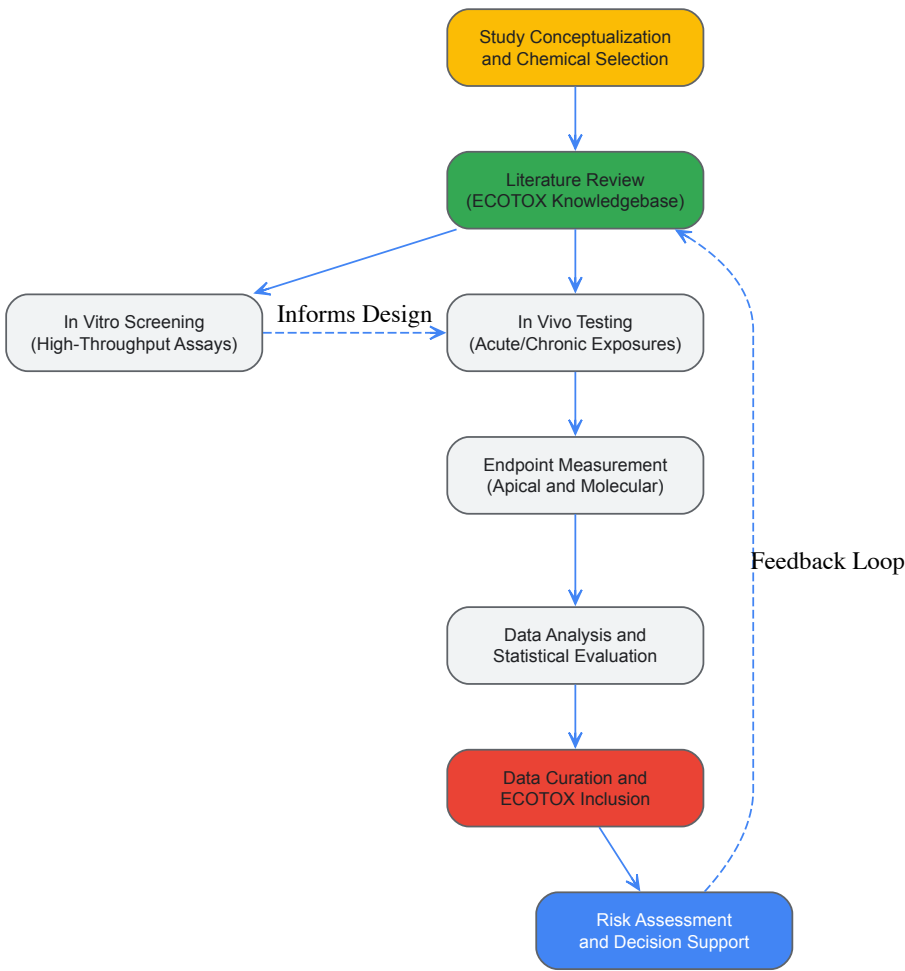


Diagram 1: Ecotoxicity Study Workflow

Adverse Outcome Pathway Visualization

The following diagram illustrates the Adverse Outcome Pathway (AOP) for aromatase inhibition leading to reproductive impairment in fish, a key mechanistic framework supported by ECOTOX data:

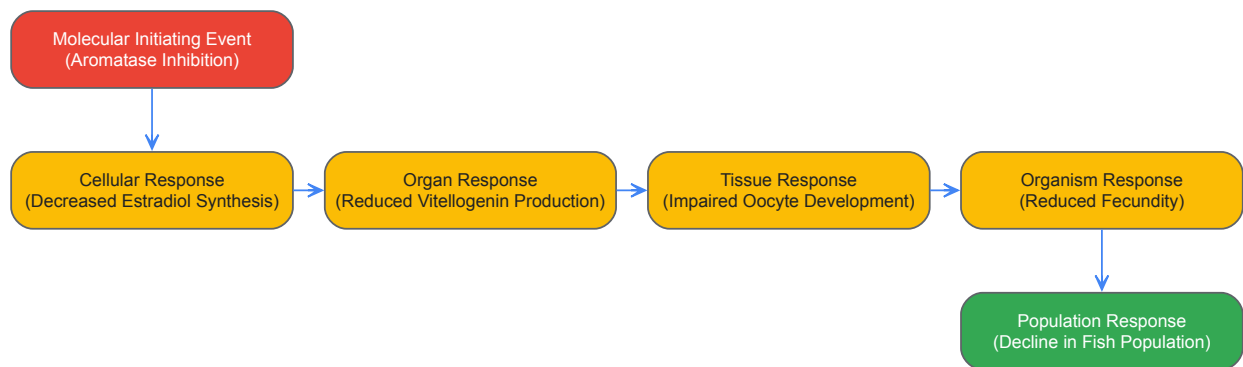


Diagram 2: AOP for Aromatase Inhibition in Fish

The ECOTOX Knowledgebase serves as a critical resource supporting three fundamental applications in environmental science and regulation: ecological risk assessment, water quality criteria development, and chemical prioritization. Through its comprehensive collection of curated ecotoxicity data, systematic review procedures, and interoperability with other tools and databases, ECOTOX enables evidence-based decision-making that protects ecological systems from chemical stressors. The integration of traditional toxicity data with emerging approaches, including NAMs and AOP frameworks, positions ECOTOX as a evolving resource that will continue to support innovative approaches to chemical assessment and management. As regulatory requirements evolve and the number of chemicals in commerce grows, the role of curated, accessible toxicity databases like ECOTOX will become increasingly vital for efficient and effective environmental protection.

Maximizing ECOTOX Efficiency: Expert Tips and Support Resources

This technical guide provides researchers and scientists with advanced methodologies for leveraging duration filters and output field customization within the Ecotoxicology (ECOTOX) Knowledgebase. As the world’s largest compilation of curated ecotoxicity data, containing over **1.1 million test records** from more than **54,000 references**, ECOTOX serves as a critical resource for ecological risk assessments and chemical safety evaluations [7] [4]. This whitepaper details systematic protocols for constructing precise search queries, utilizing the platform’s data visualization tools, and exporting customized datasets for external analysis, thereby enhancing the efficiency and effectiveness of ecotoxicology research within a rapidly evolving chemical regulatory landscape.

The ECOTOX Knowledgebase, maintained by the United States Environmental Protection Agency (USEPA), is an authoritative source of curated single-chemical ecotoxicity data for ecologically relevant aquatic and terrestrial species [1] [4]. Its development was initiated in the early 1980s to provide regulatory EPA Program Offices with rapid access to toxicity data, supporting mandates under various acts including the Federal Insecticide, Fungicide, and Rodenticide Act; the Clean Water Act; and the Toxic Substances Control Act [4]. The knowledgebase has evolved into a comprehensive system that now includes over **13,000 chemicals** and nearly **14,000 species**, with continuous quarterly updates incorporating new data and chemicals of emerging concern, such as 6-PPD quinone, cyanotoxins, and PFAS compounds [7].

The platform’s recent transformation with version 5 introduced an entirely re-designed user interface with enhanced data queries and retrieval options, visualizations to aid data exploration, and customizable outputs for export and use in external applications [4]. This guide focuses specifically on mastering two of the most powerful features for advanced research: duration filters for temporal refinement of ecotoxicity data and output field customization for tailored data analysis, both essential for developing accurate ecological risk assessments and supporting regulatory decision-making processes.

ECOTOX Database Structure and Quantitative Scope

Understanding the quantitative scope and structure of the ECOTOX database is fundamental to designing effective search strategies. The following table summarizes the core data metrics available to researchers:

Table 1: ECOTOX Knowledgebase Quantitative Data Metrics

Data Category	Metric Value	Source
Total Test Records	Over 1.1 million	[7]
Number of References	Over 54,000	[7]
Number of Chemicals	Approximately 13,000	[7]
Aquatic and Terrestrial Species	Nearly 14,000	[7]
Monthly Users (2023)	Over 16,000	[7]

The data curation process follows systematic review practices consistent with contemporary systematic review methodologies, employing documented standard operating procedures for literature search, citation identification, applicability criteria, and data abstraction [4]. The database structure organizes information hierarchically, with study references containing one or more test results, each with associated chemicals, species, and measured effects. This structure enables complex filtering across multiple dimensions while maintaining data integrity and relational consistency.

Applying Duration Filters: Experimental Protocols

Understanding Duration Filtering

Duration filtering in ECOTOX allows researchers to isolate studies based on exposure periods, which is critical for assessing both acute and chronic effects of chemical stressors on ecological species. Exposure duration represents a fundamental parameter in ecotoxicology, as the temporal component of chemical exposure often determines the nature and severity of observed effects. The ECOTOX knowledgebase captures exposure duration as a discrete data field, enabling precise filtering to align with specific research objectives and regulatory frameworks that often distinguish between short-term (acute) and long-term (chronic) exposure scenarios.

Methodology for Duration Filter Application

The experimental protocol for implementing duration filters involves a systematic approach to query construction and validation:

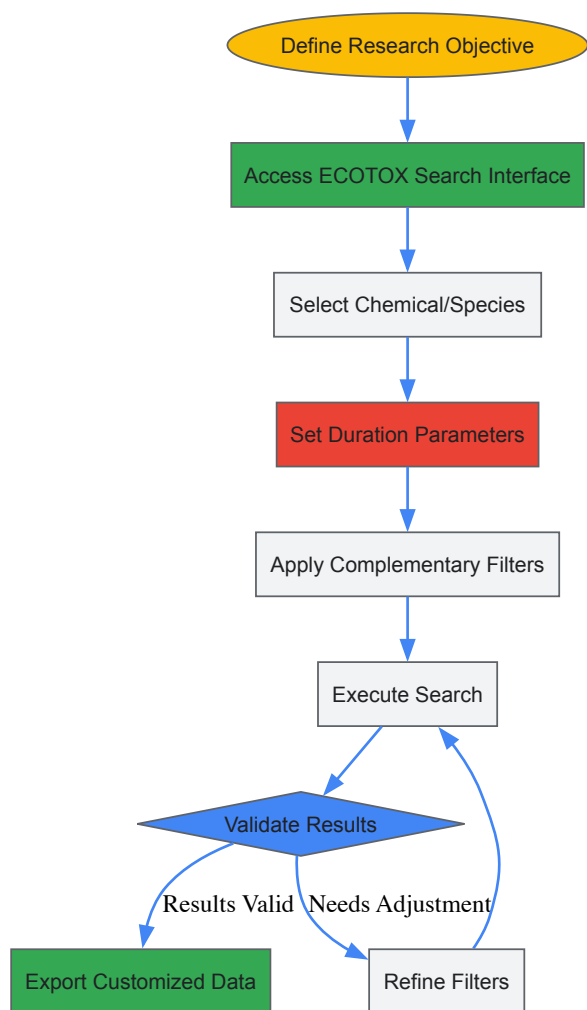
- **Access the ECOTOX Search Interface:** Navigate to the ECOTOX Knowledgebase (www.epa.gov/ecotox) and select the "Search" feature to access the query builder [1].
- **Define Initial Search Parameters:** Establish primary search criteria by selecting a specific chemical or species of interest. The chemical search integrates with the CompTox Chemicals Dashboard, providing additional chemical information and identifiers [1].
- **Locate Duration Filter Parameters:** Within the search interface, identify the exposure duration filter field. This may be located under "Test Conditions" or "Advanced Filters" sections.
- **Set Duration Thresholds:** Input minimum and/or maximum exposure duration values using standardized temporal units (e.g., hours, days). For example:
 - Acute toxicity studies: 1-96 hours
 - Chronic toxicity studies: >96 hours
 - Life-cycle assessments: Species-specific full or partial life cycle durations
- **Apply Complementary Filters:** Combine duration filters with other relevant parameters such as:
 - Effect measurements (mortality, growth, reproduction)
 - Endpoint types (LC50, EC50, NOEC)
 - Species taxonomic groups
 - Test environment (freshwater, marine, terrestrial)
- **Execute Search and Validate Results:** Run the query and review returned records for consistency with duration parameters. Adjust filters iteratively based on result yield and relevance.
- **Document Filter Strategy:** Maintain a record of all filter parameters applied, including duration ranges, for methodological transparency and research reproducibility.

Technical Considerations for Duration Filtering

When implementing duration filters, researchers should account for several technical considerations to ensure data quality and relevance:

- **Unit Consistency:** The database standardizes duration values, but original studies may report varying units that have been converted during curation.
- **Life Stage Specificity:** Consider combining duration filters with life stage parameters, as sensitivity to chemicals may vary across developmental stages.
- **Test Type Alignment:** Duration significance differs across test types (static, renewal, flow-through), which may affect chemical exposure concentrations over time.
- **Species-Specific Considerations:** Appropriate duration thresholds may vary significantly across species with different life histories and generation times.

The workflow below illustrates the systematic process for applying duration filters within ECOTOX search strategies:



Output Field Customization: Methodologies for Data Extraction

Principles of Output Field Selection

Output field customization enables researchers to tailor extracted data to specific analytical needs, enhancing both the efficiency of data processing and the relevance of exported datasets. ECOTOX version 5 provides access to over 100 data fields that can be selected, organized, and exported based on research requirements [1]. The selection of appropriate output fields should align with both immediate analytical objectives and potential future research needs, balancing comprehensiveness with practical manageability of datasets.

Protocol for Output Customization

The methodology for customizing output fields involves a structured approach to field selection and export configuration:

- **Access Output Customization Interface:** After executing a search, locate the "Customize Output" or "Output Options" feature within the results interface [1].
- **Review Available Field Categories:** Navigate through the available field categories, which typically include:
 - Chemical identification fields (name, CASRN, SMILES)
 - Species information (scientific name, common name, taxonomy)
 - Test conditions (duration, route, exposure type)
 - Effect measurements (endpoint, value, units)
 - Study metadata (citation, year, curator)
- **Select Relevant Fields:** Methodically select fields based on research questions:
 - For dose-response modeling: chemical concentration, effect values, measurement units
 - For species sensitivity distributions: species taxonomy, effect values, test conditions
 - For meta-analyses: full citation information, test methodologies, quality indicators

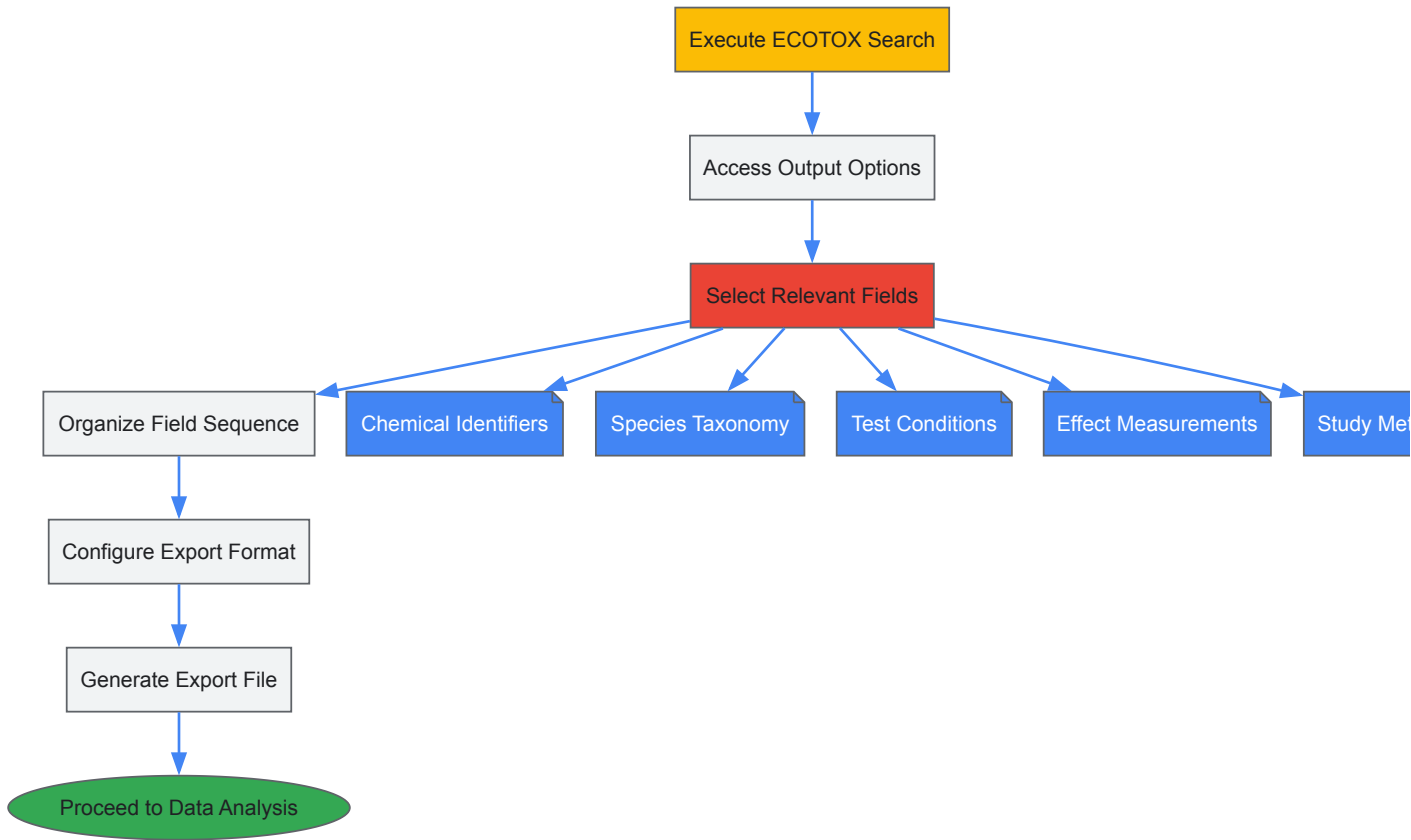
- **Organize Field Sequence:** Arrange selected fields in a logical sequence that supports downstream analysis, typically grouping related parameters (e.g., chemical identifiers together, test conditions together).
- **Configure Export Format:** Select appropriate export formats based on intended use:
 - CSV for statistical analysis in R, Python, or similar platforms
 - Excel for preliminary review and visualization
 - XML for structured data interchange
- **Execute Export and Validate Structure:** Generate the export file and verify that all selected fields are present and properly formatted.

Advanced Output Strategies

For complex research applications, several advanced output strategies enhance the utility of exported data:

- **Comparative Analyses:** Select parallel fields for multiple chemicals or species to enable direct comparison.
- **Model Input Preparation:** Customize outputs to match specific input requirements for quantitative structure-activity relationship (QSAR) models or other predictive tools.
- **Interoperability Optimization:** Include fields that facilitate linkage with complementary databases, such as chemical identifiers that connect to the CompTox Chemicals Dashboard [4].
- **Visualization-Ready Outputs:** Structure data to streamline import into visualization platforms, with appropriate grouping variables and numerical formats.

The relationship between output customization and subsequent data analysis phases can be visualized as follows:



Integrated Workflow: Combining Duration Filters and Output Customization

Experimental Design for Comprehensive Data Extraction

The most powerful ECOTOX search strategies integrate both precise duration filtering and targeted output customization to create streamlined workflows for specific research applications. The following table outlines representative research scenarios with recommended filter and output configurations:

Table 2: Integrated Filter and Output Configurations for Research Applications

Research Application	Recommended Duration Filters	Essential Output Fields	Complementary Filters
Acute Aquatic Toxicity Assessment	24-96 hours	Chemical name, CASRN, Species, Effect, Endpoint, Value, Units	Freshwater environment, Mortality effects
Chronic Terrestrial Plant Growth	>96 hours up to full life cycle	Chemical identity, Species family, Exposure route, Effect measurement, Statistical significance	Soil exposure, Growth/reproduction endpoints
Species Sensitivity Distribution for Regulatory Criteria	Full range with acute/chronic differentiation	Chemical identifier, Species taxonomy, Effect concentration, Confidence intervals, Test conditions	Peer-reviewed studies, Specific taxonomic groups
Chemical Prioritization Based on Potency	Multiple duration tiers	Chemical properties, Most sensitive endpoint, Test organism life stage, Effect values	Multiple trophic levels, Critical life stages

Validation and Quality Assurance Protocols

To ensure the reliability of data extracted using customized search strategies, implement the following validation protocols:

- **Cross-Reference Validation:** Verify a subset of extracted records against original publications to confirm accuracy of abstracted data.
- **Completeness Assessment:** Document the percentage of records containing complete data for critical fields identified in output customization.
- **Sensitivity Analysis:** Test slight variations in duration thresholds to assess potential impact on results and conclusions.
- **Peer Review of Strategy:** Submit search methodology, including duration parameters and output fields, for peer review to identify potential biases or omissions.

The Scientist's Toolkit: Research Reagent Solutions

Ecotoxicology research relies on both digital and physical research tools. The following table details essential components of the ecotoxicology researcher's toolkit, with particular emphasis on resources relevant to ECOTOX database utilization:

Table 3: Research Reagent Solutions for Ecotoxicology Investigations

Tool Category	Specific Solution	Function in Research	Application Context
Database Platforms	ECOTOX Knowledgebase	Curated repository of ecotoxicity test results	Primary source for historical toxicity data for chemical assessments [1] [4]
Search Optimization Tools	Boolean Operators	Combine search terms using AND, OR, NOT to refine results	Creating precise search queries in ECOTOX and literature databases [32]
Chemical Identification	CompTox Chemicals Dashboard	Chemical information, properties, and identifiers	Cross-referencing chemical data and structure-property relationships [1]
Statistical Analysis	R/Python with ecotoxicology packages	Statistical analysis of toxicity data, dose-response modeling, SSDs	Analyzing exported ECOTOX data for trend identification and benchmark derivation
Data Visualization	ChartExpo, Powerdrill AI	Creating charts and graphs for quantitative data presentation	Visualizing ECOTOX data patterns, trends, and relationships [33] [34]
Quality Assessment	Systematic Review Protocols	Framework for evaluating study quality and relevance	Assessing reliability of studies retrieved through ECOTOX searches [4]
Reference Management	Zotero, EndNote	Organization of literature references and citations	Managing publications identified through ECOTOX and complementary searches

Mastering duration filters and output field customization within the ECOTOX Knowledgebase enables researchers to conduct more precise, efficient, and reproducible ecotoxicology assessments. The systematic approaches outlined in this guide—from fundamental filter application to advanced output strategies—support the evolving needs of chemical safety evaluation and ecological risk assessment. As ECOTOX continues to grow with quarterly updates and expanding data coverage, these search methodologies will remain essential for transforming extensive toxicity information into actionable scientific insights. By implementing these protocols, researchers and regulatory professionals can enhance the quality and defensibility of chemical assessments while maximizing the utility of this comprehensive knowledgebase.

Within ecotoxicology, the reliable interpretation of data is fundamental to the accurate assessment of chemical risks to ecosystems. The **ECOTOXicology Knowledgebase (ECOTOX)** stands as a pivotal repository, compiling toxicity test results for thousands of chemicals and species [35]. However, the sheer volume of data, along with its inherent variability, presents a significant challenge for risk assessors. This guide outlines best practices for interpreting this complex information, focusing on the critical understanding of **standardized units** and **test results** to ensure robust and reproducible ecological risk assessments. Adherence to standardized methods and rigorous data evaluation criteria is essential for generating reliable data that can support regulatory decision-making [36] [10].

Foundational Concepts in Toxicity Testing

Standardized Toxicity Tests

Standardized toxicity tests are the backbone of ecotoxicology, providing consistent methods that ensure data comparability across different laboratories and studies. These tests are governed by internationally accepted guidelines, which are crucial for regulatory acceptance [36].

- **OECD Guidelines:** Developed by the Organisation for Economic Co-operation and Development, these provide the most relevant internationally agreed testing methods for a variety of endpoints and test organisms (e.g., fish, invertebrates, plants) [36].
- **EPA Protocols:** The United States Environmental Protection Agency's Office of Chemical Safety and Pollution Prevention (OCSPP) develops and implements test guidelines for pesticides and toxic substances, covering both acute and chronic toxicity tests for aquatic and terrestrial organisms [36].

The importance of these standardized methods cannot be overstated. They ensure **reproducibility** and **comparability** by minimizing variability through consistent test conditions (e.g., temperature, light, media). This consistency allows for the direct comparison of toxicity data for different chemicals or environmental samples and is typically essential for data to be accepted for regulatory purposes and risk assessment [36].

Key Toxicity Endpoints and Biomarkers

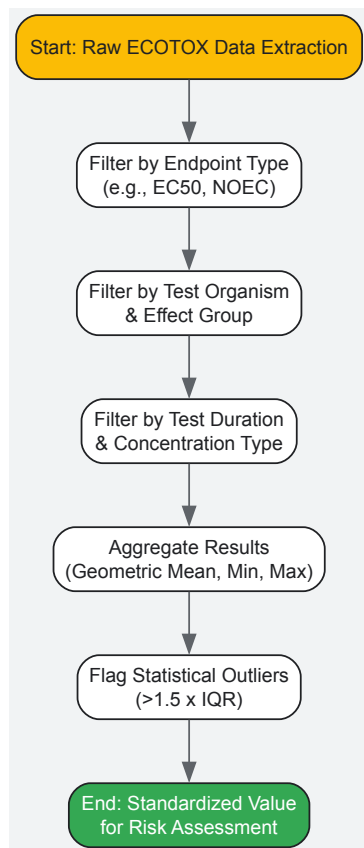
Toxicity tests measure a range of endpoints, from overt lethal effects to more subtle sublethal and cellular changes, providing a comprehensive picture of a substance's toxicological profile [36].

Table 1: Categories of Toxicity Endpoints

Endpoint Category	Description	Common Metrics	Interpretation
Lethal Endpoints	Measure death of test organisms, typically in acute toxicity tests.	LC50 (Lethal Concentration 50%), LD50 (Lethal Dose 50%) [36].	Lower LC50/LD50 values indicate higher acute toxicity [36].
Sublethal Endpoints	Reveal long-term impacts on organism health, survival, and fitness.	Growth inhibition, reproductive effects (e.g., number of offspring, embryo development), behavioral changes [36].	Indicates chronic effects and potential impacts on population dynamics [36].
Biomarkers & Cellular Effects	Provide measurable biological responses indicating exposure or early effects.	Biochemical (enzyme activity, gene expression), physiological (respiration rate), histopathology (tissue damage), genotoxicity (DNA damage) [36].	Serve as early warning signals before adverse effects are apparent at the organismal level [36].

Data Interpretation and Standardization Workflow

Interpreting ecotoxicity data requires a structured process to transform raw, often variable, test results into a reliable, aggregated value suitable for risk assessment. The following workflow, adapted from initiatives like Standartox, outlines this multi-stage process [35].



Data Filtering and Quality Control

The initial phase involves applying stringent filters to the raw data to ensure quality and relevance. This aligns with the **U.S. EPA's Evaluation Guidelines**, which specify criteria for accepting open literature toxicity data [10].

- **Filter by Endpoint Type:** Restrict data to commonly used and interpretable endpoints, such as **half-maximal effective concentrations (EC50)** or **no-observed-effect concentrations (NOEC)** [35]. This ensures consistency in the type of effect being analyzed.
- **Filter by Test Organism and Effect:** Refine data by taxonomic group, habitat (e.g., freshwater, marine, terrestrial), and the specific biological effect measured (e.g., mortality, growth, reproduction) [10] [35]. This allows for targeted assessments.
- **Filter by Test Conditions:** Apply filters based on test duration, concentration type (e.g., active ingredient vs. formulation), and other experimental parameters to enhance comparability [35].

The U.S. EPA guidelines mandate that accepted studies must meet minimum criteria, including: toxic effects from **single chemical exposure**; effects on live, whole organisms; a reported **concurrent chemical concentration** and **explicit exposure duration**; and comparison to an **acceptable control group** [10].

Data Aggregation and Outlier Management

For a given chemical and test organism combination, multiple test results are often available and can exhibit significant variability [35]. Aggregation is essential to derive a single, robust value.

- **Calculate Geometric Mean:** The **geometric mean** is the preferred method of aggregation as it is less influenced by outliers and is suitable for skewed data distributions, which are common in ecotoxicity data [35]. It is considered more reliable than the median for small datasets [35].
- **Identify Statistical Outliers:** As part of the aggregation process, outliers that exceed **1.5 times the interquartile range** should be flagged. This alerts the reviewer to data points that may require closer scrutiny, though they are typically still included in the geometric mean calculation due to its robustness [35].

This standardized aggregation process, as implemented by tools like **Standartox**, reduces the uncertainty in risk assessments that arises from the subjective selection of different ecotoxicological test data from the vast array available [35].

The Researcher's Toolkit for Data Evaluation

Key Research Reagent Solutions

The following table details essential materials and tools used in the evaluation and interpretation of ecotoxicity data.

Table 2: Essential Tools for Ecotoxicity Data Evaluation

Tool / Material	Function in Data Interpretation
ECOTOX Knowledgebase	Primary public database for sourcing single-chemical ecotoxicity test results for aquatic and terrestrial plants and animals [10] [35].
Standardized Test Organisms	Well-established species (e.g., <i>Daphnia magna</i> , <i>Raphidocelis subcapitata</i>) used in guideline tests to provide a consistent basis for comparing chemical toxicity [35].
Standartox Tool/Database	A processing tool that standardizes and aggregates ECOTOX data, calculating geometric means for specific chemical-organism test combinations to reduce variability [35].
Quality Control (QC) Criteria	A defined set of rules (e.g., EPA acceptance criteria) used to screen studies for reliability, verifiability, and relevance before inclusion in an assessment [10].

Visualizing Data for Interpretation

Effective data visualization is key to understanding patterns, trends, and outliers in quantitative ecotoxicity data. Choosing the right chart type is critical for clear communication.

- **Box Plots for Data Distribution:** A **box plot** (or box-and-whisker plot) is ideal for showing the distribution of ecotoxicity values for a particular chemical. It visually displays the median, quartiles, and potential outliers, allowing for easy comparison of the central tendency and variability of multiple datasets [34].
- **Bar Charts for Categorical Comparison:** **Bar charts** are excellent for comparing aggregated toxicity values (e.g., geometric mean EC50) across different chemical groups or taxonomic families [37] [34].
- **Line Charts for Trends Over Time:** While less common for single endpoint comparison, line charts are useful for illustrating how toxicity parameters or risk indicators change over continuous time intervals, such as throughout a long-term monitoring study [34].

When creating any visualization, ensure **sufficient color contrast** between foreground elements (text, lines) and their background to meet accessibility standards and facilitate legibility [38].

The path from raw ecotoxicity data to a reliable risk assessment is paved with rigorous standardization and meticulous interpretation. By adhering to established test guidelines, implementing a systematic workflow for data filtering and aggregation, and utilizing appropriate tools and visualizations, researchers can effectively navigate the complexities of databases like ECOTOX. Mastering these best practices for understanding standardized units and test results is not merely an academic exercise; it is the foundation for producing the consistent, high-quality, and reproducible science that is essential for protecting ecosystems and informing sound regulatory policy. Ongoing efforts, such as the development of good practices for omics data interpretation, indicate that this field continues to evolve to incorporate new data types and methodologies for regulatory application [39].

The **U.S. Environmental Protection Agency (EPA) New Approach Methods (NAMs) Training Program** represents a strategic initiative to equip researchers, risk assessors, and regulatory professionals with the expertise required to implement modernized toxicological approaches. Established under the EPA's NAMs Work Plan, this program directly supports the agency's mandate to **reduce vertebrate animal testing** while continuing to deliver robust chemical safety assessments [40]. The program provides comprehensive training on computational tools, bioinformatics, and predictive modeling that form the foundation of 21st-century toxicology.

Training focuses on the practical application of EPA's suite of computational toxicology resources, with the **ECOTOX Knowledgebase** representing a cornerstone tool for ecological risk assessment. By providing accessible, structured training on these resources, the EPA empowers scientists to efficiently generate and interpret data for chemical evaluations, regulatory submissions, and research applications. The demonstrated impact is substantial, with individual training sessions attracting hundreds of participants—including 575 attendees for a February 2023 ECOTOX session—highlighting the scientific community's strong demand for these specialized skill development opportunities [40].

ECOTOX Knowledgebase: Foundation and Significance

The **Ecotoxicology (ECOTOX) Knowledgebase** is a comprehensive, publicly accessible database providing curated single-chemical toxicity data for ecologically relevant species. It stands as the **world's largest compilation of curated ecotoxicity data**, integrating information from over 53,000 references to deliver more than one million test records covering **over 13,000 aquatic and terrestrial species** and **12,000 chemicals** [1] [4]. This extensive knowledgebase evolved from three originally independent databases—AQUIRE, PHYTOTOX, and TERRETOX—creating a unified system that supports diverse assessment needs [41].

ECOTOX functions as a critical resource for addressing multiple regulatory and research objectives. Its applications include supporting the development of **chemical benchmarks** for water and sediment quality assessments, informing **ecological risk assessments** for chemical registration and reregistration, and aiding in the **prioritization of chemicals** under the Toxic Substances Control Act (TSCA) [1]. Furthermore, the

database plays an essential role in advancing New Approach Methodologies by providing the empirical in vivo data necessary for developing and validating computational models, including quantitative structure-activity relationship (QSAR) models and cross-species extrapolation tools [1] [4].

Systematic Data Curation Methodology

The scientific integrity of ECOTOX relies on a rigorous, systematic literature review and data curation pipeline that aligns with contemporary systematic review practices. The process follows a defined sequence of operations with standard operating procedures for each phase, ensuring transparency, consistency, and reliability in the captured data [4].

Table: ECOTOX Knowledgebase Data Curation Pipeline

Pipeline Stage	Key Activities	Quality Assurance Measures
Literature Identification	Comprehensive searches of peer-reviewed and gray literature using chemical-specific search terms	Chemical and species verification against standardized vocabularies and authoritative sources
Study Screening	Title/abstract review followed by full-text assessment against applicability criteria	Evaluation against pre-defined inclusion criteria for ecologically relevant species, exposure concentration, and duration
Data Abstraction	Extraction of detailed chemical, species, test condition, and result information using controlled vocabularies	Documentation of experimental controls and measured endpoints to ensure data acceptability
Data Maintenance	Quarterly updates with new data and features; continuous quality control checks	Regular updates to standard operating procedures; interoperability enhancements with related tools

The curation methodology requires that all studies include documented controls and explicitly reported endpoints to be included in the knowledgebase. This systematic approach ensures that ECOTOX data adheres to the **FAIR principles** (Findable, Accessible, Interoperable, and Reusable), making it suitable for regulatory decision-making and advanced research applications [4].

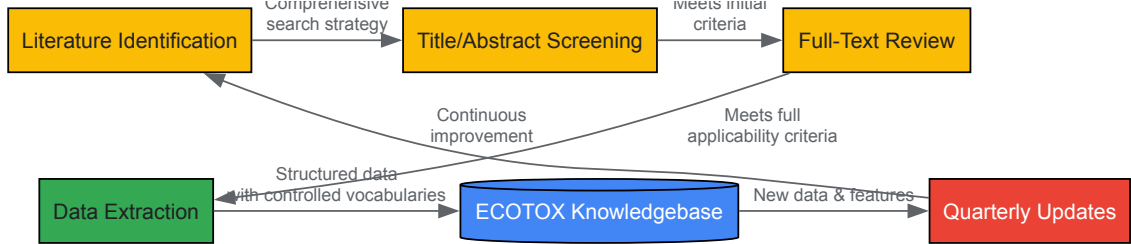


Figure: The systematic data curation workflow ensures all ECOTOX data undergoes rigorous quality control before inclusion.

Comprehensive Training Resource Catalog

The EPA NAMs training program provides multiple resource types tailored to different learning preferences and experience levels. These materials are systematically organized on the EPA's dedicated NAMs training portal, allowing users to filter content by topic, subtopic, resource type, or year to locate their required materials efficiently [42].

Table: ECOTOX and Related NAMs Training Resources (2024-2025)

Year	Topic Area	Resource Type	Title/Description	Key Presenters/Focus
2025	Ecotoxicology	Video & Slides	SeqAPASS Virtual Training	Carlie LaLone & Peter Schumann: Tool interface demonstration
2025	CompTox Dashboard	Video & Slides	CompTox Chemicals Dashboard Virtual Training	Nisha Sipes: Risk evaluation components & case study
2024	ECOTOX	Tool Tips Videos	ECOTOX Knowledgebase Tool Tips Series	Short videos on basic functionality
2024	ECOTOX	Slide Deck	ECOTOX Knowledgebase - NAMs Workshop	Jennifer Olker: Core functionality and applications
2024	Exposure	User Guide	ChemExpo Knowledgebase User Guide	Overview of data, functionality, and use cases

Year	Topic Area	Resource Type	Title/Description	Key Presenters/Focus
2024	ECOTOX	Video & Slides	Introduction to ECOTOX Knowledgebase	Jennifer Olker: Foundational concepts

Specialized Training Modules

Beyond foundational resources, the EPA offers advanced training materials addressing specific applications and integrated workflows:

- **Integrated Case Studies:** The 2024 NAMs Workshop featured a comprehensive case study integrating **ToxCast, SeqAPASS, and ECOTOX** tools, demonstrating how these resources can be combined to address complex assessment scenarios through complementary data streams [42].
- **Chemical Transformation Prediction:** A June 2024 webinar covered the **Chemical Transformation Simulator (CTS)**, a web-based tool for predicting transformation pathways and physicochemical properties of organic chemicals, with presenter Caroline Stevens explaining its application for addressing data gaps in chemical registration and assessment [42].
- **PFAS-Focused Training:** A December 2024 webinar specifically addressed **ECOTOX updates for PFAS chemicals**, providing researchers and risk assessors with targeted strategies for evaluating this priority chemical class using the knowledgebase's curated data [43].

Experimental Protocol: Systematic Search and Data Extraction

The operational foundation of the ECOTOX Knowledgebase relies on a rigorous, multi-stage protocol for identifying, evaluating, and extracting ecotoxicity data from the scientific literature. This systematic approach ensures the consistency, reliability, and transparency of the database content.

Literature Search Strategy

The initial phase involves comprehensive literature identification through systematic search operations:

- **Search Query Development:** Formulate chemical-specific search strings incorporating Chemical Abstracts Service (CAS) Registry Numbers, preferred chemical names, and synonymous terms to maximize retrieval of relevant studies.
- **Database Interrogation:** Execute searches across multiple scientific literature databases, including those indexing peer-reviewed publications and relevant gray literature sources.
- **Reference Compilation:** Aggregate results from all sources into a unified reference library while eliminating duplicate entries to create an efficient screening workflow.

Study Screening and Eligibility Assessment

The screening process employs a two-tiered approach to identify studies meeting inclusion criteria:

- **Title/Abstract Screening:** Initially assess references for general relevance based on predefined criteria including:
 - Investigation of single chemical stressors (excluding complex mixtures)
 - Use of ecologically relevant aquatic or terrestrial species
 - Reporting of exposure concentrations and duration
 - Measurement of adverse effects or toxicity endpoints
- **Full-Text Review:** Conduct comprehensive evaluation of potentially relevant studies against refined eligibility criteria:
 - Adequate biological species identification (genus and species)
 - Clear methodological documentation including exposure regime
 - Appropriate experimental controls and statistical analyses
 - Measured effects relevant to ecological risk assessment

For studies meeting all eligibility criteria, trained reviewers extract detailed information using standardized processes:

- **Chemical Information:** Record chemical identity, purity, formulation, and verification against DSSTox substance records to ensure accurate chemical representation.
- **Test Organism Details:** Document species name, life stage, sex, source, and acclimation conditions using controlled taxonomic vocabularies.
- **Experimental Methodology:** Capture exposure system, route, duration, medium characteristics, and temperature conditions employing standardized descriptive terms.
- **Results and Endpoints:** Extract quantitative toxicity values (e.g., LC50, EC50, NOEC), statistical measures, and observed effects using consistent endpoint terminology.

This meticulous protocol ensures that all data incorporated into ECOTOX meets quality standards sufficient for regulatory application and scientific research.

Effectively utilizing the ECOTOX Knowledgebase and related NAMs often requires familiarity with a suite of complementary computational tools and resources. The EPA's Computational Toxicology and Exposure program provides this integrated toolkit to support comprehensive chemical safety assessment.

Table: Essential Computational Tools for Ecotoxicology Research

Tool Name	Resource Type	Primary Function	Application in Research
CompTox Chemicals Dashboard	Data Repository	Chemistry, toxicity, and exposure data for >1 million chemicals	Chemical identification, property data, and hazard screening
SeqAPASS	Prediction Tool	Cross-species susceptibility extrapolation	Predicting chemical susceptibility for data-limited species
Web-ICE	Estimation Model	Acute toxicity estimation for aquatic and terrestrial organisms	Screening-level risk assessment using species sensitivity distributions
Chemical Transformation Simulator	Prediction Tool	Transformation pathway and property prediction	Identifying potential degradation products and their properties
ToxCast	Bioactivity Database	High-throughput screening bioactivity data	Mechanistic insight into potential toxicity pathways
httk R Package	Modeling Tool	Toxicokinetic modeling and dosimetry	Predicting in vivo toxicity from in vitro bioactivity data

Tool Integration Workflow

The power of these resources is magnified when applied in coordinated workflows. A typical assessment might begin with chemical characterization using the **CompTox Chemicals Dashboard**, progress to effects evaluation using **ECOTOX** and **ToxCast** data, and incorporate **SeqAPASS** to extend predictions to species lacking testing data. The **httk R package** then facilitates translation of in vitro concentrations to in vivo dosimetry for more meaningful interpretation of bioactivity data [42] [44].

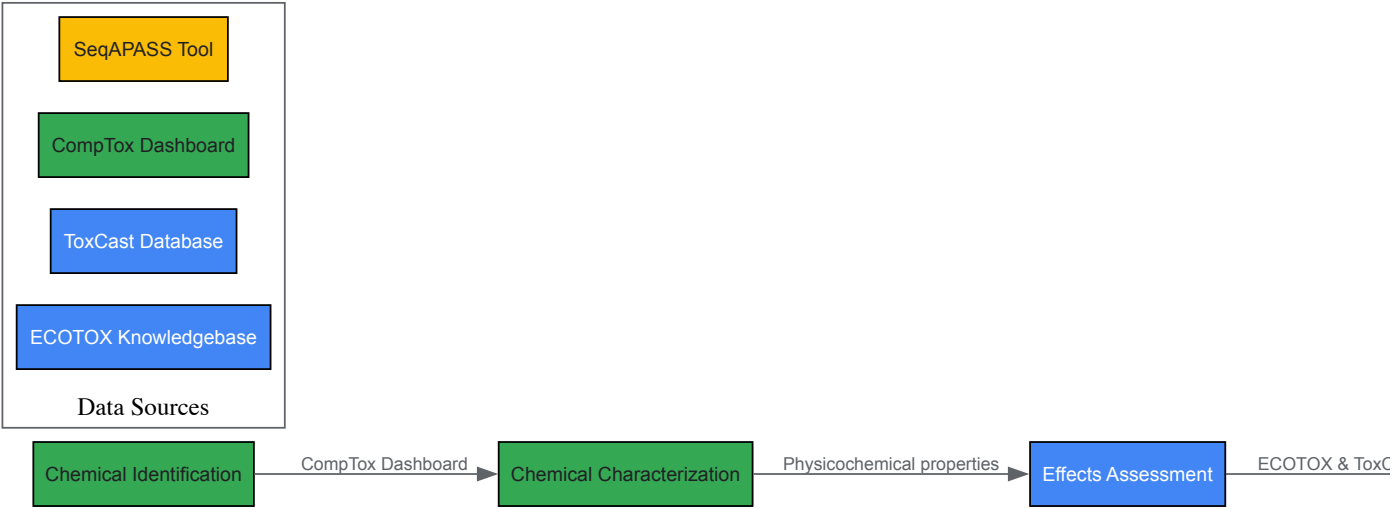


Figure: Integrated workflow showing how EPA's computational tools combine to support comprehensive chemical assessment.

Accessing and Utilizing Training Materials

Researchers have multiple pathways to access NAMs training resources, structured to accommodate different learning preferences and experience levels.

Centralized Training Portal

The primary access point for NAMs training is the **EPA's NAMs Training page** [42], which provides:

- **Filterable Resource Catalog:** Searchable table of training materials filterable by topic, subtopic, type, and year
- **Multiple Format Options:** Videos, slide decks, worksheets, and user guides catering to different learning styles
- **Progressive Difficulty Levels:** Beginner to advanced content structured to build competency systematically

Beyond the core training portal, several specialized resource types enhance the learning experience:

- **Hands-On Worksheets:** Practical exercises with answer keys for self-directed learning, available for tools including ECOTOX, CompTox Chemicals Dashboard, GenRA, and the htk R package [45].
- **Tool Tips Video Series:** Brief, focused videos (typically 2-5 minutes) demonstrating specific functionalities, such as navigating the ECOTOX application interface or exporting data for external analysis [46].
- **Live Training Opportunities:** Scheduled virtual training sessions, such as the CompTox Chemicals Dashboard training scheduled for June 12, 2025, which offer opportunities for real-time interaction with EPA subject matter experts [42] [44].

Technical Support Channels

The EPA maintains dedicated support channels for researchers requiring additional assistance:

- **ECOTOX Technical Support:** Specialized assistance available via ecotox.support@epa.gov for questions specific to the ECOTOX Knowledgebase [1].
- **General NAMs Inquiries:** The NAM@epa.gov mailbox addresses broader questions about the training program and resource materials [45] [46].
- **Center for Computational Toxicology and Exposure:** Expert consultation for tool selection and application strategies through the centralized contact point for computational toxicology resources [44].

This multi-faceted support structure ensures that researchers can efficiently overcome technical challenges and maximize the utility of EPA's NAMs resources in their scientific and regulatory workflows.

The **Ecotoxicology (ECOTOX) Knowledgebase** is a comprehensive, publicly available resource developed by the U.S. Environmental Protection Agency (EPA). It serves as the world's largest curated repository of single chemical ecotoxicity data, providing critical information on adverse effects of chemical stressors to ecologically relevant aquatic and terrestrial species [1] [5]. This knowledgebase supports environmental research, ecological risk assessments, and regulatory decision-making by offering systematically curated toxicity data compiled from peer-reviewed scientific literature [5].

As of its latest version, ECOTOX contains an extensive collection of over **one million test results** derived from more than **53,000 references**, covering **12,000 chemicals** and **13,000 species** [1]. The database is updated quarterly with new data and features, ensuring researchers have access to the most current ecotoxicological information [1]. The primary user base includes researchers, risk assessors, and decision-makers at local, state, and tribal governments who require reliable toxicity data for chemical safety evaluations and environmental protection initiatives [1].

Technical Support and Contact Information

Official Support Channel

For technical assistance with the ECOTOX Knowledgebase, users should direct their inquiries to the dedicated support team via email:

- **Primary Support Email:** ecotox.support@epa.gov [1]

This official channel is monitored by technical specialists who can address issues ranging from data access problems to functionality questions. Users experiencing technical difficulties, such as browser compatibility issues (the Knowledgebase may not function properly in Chrome in some instances), are advised to clear their browsing data, cache, or history as an initial troubleshooting step [12].

Alternative Contact Methods

While the email support remains the primary contact method, the EPA also provides a broader contact form for questions about its CompTox tools suite, of which ECOTOX is a component. This can be accessed through the "Contact Us About CompTox Tools" link available on the relevant EPA resource pages [1] [12].

Structured Training Programs

The EPA has developed comprehensive training resources for ECOTOX through its **New Approach Methods (NAMs) Training Program** [1] [42]. This program houses various training materials designed to assist users with EPA NAMs tools, including the ECOTOX Knowledgebase.

Table: ECOTOX Knowledgebase Training Resources

Year	Topic	Type	Title	Key Content
2024	ECOTOX Knowledgebase	Tool Tips	ECOTOX Knowledgebase Tool Tips [42]	Series of short videos demonstrating basic functionality
2024	ECOTOX Knowledgebase	Slide Deck	ECOTOX Knowledgebase - NAMs Workshop [42]	Overview of system capabilities and use cases
2024	ECOTOX Knowledgebase	Video	Introduction to ECOTOX Knowledgebase [42]	Basic orientation for new users
2022	ECOTOX Knowledgebase	Journal Article	The ECOTOXicology Knowledgebase: A Curated Database [5]	Comprehensive peer-reviewed description of database methodology

Experimental Protocols and Data Curation Methodology

The ECOTOX Knowledgebase employs rigorous, systematic review procedures for data identification and curation that align with contemporary evidence-based toxicology practices [\[5\]](#). The protocol involves:

- **Literature Identification:** Comprehensive searches of peer-reviewed literature using standardized search strategies [\[5\]](#)
- **Study Evaluation:** Application of pre-defined criteria for relevance and acceptability of toxicity tests [\[5\]](#)
- **Data Extraction:** Systematic abstraction of methodological details and results using controlled vocabularies [\[5\]](#)
- **Quality Assurance:** Implementation of quality control measures throughout the curation pipeline [\[5\]](#)

This methodological rigor ensures the data within ECOTOX meets high standards of reliability and usability for research and regulatory applications. The curation team extracts pertinent information on species, chemicals, test methods, and results as presented by the original authors, maintaining consistency through well-established controlled vocabularies [\[5\]](#).

ECOTOX Knowledgebase Functionality and Features

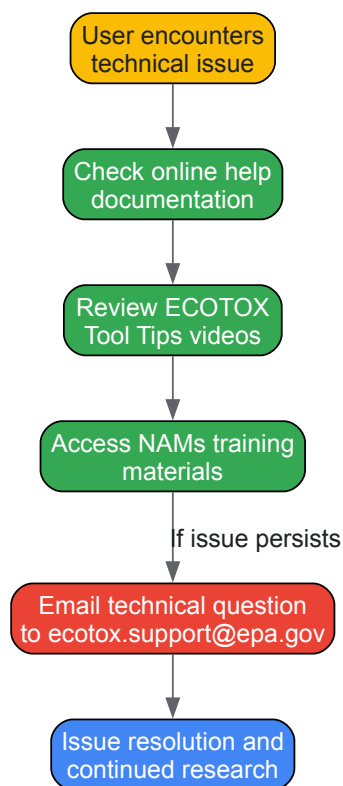
Core Functional Capabilities

The ECOTOX Knowledgebase provides three primary modes of interaction with its data, each designed to support different user needs and levels of familiarity with the system [\[1\]](#):

- **Search Feature:** Allows targeted queries for specific chemicals, species, effects, or endpoints. Users can refine searches by 19 different parameters and customize output selections from over 100 data fields. This feature includes direct links to the CompTox Chemicals Dashboard for additional chemical information [\[1\]](#).
- **Explore Feature:** Designed for users who may not have exact search parameters, this feature enables broader investigation by chemical, species, or effects. Additional data fields allow customization of output results for import into other analytical tools [\[1\]](#).
- **Data Visualization Feature:** Provides interactive plots for data exploration, allowing users to hover over data points, zoom in on specific sections, and retrieve detailed information of interest [\[1\]](#).

Workflow Integration Pathway

The following diagram illustrates the systematic workflow for utilizing ECOTOX support resources and technical documentation:



Research Applications and Interoperability

Key Research Applications

The ECOTOX Knowledgebase supports diverse research and regulatory applications, including:

- **Chemical Benchmark Development:** Creating benchmarks for water and sediment quality assessments [1]
- **Aquatic Life Criteria:** Informing the design of criteria to protect freshwater and saltwater organisms from short-term and long-term chemical exposure [1]
- **Ecological Risk Assessment:** Supporting chemical registration and re-registration evaluations [1]
- **Chemical Prioritization:** Aiding in the assessment of chemicals under the Toxic Substances Control Act (TSCA) [1]
- **Model Development and Validation:** Enabling the creation of quantitative structure-activity relationship (QSAR) models and extrapolation from in vitro to in vivo effects [1]

Table: Key Computational Toxicology Resources for Ecotoxicology Research

Resource Name	Type	Function	Access Point
CompTox Chemicals Dashboard [22]	Data Resource	Provides chemistry, toxicity, and exposure data for over one million chemical substances	https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard-resource-hub
ToxValDB [47]	Database	Compilation of in vivo toxicology data and derived toxicity values covering 41,769 unique chemicals	Downloadable via Computational Toxicology Data
SeqAPASS [42]	Screening Tool	Online tool for extrapolating toxicity information across species using protein sequence similarity	NAMs Training Site
EnviroTox Database [48]	Database	Curated aquatic toxicity database supporting ecological risk assessment and model development	https://envirotoxdatabase.org

Quantitative Data Landscape

The following table summarizes the comprehensive data available within the ECOTOX Knowledgebase, highlighting its extensive coverage of chemicals, species, and test results:

Table: ECOTOX Knowledgebase Quantitative Data Summary

Data Category	Count	Scope and Coverage
Test Records	1,000,000+	Single chemical toxicity tests [1]
Chemical Substances	12,000+	Primarily industrial compounds, pesticides, and contaminants [1]
Ecological Species	13,000+	Aquatic and terrestrial organisms [1]
Source References	53,000+	Peer-reviewed literature sources [1]

The field of ecotoxicology continues to evolve with an increasing emphasis on **New Approach Methodologies (NAMs)** that reduce reliance on animal testing while maintaining scientific rigor [\[48\]](#). The ECOTOX Knowledgebase is positioned to support this transition by providing curated in vivo data necessary for validating alternative methods and computational models [\[5\]](#).

Recent initiatives within the scientific community highlight the importance of updating statistical practices in ecotoxicology [\[49\]](#) and developing integrated testing strategies that combine traditional and novel approaches [\[48\]](#). The ECOTOX team continues to enhance the system's interoperability with other relevant resources, following FAIR principles (Findable, Accessible, Interoperable, and Reusable) to maximize data utility [\[5\]](#).

For researchers, maintaining awareness of ongoing updates to the ECOTOX Knowledgebase—including quarterly data additions and feature enhancements—ensures access to the most current ecotoxicological information. The technical support channels and documentation resources detailed in this guide provide multiple pathways for overcoming operational challenges and maximizing the research value of this critical environmental database.

The **Ecotoxicology (ECOTOX) Knowledgebase** is a comprehensive, publicly available resource developed by the United States Environmental Protection Agency (EPA). It provides curated information on the adverse effects of single chemical stressors to ecologically relevant aquatic and terrestrial species [\[1\]](#). The database is an essential tool for ecological risk assessments, the design of aquatic life criteria, and the prioritization of chemicals, supporting researchers, risk assessors, and decision-makers [\[1\]](#).

Integrating this knowledgebase with the R programming environment unlocks powerful capabilities for reproducible data retrieval, custom analysis, and the generation of publication-quality visualizations. This guide provides a technical overview of methods for accessing ECOTOX data, transforming it within R, and applying advanced visualization techniques to derive ecotoxicological insights.

Data Retrieval and Export from ECOTOX

Accessing the ECOTOX Knowledgebase

The ECOTOX Knowledgebase can be accessed directly through its online interface. The system offers several pathways for data discovery and extraction [\[1\]](#):

- **Search Feature:** Allows users to find data using specific parameters such as chemical name, species, or effect endpoint. Results can be refined using 19 different filters, and outputs can be customized from over 100 available data fields [\[1\]](#).
- **Explore Feature:** Useful when exact search parameters are unknown, enabling a more flexible discovery of relevant data by chemical, species, or effects [\[1\]](#).
- **Data Visualization Feature:** Provides interactive plots for initial data exploration within the web application, allowing users to hover over data points and zoom into specific sections [\[1\]](#).

The database is updated quarterly and contains over one million test records compiled from more than 53,000 references, covering over 13,000 species and 12,000 chemicals [\[1\]](#).

Manual Export and Download

Users can export data directly from the ECOTOX web interface:

- Perform a search using the "Search" or "Explore" modules.
- Refine the results using available filters (e.g., species group, effect, exposure duration).
- Customize the output columns to include relevant data fields for your analysis.
- Use the export functionality to download the results in a structured format (e.g., CSV, TXT) for import into R [\[46\]](#).

Table 1: Core ECOTOX Data Tables and Descriptions

Table Name	Key Contents	Primary Key
Species	Taxonomic information (species, genus, family, etc.) and ecological group.	species_number

Table Name	Key Contents	Primary Key
Tests	Experimental design details (test duration, exposure type, test location).	test_id
Results	Experimental outcomes, including effect and endpoint values (e.g., LC50, EC50).	result_id
Chemicals	Chemical identifiers (CAS, DTXSID, Name) and structures.	cas_number, dsstox_substance_id

Programmatic Access and Data Curation in R

Using the ECOTOXr Package

For reproducible and large-scale analysis, the `ECOTOXR` R package offers a programmatic alternative to manual downloads. This package facilitates transparent retrieval and curation of data directly from the ECOTOX database into the R environment [24].

Experimental Protocol: Data Retrieval with ECOTOXr

- Installation:** The `ECOTOXR` package can be installed from CRAN or its development repository.
- Database Setup:** The package may require downloading the latest ECOTOX ASCII files and building a local SQLite database for efficient querying.
- Query Construction:** Formulate a query to extract specific data. For example, to retrieve acute toxicity data for a chemical:
- Data Curation:** The retrieved data requires cleaning and harmonization before analysis. Key steps include:
 - Unit Standardization:** Converting all concentration values to a common unit (e.g., mg/L or mol/L).
 - Identifier Harmonization:** Using robust chemical identifiers like DTXSID or InChIKey to resolve ambiguities in chemical naming [11].
 - Filtering:** Applying criteria for test duration, life stage, and data quality based on the research question.

The ADORE Benchmark Dataset

Researchers can also utilize pre-compiled benchmark datasets like **ADORE** (Aquatic Toxicity Datasets for Organic Chemicals), which is derived from ECOTOX. ADORE provides a curated subset focused on acute aquatic toxicity for fish, crustaceans, and algae, and includes additional chemical and species-specific features suitable for machine learning [11].

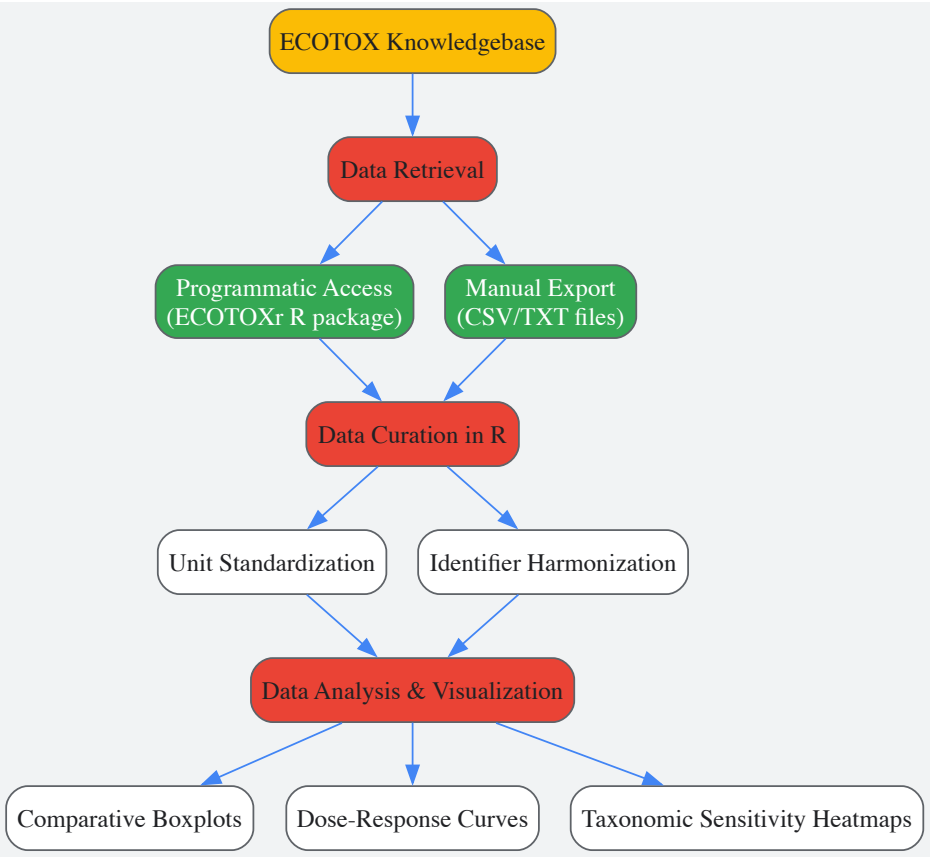
Table 2: Key Features of the ADORE Benchmark Dataset

Feature Category	Example Variables	Utility in Analysis
Ecotoxicology Core	Chemical, Species, Endpoint (e.g., LC50, EC50), Effect (e.g., MOR, ITX), Duration	Primary toxicity response data.
Chemical Properties	Molecular weight, LogP, SMILES, DTXSID, InChIKey	For QSAR modeling and chemical similarity analysis.
Species Traits	Taxonomic family, genus, species, phylogenetic data	For cross-species extrapolation and understanding taxonomic sensitivity.

Data Visualization and Analysis in R

Creating Informative Visualizations

Once data is imported and curated in R, it can be visualized to reveal patterns and relationships. The following diagram illustrates a recommended workflow for data handling and visualization.



Visualization Workflow

Comparative Graphics for Ecotoxicological Data

When comparing quantitative data across different groups (e.g., species, chemicals), several graph types are particularly effective [50]:

- **Boxplots (Parallel Boxplots):** Ideal for summarizing the distribution of a toxicity endpoint (e.g., LC50) across multiple groups. They display the median, quartiles, and potential outliers, facilitating quick comparison of central tendency and variability [50].
- **2-D Dot Charts:** Useful for smaller datasets, showing individual data points. Points can be stacked or jittered to prevent overplotting [50].
- **Back-to-back Stemplots:** A historical method for comparing distributions between two groups, retaining the original data values [50].

Applying Color in R Visualizations

Color is a critical preattentive feature for effectively distinguishing groups in data visualizations [51]. R offers several robust color palette packages and functions.

Recommended Color Palettes and Functions:

- **Viridis Palettes:** Perceptually uniform, robust to colorblindness, and printer-friendly. The `viridis` package provides scales like "viridis", "magma", "plasma", and "inferno" [52].
- **RColorBrewer Palettes:** The `RColorBrewer` package offers curated palettes. Use `display.brewer.all(colorblindFriendly = TRUE)` to view accessible options. Qualitative palettes (e.g., "Set2", "Dark2") are suitable for categorical data [52].
- **Base R HCL Palettes:** Since version 4.0.0, R includes improved default palettes. The `hcl.colors()` function provides a wide range of perceptually-based qualitative, sequential, and diverging palettes [51].

Table 3: Essential R Packages for Ecotoxicology Data Analysis

Package	Category	Primary Function	Research Reagent Solution
ECOTOXr	Data Access	Programmatic retrieval and curation of ECOTOX data.	Enables reproducible data import.
ggplot2	Visualization	Creates sophisticated, multi-layered graphs.	The core grammar of graphics for R.

Package	Category	Primary Function	Research Reagent Solution
viridis	Visualization	Provides colorblind-friendly, perceptually uniform color scales.	Ensures accessible and accurate color representation.
dplyr	Data Wrangling	Data manipulation (filtering, summarizing, transforming).	Essential for data cleaning and transformation.
RColorBrewer	Visualization	Offers a collection of color palettes for charts and maps.	Provides pre-tested color schemes.

The integration of the EPA's ECOTOX Knowledgebase with the analytical power of R creates a robust framework for advancing ecotoxicological research. By leveraging tools like the `ECOTOXR` package for reproducible data retrieval, adhering to careful data curation protocols, and applying principled visualization techniques with perceptually appropriate color palettes, researchers can efficiently transform raw ecotoxicity data into actionable scientific insights. This workflow not only enhances the rigor of data analysis but also promotes transparency and reproducibility in the field.

Evidence and Impact: Validating ECOTOX in Regulatory and Research Contexts

The **Ecotoxicology (ECOTOX) Knowledgebase** stands as a pivotal, authoritative source of curated ecotoxicity data, directly supporting U.S. Environmental Protection Agency (EPA) regulatory mandates for over three decades. Developed starting in the 1980s, ECOTOX has evolved into the world's largest compilation of curated single-chemical ecotoxicity data, providing systematic and transparent data for ecological risk assessments [5]. Its primary function is to supply rigorously curated data that informs the development of **water quality criteria** under the Clean Water Act and **pesticide risk assessments** under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA) [5]. The knowledgebase embodies a long-term commitment to data quality, systematic review, and interoperability, making it an indispensable tool for regulatory scientists, risk assessors, and environmental researchers tasked with protecting ecological health from chemical stressors.

Quantitative Scope of the ECOTOX Knowledgebase

The ECOTOX Knowledgebase provides an unparalleled volume of curated ecotoxicity data, abstracted from the peer-reviewed literature through a rigorous, standardized process. The tables below summarize the extensive quantitative scope of the data available to support environmental research and regulatory decisions.

Table 1: Overall Data Volume in ECOTOX (as of 2022) [5]

Component	Quantity
Scientific References	>53,000
Test Results	>1 million
Ecological Species	>13,000 (aquatic & terrestrial)
Unique Chemicals	>12,000

Table 2: Data Applications in EPA Regulatory Contexts [1] [5]

Regulatory Program	Use of ECOTOX Data
Clean Water Act	Developing Aquatic Life Criteria for freshwater and saltwater organisms.
FIFRA	Informing ecological risk assessments for pesticide registration and reregistration.
Toxic Substances Control Act (TSCA)	Aiding in the prioritization and assessment of chemicals.

Systematic Data Curation and Review Methodologies

The regulatory utility of ECOTOX is rooted in its **systematic, evidence-based review protocols** for data acquisition and curation, which align with contemporary systematic review practices [5].

Literature Search and Acquisition

The process begins with comprehensive searches of the open scientific literature to identify relevant peer-reviewed journal articles reporting results from toxicity tests on single chemical stressors. The search strategy is designed to be exhaustive, ensuring all potentially relevant studies are captured

for subsequent evaluation [5].

Data Extraction and Curation

Relevant information from accepted studies is abstracted into the knowledgebase using well-established **controlled vocabularies** to ensure consistency. The following key data is systematically extracted from each study:

- **Test Substance Information:** Chemical identity, purity, and formulation.
- **Test Organism Details:** Species name, life stage, source, and maintenance conditions.
- **Experimental Methodology:** Exposure pathway (e.g., water, diet, sediment), duration, and test endpoints measured.
- **Results Data:** Quantitative results (e.g., LC50, NOEC, LOEC) and statistical measures [5].

This meticulous curation process transforms disparate experimental data from thousands of sources into a structured, harmonized, and reusable format suitable for robust regulatory analysis.

Quality Assurance and Transparency

The entire pipeline for literature review and data curation is designed to be **transparent and objective**, featuring standardized operating procedures. The recent release of ECOTOX Version 5 represents a significant advancement in making both the data and the processes behind it more accessible and aligned with FAIR principles (Findable, Accessible, Interoperable, and Reusable) [5].

Application in Water Quality Criteria Development

The ECOTOX Knowledgebase is fundamentally embedded within the EPA's science-based process for establishing **National Recommended Water Quality Criteria** under Section 304(a) of the Clean Water Act [53] [54]. These criteria provide guidance to states and Tribes for setting water quality standards that protect aquatic life and human health [54].

The data from ECOTOX is crucial for deriving criteria that protect the majority of species in a given aquatic environment from the adverse effects of pollutants. The database's vast holdings on **acute and chronic toxicity** across thousands of species and chemicals allow EPA scientists to perform robust species sensitivity distributions and determine the highest concentration of a pollutant not expected to pose significant risk [1]. Furthermore, ECOTOX data supports the development of **numeric nutrient criteria** for nitrogen and phosphorus, which are essential for restoring water bodies impaired by nutrient pollution [53].

Role in Pesticide Registration and Review

Within the EPA's pesticide regulatory programs, ECOTOX provides critical data for **ecological risk assessments** that support the registration and periodic re-evaluation (registration review) of pesticides [1] [5].

Informing Registration Review

The EPA is mandated to review each registered pesticide at least every **15 years** to ensure it can still perform its intended function without creating unreasonable adverse effects on the environment [55]. The compilation of ecotoxicity data in ECOTOX is uniquely suited for linking traditional biological effects used in these regulatory risk assessments with mechanistic responses across species [1]. For instance, the registration review process for chemicals like **Atrazine** (Case #62) and **Acephate** (Case #42) relies on curated ecotoxicity data to draft proposed interim decisions [55].

Supporting Process Modernization

Recent third-party audits of the EPA's Office of Pesticide Programs (OPP) have identified challenges such as process fragmentation and knowledge gaps among reviewers [56]. Authoritative resources like ECOTOX help address these issues by providing a **standardized, centralized source of toxicity data**, which fosters more consistent review procedures and decision-making across the agency [56].

Leveraging the ECOTOX Knowledgebase effectively often involves using it in concert with other specialized tools and databases. The following table details essential resources for researchers and regulatory professionals in this field.

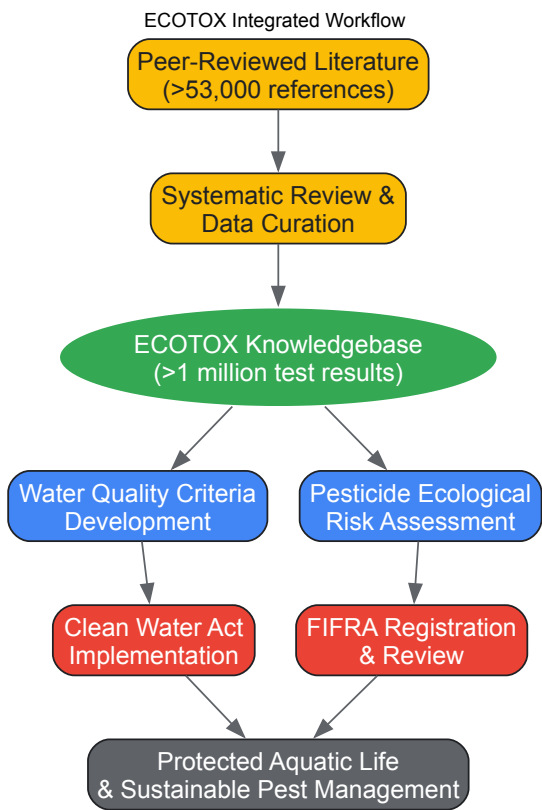
Table 3: Essential Research Tools and Resources for Ecotoxicology

Tool/Resource Name	Function and Utility
CompTox Chemicals Dashboard	Provides access to chemical structures, properties, and additional toxicity data, and is directly linked from ECOTOX chemical searches [1] [12].
ToxCast Data	Offers high-throughput screening data on thousands of chemicals, which can be used alongside traditional ecotoxicity data for predictive modeling [12].

Tool/Resource Name	Function and Utility
ToxRefDB (Toxicity Reference Database)	Contains in vivo animal toxicity data from guideline studies, useful for bridging mammalian and ecological toxicology [12].
Abstract Sifter	A literature mining tool that enhances PubMed searches, helping researchers quickly triage and find articles of interest for deeper analysis [12].
SHEDS-HT & SEEM Models	Provide high-throughput exposure predictions and estimates, which can be combined with hazard data from ECOTOX for risk-based prioritization [12].

Integrated Workflows: From Data Curation to Regulatory Application

The pathway from primary scientific literature to informed regulatory decisions is a multi-stage process. The following diagram visualizes the integrated workflow of the ECOTOX Knowledgebase, highlighting its central role in supporting both water quality and pesticide assessment programs.



For over three decades, the ECOTOX Knowledgebase has proven to be an indispensable asset for environmental regulation, directly supporting the scientific foundation of EPA's water quality criteria and pesticide assessments. Its enduring value lies in the **rigorous, systematic curation** of ecologically relevant toxicity data, its vast and growing **quantitative scope**, and its increasing **interoperability** with modern computational toxicology tools. As chemical challenges and regulatory science continue to evolve, ECOTOX remains a critical resource for transforming primary ecotoxicological research into actionable, defensible regulatory decisions that protect our nation's ecosystems.

Ecotoxicology databases are foundational tools for environmental research and chemical risk assessment, providing critical data on the adverse effects of chemical stressors on ecological species. The need for assembled, high-quality toxicity data has accelerated as thousands of chemicals enter commerce and regulatory mandates require safety assessments. Among these resources, the **ECOTOXicology Knowledgebase (ECOTOX)** stands as a comprehensive, curated database of ecologically relevant toxicity tests supporting environmental research and risk assessment [5]. This analysis provides a technical comparison between ECOTOX and other prominent ecotoxicological resources, examining their scope, data curation methodologies, functionality, and application in regulatory and research contexts.

The evolution of toxicity testing toward **New Approach Methodologies (NAMs)** and the increasing mandate to reduce animal testing have further emphasized the importance of accessible, well-curated historical data [5] [47]. Such databases provide the empirical foundation necessary for developing and validating computational models, including quantitative structure-activity relationship (QSAR) models and adverse outcome pathways (AOPs) [57]. This whitepaper examines the technical architecture and applicative strengths of ECOTOX relative to other resources, providing researchers and regulatory professionals with a guide for optimal database selection and use.

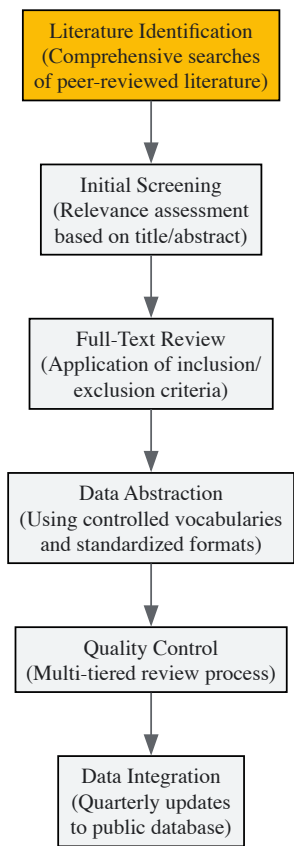
Architecture and Scope

ECOTOX, maintained by the U.S. Environmental Protection Agency (EPA), is the world's largest compilation of curated ecotoxicity data. As of its latest peer-reviewed description in 2022, the knowledgebase contains over **one million test results** covering more than **12,000 chemicals** and **13,000 aquatic and terrestrial species**, abstracted from over **53,000 scientific references** [5] [1]. The database provides information on single chemical stressors to ecologically relevant species, with data updated quarterly to include new findings and features [1].

The system is built on a structured curation pipeline that aligns with contemporary **systematic review practices**, ensuring transparency and objectivity in data collection and evaluation [5]. ECOTOX serves as an authoritative source for regulatory applications under multiple legislative frameworks, including the **Clean Water Act**, the **Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA)**, and the **Toxic Substances Control Act (TSCA)** [5]. Its data is explicitly designed to support the development of water quality criteria, ecological risk assessments, and the prioritization of chemicals for further evaluation.

Data Curation Methodology

ECOTOX employs a rigorous, multi-stage process for data harvesting and validation, adhering to systematic review principles for identifying, evaluating, and extracting toxicity data from the scientific literature.



ECOTOX Literature Review and Data Curation Pipeline

The curation process begins with comprehensive literature searches across multiple scientific platforms. Relevant studies are identified through systematic queries, and retrieved articles undergo a multi-stage evaluation. Pertinent information on species, chemical, test methods, and results presented by the authors is abstracted into the knowledgebase using **well-established controlled vocabularies** [5]. This controlled vocabulary enhances data consistency and enables precise querying. A key feature of the ECOTOX approach is the **extraction of detailed methodological metadata**, including test organism life stage, exposure conditions, and statistical analyses, allowing for sophisticated filtering and relevance assessment during data retrieval [5].

Comparative Analysis of Database Content and Coverage

Quantitative Comparison of Database Scope

The table below provides a quantitative comparison of ECOTOX and other frequently used toxicology databases, highlighting differences in their primary focus and data volume.

Table 1: Comparative Scope of Ecotoxicology and Toxicology Databases

Database	Primary Focus	Chemical Coverage	Species/Organism Coverage	Data Records	Key Strengths
ECOTOX	Ecological toxicity	>12,000 chemicals [1]	>13,000 aquatic & terrestrial species [1]	>1 million test results [1]	Comprehensive ecological focus, detailed test metadata, regulatory application
ToxValDB	Human health toxicity	41,769 unique chemicals (v9.6.1) [47]	Mammalian species & models	242,149 records (v9.6.1) [47]	Standardized toxicity values, exposure guidelines, integrated with CompTox Dashboard
CompTox Chemicals Dashboard	Aggregated chemical data	>1,000,000 chemicals [58]	Multiple species types	Integrated from multiple sources [12]	One-stop access to chemical properties, hazard, exposure data
TOXLINE	General toxicology	Not specified	Not specified	Not specified	Bibliographic information on toxicology [59]
Agricultural & Environmental Science Database	Environmental science	Not specified	Not specified	Not specified	Interdisciplinary coverage including pollution impacts [59]

Data Quality and Curation Approaches

A critical differentiator among toxicological databases is their approach to data quality and curation. ECOTOX employs a **manual curation process** with explicit quality considerations, where data are abstracted by trained curators following standardized protocols [5]. This human-centric approach, while resource-intensive, helps address inconsistencies that can be present in larger automated aggregations. In fact, one independent assessment noted that the US EPA Ecotox database (ECOTOX) "unfortunately has lots of inconsistencies and can't be used without checks," suggesting users should implement their own quality verification procedures when working with the data [60].

In contrast, **ToxValDB** utilizes a hybrid approach with both manual and programmatic data extraction, followed by a standardization phase that maps data from multiple sources onto a common structure and vocabulary [47]. The **CompTox Chemicals Dashboard** operates primarily as an aggregator and integrator, bringing together data from multiple sources including ToxCast, ToxRefDB, and DSSTox, with varying levels of curation for different data streams [12] [58].

Functionality and User Interface Comparison

ECOTOX Search and Visualization Capabilities

The ECOTOX Knowledgebase offers three primary functional modalities for data access and analysis:

- **Search Feature:** Allows targeted queries for specific chemicals, species, effects, or endpoints. Users can refine searches using 19 different parameters and customize outputs from over 100 data fields. A key functionality is the direct linkage to the **CompTox Chemicals Dashboard** for additional chemical information [1].
- **Explore Feature:** Designed for use when exact search parameters are unknown, providing a more flexible discovery interface. Users can search by chemical, species, or effects and customize output results for import into external analytical tools [1].
- **Data Visualization Feature:** Includes interactive plotting capabilities that allow users to hover over data points and zoom into specific sections of data to retrieve detailed information. These visualizations aid in data exploration and pattern recognition [1].

Alternative Database Functionalities

The **CompTox Chemicals Dashboard** provides a distinctly different user experience focused on chemical-centric data aggregation. Its interface enables searching by chemical name, structure, or property and provides access to a wide array of data including physicochemical properties, environmental fate and transport, hazard, exposure, and bioactivity data [58] [12]. The dashboard is particularly strong in its **batch search capabilities** and **structure-search functionality**, making it valuable for screening large chemical inventories.

TOXLINE and the **Agricultural & Environmental Science Database** function primarily as bibliographic databases, providing access to the scientific literature rather than curated experimental data [59]. These resources are valuable for comprehensive literature reviews but require significant manual effort to extract and standardize toxicity data for comparative analyses.

Research and Regulatory Applications

ECOTOX in Environmental Decision-Making

ECOTOX has been extensively used for more than 20 years as a rapid source for toxicity data to develop chemical benchmarks for water and sediment quality assessments [1]. Its primary regulatory applications include:

- **Development of aquatic life criteria** to protect freshwater and saltwater organisms from short-term and long-term exposure
- **Informing ecological risk assessments** for chemical registration and reregistration
- **Prioritization and assessment of chemicals** under the Toxic Substances Control Act (TSCA)
- **Site-specific criteria development** by local, state, and tribal governments for interpreting monitoring data

The database is particularly valued for its ability to link traditional biological effects used in regulatory risk assessments with **mechanistic responses** across multiple levels of biological organization and different species [1].

Application in Predictive Model Development

Both ECOTOX and resources within the CompTox Chemicals Dashboard are increasingly used to develop and validate predictive models in ecotoxicology. ECOTOX data supports:

- **Quantitative Structure-Activity Relationship (QSAR) models** that predict toxicity based on physical characteristics of a chemical's structure
- **Cross-species extrapolation models** that estimate effects across taxonomic groups
- **Adverse Outcome Pathway (AOP) development** by providing empirical data linking molecular initiating events to organism-level outcomes
- **Data gap and meta-analyses** to guide future research and testing strategies [1]

A recent research initiative demonstrated the utility of ECOTOX data in creating a curated dataset of effect concentrations and mode of action information for over 3,300 environmentally relevant chemicals, highlighting its value for chemical grouping and read-across approaches [57].

Integrated Workflows and Database Interoperability

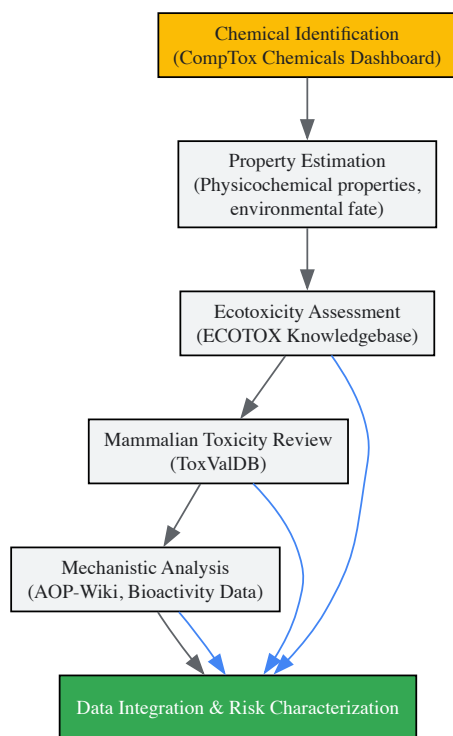
The Scientist's Toolkit: Essential Research Reagent Solutions

Table 2: Key Computational Tools for Ecotoxicology Research

Tool/Resource	Function	Application in Research
ECOTOX Knowledgebase	Curated ecotoxicity data repository	Source of experimental toxicity data for baseline assessments and model training
CompTox Chemicals Dashboard	Chemical data aggregation platform	Access to physicochemical properties, exposure predictions, and bioactivity data
QSAR Models	Quantitative Structure-Activity Relationship prediction	Estimating toxicity for data-poor chemicals based on structural similarity
SeqAPASS	Sequence alignment to predict susceptibility	Cross-species extrapolation based on protein sequence similarity
ToxValDB	Curated in vivo toxicology database	Source of mammalian toxicity data for comparative analyses
AOP-Wiki	Adverse Outcome Pathway knowledgebase	Organizing mechanistic toxicity information for hypothesis testing

Complementary Database Usage Workflow

Modern ecotoxicology research often requires the integration of multiple database resources to address complex questions. The following workflow illustrates how ECOTOX can be integrated with other databases in a complementary research strategy.



Integrated Ecotoxicology Assessment Workflow

This workflow begins with chemical identification and characterization using the **CompTox Chemicals Dashboard**, which provides access to predicted and experimental physicochemical properties [12] [58]. Researchers then query **ECOTOX** for species-specific toxicity data, which serves as the core ecological effects dataset [1] [5]. For a comprehensive hazard assessment, this ecological data is complemented with mammalian toxicity information from **ToxValDB** [47] [12]. Mechanistic context from bioassay data and Adverse Outcome Pathways helps interpret results and identify potential modes of action [57]. Finally, data from all sources are integrated to form a complete risk characterization.

ECOTOX represents a unique resource in the landscape of ecotoxicology databases due to its **comprehensive ecological focus, rigorous curation methodology**, and **direct applicability to regulatory decision-making**. While other databases like the CompTox Chemicals Dashboard offer broader chemical coverage and ToxValDB provides more extensive human health toxicity values, ECOTOX remains the preeminent resource for curated ecological effects data.

The future of ecotoxicology databases lies in enhanced **interoperability** between resources, continued refinement of **data quality assurance** processes, and development of more sophisticated **tools for data extraction and analysis**. As the field continues to evolve toward greater use of NAMs and computational toxicology approaches, the empirical data provided by ECOTOX and similar resources will remain essential for model validation and interpretation. Researchers and regulators would benefit from strategic integration of ECOTOX with complementary databases to leverage the respective strengths of each resource in comprehensive chemical assessment and management.

The **Ecotoxicology (ECOTOX) Knowledgebase**, maintained by the U.S. Environmental Protection Agency (EPA), serves as a critical computational tool for addressing the complex challenges posed by **Per- and Polyfluoroalkyl Substances (PFAS)** and other **emerging contaminants**. As a comprehensive, curated database containing over one million test records for more than 12,000 chemicals and 13,000 species, ECOTOX provides researchers and risk assessors with systematically reviewed toxicity data essential for identifying ecological hazards, supporting regulatory decisions, and prioritizing future research [1] [5]. This technical guide examines ECOTOX's application within the contemporary regulatory landscape, where over 350,000 chemical substances potentially enter the environment, yet less than 1% are governed by international regulations [61]. By detailing ECOTOX's architecture, functionality, and practical implementation in PFAS assessment, this review underscores its indispensable role in bridging critical data gaps and advancing ecological risk assessment paradigms.

Emerging environmental contaminants (ENCs) represent a formidable scientific and regulatory challenge characterized by pervasive environmental presence, potential for significant ecological and human health impacts, and absence of comprehensive regulatory frameworks [61]. **Per- and Polyfluoroalkyl Substances (PFAS)** exemplify these challenges with their extreme persistence, bioaccumulation potential, and documented health risks including carcinogenicity, reproductive dysfunction, and immune system effects [62] [61]. The environmental assessment landscape is further complicated by the staggering disparity between the number of chemicals in commerce and those adequately regulated – approximately 350,000 widely used chemical substances versus less than 1% covered by international conventions [61].

Traditional toxicity testing approaches are insufficient to address this scale, with estimates suggesting that conventional ecotoxicity testing for 10,000 chemicals would cost approximately \$1.18 billion and require a century to complete [61]. This assessment crisis has accelerated the adoption of **New Approach Methodologies (NAMs)** and computational tools that can efficiently leverage existing empirical data. Within this context, the ECOTOX

Knowledgebase has evolved as an authoritative source of curated ecotoxicity data, fulfilling a critical need for accessible, high-quality toxicity information to support evidence-based decision-making [5].

ECOTOX Knowledgebase: Architecture and Capabilities

Initiated in the 1980s, ECOTOX has transformed from multiple ecosystem-specific databases into a unified knowledgebase supporting regulatory programs under statutes including the **Clean Water Act, Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), and Toxic Substances Control Act (TSCA)** [5]. The recently released Version 5 represents a significant advancement in user interface design, data interoperability, and accessibility, aligning with FAIR (Findable, Accessible, Interoperable, and Reusable) data principles [5].

Data Curation and Quality Assurance

ECOTOX employs systematic literature review and data curation processes consistent with standardized guidelines for systematic reviews and evidence maps [5]. The knowledgebase integrates toxicity data through a rigorous pipeline:

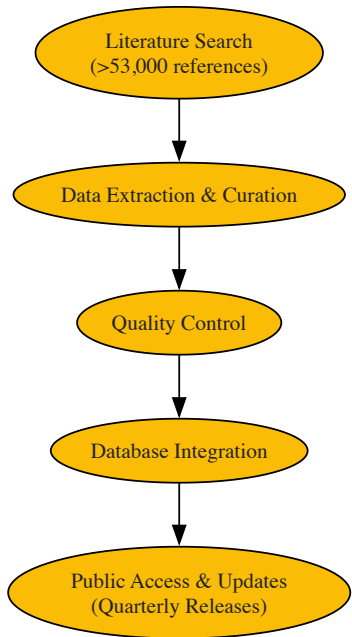


Figure 1: ECOTOX Data Curation Workflow

The curation process abstracts pertinent methodological details and results using controlled vocabularies, with newly extracted toxicity data added quarterly to the public website [5]. This systematic approach ensures data quality and consistency for risk assessment applications.

Scale and Scope of Data Holdings

Table 1: ECOTOX Knowledgebase Data Inventory (as of 2025)

Data Category	Volume	Description
Test Records	>1 million	Individual toxicity test results
Chemical Substances	>12,000	Single chemical stressors
Ecological Species	>13,000	Aquatic and terrestrial organisms
References	>53,000	Peer-reviewed literature sources
Data Fields	>100	Available for search customization

[1] [5] [6]

Functionality and User Interface

ECOTOX provides three primary functional modules for data access and analysis:

- **SEARCH Feature:** Enables targeted queries for specific chemicals, species, effects, or endpoints with filtering across 19 parameters [1]
- **EXPLORE Feature:** Supports exploratory searches when exact parameters are unknown [1]

- **DATA VISUALIZATION:** Interactive plotting tools for data exploration and analysis [1]

The system's interoperability with other EPA computational toxicology resources, including the **CompTox Chemicals Dashboard**, enhances its utility for integrated assessment approaches [1] [12].

ECOTOX in PFAS Risk Assessment: Applications and Protocols

PFAS Ecological Risk Assessment Framework

PFAS assessment presents unique challenges including complex fate and transport behavior, bioaccumulation potential, and mixture toxicity concerns [62]. ECOTOX supports multiple stages of the ecological risk assessment paradigm through its curated data holdings.

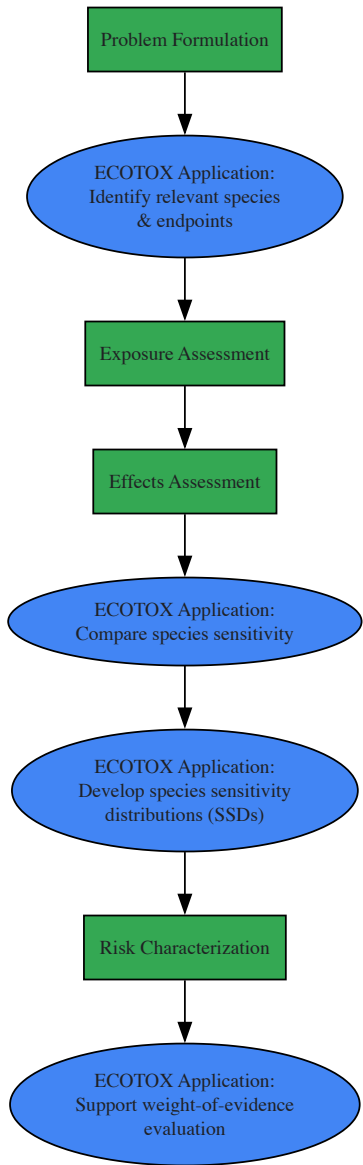


Figure 2: ECOTOX Integration in Ecological Risk Assessment

Experimental Data and Toxicity Value Derivation

ECOTOX provides critical underlying data for deriving toxicity values used in PFAS risk assessments. The database supports the evaluation of both noncancer and cancer endpoints, which is particularly relevant for PFAS with carcinogenicity classifications.

Table 2: PFAS Toxicity Values for Human Health Risk Assessment

PFAS Compound	Toxicity Value (ng/kg-day)	Basis	Source
PFOA	0.03 (RfD)	Decreased antibody response in children, low birth weight, increased cholesterol	USEPA Office of Water [62]
PFOS	0.1 (RfD)	Low birth weight, increased cholesterol in humans	USEPA Office of Water [62]
PFOA	3 (MRL)	Behavioral and skeletal effects in mice (developmental exposure)	ATSDR [62]
PFOS	2 (MRL)	Delayed eye opening, decreased pup weight in rats	ATSDR [62]

The variability in toxicity values derived by different agencies highlights the importance of accessing primary toxicity data in ECOTOX to support critical evaluation of assessment endpoints [62].

Assessment Workflows and Methodologies

Protocol 1: Systematic Literature Review for PFAS Toxicity

- **Search Strategy Development:** Define search terms including specific PFAS compounds (PFOA, PFOS, GenX, etc.) and relevant ecological taxa
- **ECOTOX Query Execution:** Implement structured searches using chemical identifiers (CAS RN, DTXSID) and taxonomic classifications
- **Data Extraction:** Abstract relevant test parameters including:
 - Test organisms (species, life stage)
 - Exposure conditions (duration, route, media)
 - Endpoints measured (lethal, sublethal, reproductive)
 - Result metrics (LC50, EC50, NOEC, LOEC)
- **Quality Assessment:** Evaluate studies based on Klimisch criteria or similar quality grading systems
- **Data Synthesis:** Organize results for quantitative analysis or weight-of-evidence evaluation

Protocol 2: Species Sensitivity Distribution (SSD) Development

- **Data Collection:** Retrieve acute and chronic toxicity values for multiple species from ECOTOX
- **Data Filtering:** Apply quality criteria and exclude unreliable data
- **Taxonomic Grouping:** Organize data by relevant taxonomic groups (fish, invertebrates, amphibians)
- **Statistical Modeling:** Fit probability distributions to toxicity values using EPA's **Species Sensitivity Distribution (SSD) Toolbox** [63]
- **Derive Protective Concentrations:** Calculate Hazardous Concentrations (HC5) protective of most species

Protocol 3: Cross-Species Extrapolation Using SeqAPASS

- **Identify Molecular Targets:** Determine potential protein targets for specific PFAS compounds
- **Sequence Similarity Analysis:** Utilize EPA's **Sequence Alignment to Predict Across Species Susceptibility (SeqAPASS)** tool to evaluate conservation of molecular targets across species [63]
- **Susceptibility Prediction:** Infer relative sensitivity of untested species based on taxonomic relatedness and target conservation
- **Data Gap Filling:** Apply predictions to augment empirical data from ECOTOX for risk assessment

Table 3: Computational Tools for Ecotoxicology Assessment

Tool/Resource	Function	Application in PFAS Assessment
ECOTOX Knowledgebase	Curated ecotoxicity data repository	Primary source for toxicity data across species [1]
CompTox Chemicals Dashboard	Chemical property and bioactivity data	PFAS structure, properties, and in vitro bioactivity [12]
SeqAPASS	Cross-species susceptibility prediction	Estimating sensitivity of protected species to PFAS [63]
Species Sensitivity Distribution (SSD) Toolbox	Statistical analysis of species sensitivity	Deriving protective benchmarks for PFAS [63]
Markov Chain Nest (MCnest)	Avian population modeling	Assessing PFAS impacts on bird reproduction [63]

Tool/Resource	Function	Application in PFAS Assessment
Virtual Tissue Models	In silico simulation of tissue responses	Predicting organ-level effects of PFAS exposure [12]
EnviroTox Database	Aggregated toxicity data for ecoTTC	Screening-level risk assessment for data-poor PFAS [48]

Regulatory Context and Future Directions

Evolving Regulatory Landscape

The regulatory environment for chemical assessment is rapidly evolving, with significant implications for PFAS and emerging contaminant evaluation. The European Union's upcoming **REACH revision** ("REACH 2.0") includes provisions for:

- 10-year validity for chemical registrations
- Introduction of a **Mixture Assessment Factor (MAF)**
- Notification and registration requirements for polymers
- Digitalization of safety data sheets and supply chain communication [64]

Similarly, the **CLP regulation revision** incorporates "stop-the-clock" mechanisms to provide industry additional time for implementation of new hazard classification and labeling requirements [64]. These regulatory developments underscore the growing importance of accessible, curated data resources like ECOTOX to support compliance and informed decision-making.

Integration with New Approach Methodologies (NAMs)

ECOTOX plays a critical role in the verification and validation of **New Approach Methodologies (NAMs)** by providing empirical in vivo data for comparison with in vitro and in silico predictions [5]. Research initiatives within EPA and collaborative organizations like the **Health and Environmental Sciences Institute (HESI)** are focusing on:

- Developing alternatives to chronic fish tests [48]
- Advancing in vitro biotransformation assays for birds and fish [48]
- Creating integrated testing strategies for endocrine disruption [48]
- Establishing ecological thresholds of toxicological concern (eco-TTC) [48]

These approaches aim to increase assessment efficiency while reducing vertebrate animal testing, aligning with the 3Rs (Replacement, Reduction, Refinement) principles [5] [48].

Global Collaboration and Knowledge Integration

The effective governance of emerging contaminants requires enhanced international cooperation and data sharing [61]. ECOTOX supports this objective through its comprehensive coverage of global literature and interoperability with international assessment frameworks. Future enhancements should focus on:

- Expanding data holdings for poorly characterized contaminant classes
- Improving integration with exposure prediction tools
- Developing automated data extraction and curation capabilities
- Enhancing cross-platform interoperability for seamless data exchange

The ECOTOX Knowledgebase represents an indispensable resource for addressing the complex challenges posed by PFAS and other emerging contaminants. Its curated, high-quality toxicity data provides the foundation for scientifically defensible risk assessments, regulatory decisions, and research prioritization. As chemical innovation continues to outpace traditional assessment capacities, computational tools like ECOTOX will play an increasingly vital role in protecting ecological and human health. By supporting the transition to next-generation risk assessment paradigms that incorporate New Approach Methodologies, integrated testing strategies, and predictive modeling, ECOTOX helps bridge the critical gap between the rapidly expanding universe of chemical substances and our capacity to evaluate their potential impacts on living organisms and ecosystems.

The field of ecological risk assessment is undergoing a profound transformation driven by the emergence of **New Approach Methodologies (NAMs)**. These methodologies represent innovative technologies and approaches that provide information on chemical hazard and risk assessment while reducing reliance on traditional animal testing [65]. In regulatory toxicology, NAMs serve as an umbrella term encompassing **in silico (computational), in chemico (chemical), and in vitro (cell-based) assays**, as well as omics technologies and integrated testing strategies [66] [67]. The fundamental shift toward NAMs addresses critical needs in modern toxicology: evaluating the growing number of chemicals in commerce, reducing animal testing in accordance with regulatory mandates, and improving the human and ecological relevance of safety assessments [4].

The **ECOTOXicology Knowledgebase (ECOTOX)** stands as a cornerstone database enabling the development and application of NAMs for environmental risk assessment. Maintained by the U.S. Environmental Protection Agency, ECOTOX is the world's largest compilation of curated ecotoxicity data, containing over **one million test results** from more than **53,000 references**, covering **12,000 chemicals** and **13,000 aquatic and terrestrial species** [1] [4]. This comprehensive database provides essential empirical toxicity data that serves as both a resource for direct risk assessment and a foundation for developing, validating, and applying NAMs. ECOTOX continues to evolve following the FAIR principles (Findable, Accessible, Interoperable, and Reusable) to better support chemical assessments and ecological research [4].

Cross-Species Extrapolation Methodologies

Theoretical Framework and Biological Basis

Cross-species extrapolation represents a critical methodology within the NAMs paradigm, addressing the fundamental challenge of predicting chemical effects across diverse taxa without species-specific testing for every chemical. The **Adverse Outcome Pathway (AOP) framework** serves as the conceptual foundation for modern cross-species extrapolation approaches [66]. An AOP describes a sequence of events beginning with a **Molecular Initiating Event (MIE)**, where a chemical interacts with a biological target, progressing through key events at cellular, tissue, and organ levels, and culminating in an **Adverse Outcome (AO)** at the individual or population level relevant to risk assessment [66].

The scientific basis for cross-species extrapolation rests on the principle of **evolutionary conservation of biological pathways**. When biological targets, signaling pathways, and physiological processes are conserved across species, toxicity pathways are likely to be similar. The **Taxonomic Domain of Applicability** within the AOP framework defines how broadly pathway knowledge can be extrapolated across taxa based on conservation of structure and function [66]. This approach allows researchers to leverage existing toxicity data from model organisms to predict effects in diverse species of ecological concern, significantly reducing the need for additional animal testing.

Practical Implementation Approaches

Implementing cross-species extrapolation requires both qualitative and quantitative approaches. Qualitatively, researchers can establish taxonomic domains of applicability by assessing the conservation of MIEs and key events through genomic comparisons, literature review, and empirical testing across representative species [66]. Quantitatively, **Species Sensitivity Distributions (SSDs)** utilize ECOTOX data to model the variation in sensitivity to a particular chemical across multiple species, enabling estimation of concentrations protective of most species in an ecosystem [68] [4].

The **International Consortium to Advance Cross-Species Extrapolation in Regulation (ICACSER)** has been established to address key challenges in this field, including increasing knowledge on functional conservation of downstream effects, addressing multiple MIEs and biological networks, and incorporating toxicokinetic considerations [66]. Dynamic Energy Budget (DEB) parameters, which represent physiological processes, have emerged as important variables in machine learning models predicting species sensitivity, particularly for invertebrates [69]. These parameters allow for more realistic predictions of effects across species without additional animal testing by capturing essential physiological drivers of differential sensitivity.

Table 1: Key Components of Cross-Species Extrapolation

Component	Description	Application in Risk Assessment
Molecular Initiating Event (MIE)	Initial interaction between chemical and biomolecule	Identify conservation of molecular targets across species
Key Events	Measurable cellular, tissue, or organ-level responses	Establish functional conservation of pathway elements
Adverse Outcome	Population-relevant effect of regulatory concern	Bridge molecular data to ecological endpoints
Taxonomic Domain of Applicability	Defined range of species where AOP is applicable	Determine scope of extrapolation for specific pathways
Species Sensitivity Distributions (SSDs)	Statistical distribution of toxicity values across species	Derive protective benchmarks for chemical concentrations

Quantitative Structure-Activity Relationship (QSAR) Modeling

Foundations and Algorithmic Approaches

Quantitative Structure-Activity Relationship (QSAR) modeling represents a powerful in silico NAM that predicts chemical toxicity based on the principle that molecular structure determines biological activity. These models establish mathematical relationships between **chemical descriptors** (quantitative representations of molecular properties) and measured biological effects, enabling toxicity prediction for data-poor

chemicals [68] [1]. QSAR approaches are particularly valuable for screening and prioritizing chemicals when empirical toxicity data are unavailable or limited.

Modern QSAR modeling has evolved from traditional regression-based approaches to incorporate **machine learning algorithms** that can capture complex, non-linear relationships between chemical structures and biological activity. Random forest models, in particular, have demonstrated strong performance in predicting chemical- and species-specific endpoints by leveraging both chemical descriptors and physiological variables [69]. These models can integrate diverse data types, including chemical fingerprints, molecular descriptors, and **Dynamic Energy Budget (DEB) parameters**, to enhance prediction accuracy across taxonomic groups [69].

The development and validation of QSAR models for regulatory applications typically follow guidelines established by the **Organisation for Economic Co-operation and Development (OECD)**, which require defined endpoints, unambiguous algorithms, appropriate domains of applicability, measures of fit and robustness, and mechanistic interpretation where possible [68]. These principles ensure that QSAR predictions are reliable and transparent for decision-making.

Implementation and Integration with ECOTOX

The ECOTOX Knowledgebase provides essential data for QSAR development, validation, and application. With over one million curated toxicity test results, ECOTOX serves as a comprehensive source of training data for model development across numerous chemical classes and species [1] [4]. Researchers can extract high-quality, standardized toxicity data for specific taxonomic groups or endpoints to build robust prediction models.

A key advancement in QSAR modeling is the incorporation of **physiological variables** alongside traditional chemical descriptors. Research has demonstrated that machine learning models trained using both chemical fingerprints/descriptors and physiological properties represented by DEB parameters can effectively predict chemical- and species-specific endpoints [69]. For invertebrates specifically, DEB parameters have proven to be relatively important variables in these models, illuminating how physiological properties drive species sensitivity [69].

Table 2: QSAR Modeling Approaches and Applications

Model Type	Key Features	Ecological Applications
Traditional QSAR	Regression-based, 2D chemical descriptors	Predicting baseline toxicity, chemical prioritization
Machine Learning QSAR	Random forest, neural networks, support vector machines	Complex toxicity prediction across multiple species
Hybrid Physiological QSAR	Combines chemical descriptors with DEB parameters	Species-specific sensitivity prediction, cross-taxa extrapolation
Read-Across	Analogous chemical grouping based on similarity	Filling data gaps for structurally similar compounds
3D-QSAR	Incorporates spatial molecular structure	Modeling receptor-mediated toxicity mechanisms

Integrated Workflows and Experimental Protocols

Systematic Data Collection and Curation

The foundation of reliable NAMs application rests on systematic data collection and curation processes. ECOTOX employs rigorous **systematic review procedures** aligned with contemporary evidence-based toxicology practices to identify, evaluate, and extract toxicity data from the scientific literature [4]. The workflow begins with comprehensive literature searches using customized search strategies for chemicals of interest, followed by tiered screening of references based on predefined eligibility criteria.

The ECOTOX curation pipeline involves several standardized phases: initial citation identification, relevance screening based on titles and abstracts, full-text assessment for applicability, and final data abstraction using controlled vocabularies [4]. This process ensures that extracted data meets specific quality criteria, including documented controls, reported endpoints, and complete methodological descriptions. The resulting curated data supports various NAMs applications, including QSAR model development, AOP construction, and cross-species extrapolation.

Weight of Evidence and Integrated Approaches

Advanced NAMs applications typically employ **Integrated Approaches to Testing and Assessment (IATA)** that combine multiple lines of evidence within a structured, hypothesis-based framework [68] [70]. These integrated strategies leverage **weight-of-evidence** methodologies to synthesize information from in silico models, in vitro assays, and existing in vivo data, providing a comprehensive basis for regulatory decision-making [68].

For bioaccumulation assessment, as exemplified in recent webinars co-organized by the European Medicines Agency and other international bodies, IATA frameworks help evaluators collect, generate, and integrate multiple lines of evidence for clear and transparent decision-making in both aquatic

and terrestrial environments [70]. These approaches are particularly valuable for addressing data-poor situations while maintaining scientific rigor and regulatory acceptance.

The diagram below illustrates a comprehensive workflow integrating ECOTOX data with cross-species extrapolation and QSAR modeling:



Essential Research Reagents and Tools

Implementing NAMs for cross-species extrapolation and QSAR modeling requires specific research reagents and computational tools. The table below details key resources essential for conducting research in this field:

Table 3: Essential Research Reagents and Computational Tools

Tool/Reagent	Type	Function/Application	Source/Access
ECOTOX Knowledgebase	Database	Curated ecotoxicity data for model development and validation	EPA Website [1]
CompTox Chemicals Dashboard	Database	Chemical property data and structure information for QSAR modeling	EPA Website [1]

Tool/Reagent	Type	Function/Application	Source/Access
AOP-Wiki	Knowledge Repository	Adverse Outcome Pathway information for cross-species extrapolation	OECD Website [66]
DEB Parameters Database	Physiological Data	Species-specific Dynamic Energy Budget parameters for hybrid QSAR models	Research Literature [69]
QSAR Modeling Software	Computational Tools	Machine learning algorithms for toxicity prediction (e.g., Random Forest)	Various Open-Source Platforms [69]
Organ-on-a-Chip Systems	In Vitro Platform	Human-relevant toxicity screening for AOP anchoring	Commercial Providers [67]
Omics Technologies	Analytical Tools	Transcriptomics, proteomics for mechanistic toxicity assessment	Core Facilities [67]

The integration of **New Approach Methodologies** with comprehensive knowledgebases like ECOTOX represents the future of ecological risk assessment. Cross-species extrapolation leveraging the **AOP framework** and advanced **QSAR modeling** incorporating both chemical descriptors and physiological parameters enables more predictive and efficient safety evaluation while reducing animal testing. As regulatory agencies worldwide continue to endorse and implement these approaches [\[66\]](#) [\[70\]](#) [\[67\]](#), the scientific community must continue to refine these methodologies, expand their applicability domains, and demonstrate their utility in real-world risk assessment scenarios. The ongoing development of integrated workflows and collaborative consortia like ICACSER will be crucial for addressing complex challenges such as chemical mixtures and multiple stressors across diverse ecosystems [\[68\]](#) [\[66\]](#).

Modern ecotoxicology research requires the integration of diverse computational tools to effectively assess chemical risks across vast taxonomic spaces. The **ECOTOX Knowledgebase** stands as a core resource, containing over one million test records covering more than 13,000 aquatic and terrestrial species and 12,000 chemicals compiled from decades of scientific literature [\[1\]](#). Its true power, however, is unlocked through interoperability with specialized tools for **cross-species extrapolation**, **drug target conservation**, and **chemical properties data**. This integrated approach allows researchers to move beyond simple toxicity data retrieval to predictive modeling of chemical effects across species with limited or no testing data. The **SeqAPASS**, **EcoDrug**, and **CompTox Chemicals Dashboard** tools provide complementary functionalities that, when used together with ECOTOX, create a powerful framework for mechanistically-driven ecological risk assessment [\[71\]](#) [\[72\]](#) [\[21\]](#). This technical guide examines the interoperability between these resources, providing researchers with methodologies to leverage their combined capabilities for advanced ecotoxicological investigations.

Tool-Specific Capabilities and Technical Specifications

Core Functional Capabilities

Table 1: Core Capabilities of Ecotoxicology Research Tools

Tool	Primary Function	Data Coverage	Key Outputs
ECOTOX Knowledgebase	Centralized repository of empirical toxicity data	>1 million test records, 13,000 species, 12,000 chemicals [1]	Curated toxicity values, species sensitivity distributions, effect concentrations
SeqAPASS	Cross-species susceptibility prediction via protein conservation	NCBI database (153+ million proteins, 95,000+ organisms) [71]	Protein conservation predictions, susceptibility classifications, 3D protein models
EcoDrug/ECOdrug+	Drug target conservation across species	1,194 pharmaceuticals, 663 targets, 640-180 species [72] [73]	Ortholog presence/absence, taxonomic conservation patterns, drug-target interactions
CompTox Chemicals Dashboard	Chemical properties and bioactivity data integration	>1 million chemicals, 300+ chemical lists [21] [44]	Physicochemical properties, toxicity predictions, exposure data, chemical identifiers

SeqAPASS Technical Architecture and Workflow

The Sequence Alignment to Predict Across Species Susceptibility (SeqAPASS) tool employs a **tiered bioinformatics approach** to evaluate protein conservation across species. The methodology progresses through three distinct levels of analysis, each providing increasing specificity in susceptibility predictions [\[74\]](#):

- **Level 1:** Primary amino acid sequence comparison using BLASTp algorithms against NCBI protein databases

- **Level 2:** Functional domain conservation analysis using conserved domain databases (CDD) and COBALT alignment tools
- **Level 3:** Critical amino acid residue evaluation focusing on residues known to mediate chemical-protein interactions

SeqAPASS Version 8 has introduced **protein structural conservation evaluations** using I-TASSER, a protein structural prediction tool that generates 3D protein models as an additional line of evidence for susceptibility predictions [75]. This structural biology integration significantly enhances the tool's ability to predict conservation of protein function beyond simple sequence homology.

Table 2: SeqAPASS Version Evolution and Feature Development

Version	Release Timeline	Significant Features Added
Initial Release	January 2016	Primary amino acid sequence comparisons, functional domain analysis [74]
Versions 2.0-4.1	2017-2020	Individual amino acid residue comparisons, interoperability with ECOTOX, reference explorer [74]
Versions 5.0-6.1	2020-2021	Customizable heat map visualizations, decision summary reports, ECOTOX widget integration [74]
Version 8	2024	I-TASSER protein structural prediction, 3D model generation [75]

EcoDrug Platform Ortholog Prediction Methodology

The ECOdrug platform applies a **consensus approach** to ortholog prediction that integrates three established methods to improve accuracy [72] [76]:

- **Ensembl Compara:** Gene-based ortholog predictions leveraging Ensembl's comparative genomics resources
- **EggNOG:** Phylogenetic-based ortholog clustering across taxonomic levels
- **InParanoid:** Pairwise ortholog predictions between human reference proteome and other species

ECOdrug applies a **majority vote principle** for species represented in all three methods, where at least two databases must agree on ortholog presence/absence. For species in only two methods, presence in at least one database predicts ortholog presence [72]. This conservative approach improves confidence in predictions, particularly for evolutionarily distant species.

The more recent **EcoDrugPlus (EcoDrug+)** platform significantly expands these capabilities, integrating data on 7,200 pharmaceuticals, 34,000 agrochemicals, 61,000 human metabolites, and 5,800 other bioactive chemicals with target conservation information for 180 organisms across diverse phyla [73]. This expansion facilitates comprehensive 'mechanism of action-based' assessment for chemicals discharged into environments.

CompTox Chemicals Dashboard as Data Integration Hub

The CompTox Chemicals Dashboard serves as a **central chemical data resource**, providing access to physicochemical properties, environmental fate parameters, exposure data, and toxicity information for over one million chemicals [21] [44]. Its role in interoperability stems from several key functionalities:

- **Chemical identifier translation** across multiple naming conventions (CASRN, DTXSID, common names)
- **Integrated bioactivity data** from ToxCast and Tox21 high-throughput screening programs
- **Batch search capabilities** for processing chemical lists and categories
- **QSAR model predictions** for various toxicity endpoints and physicochemical parameters

The Dashboard's **Application Programming Interfaces (APIs)** enable programmatic access to chemical data, allowing researchers to integrate CompTox data directly into their analytical workflows and applications [44].

Interoperability Workflows and Experimental Protocols

Integrated Cross-Species Susceptibility Assessment

The interoperability between ECOTOX, SeqAPASS, and the CompTox Dashboard creates a powerful workflow for **mechanistically-based cross-species extrapolation**. The following protocol outlines the integrated methodology:

Step 1: Chemical Identification and Protein Target Characterization

- Query the CompTox Chemicals Dashboard using chemical identifiers to retrieve known molecular targets [21]
- Identify relevant protein targets through literature review and existing ToxCast assay data
- Determine sensitive species with known chemical susceptibility from ECOTOX records [1]

Step 2: Primary Sequence Analysis (SeqAPASS Level 1)

- Input protein sequence from sensitive species using NCBI accession numbers or FASTA format
- Execute primary amino acid sequence comparison against all species in NCBI databases
- Apply initial susceptibility prediction based on overall sequence similarity thresholds [74]

Step 3: Functional Domain Evaluation (SeqAPASS Level 2)

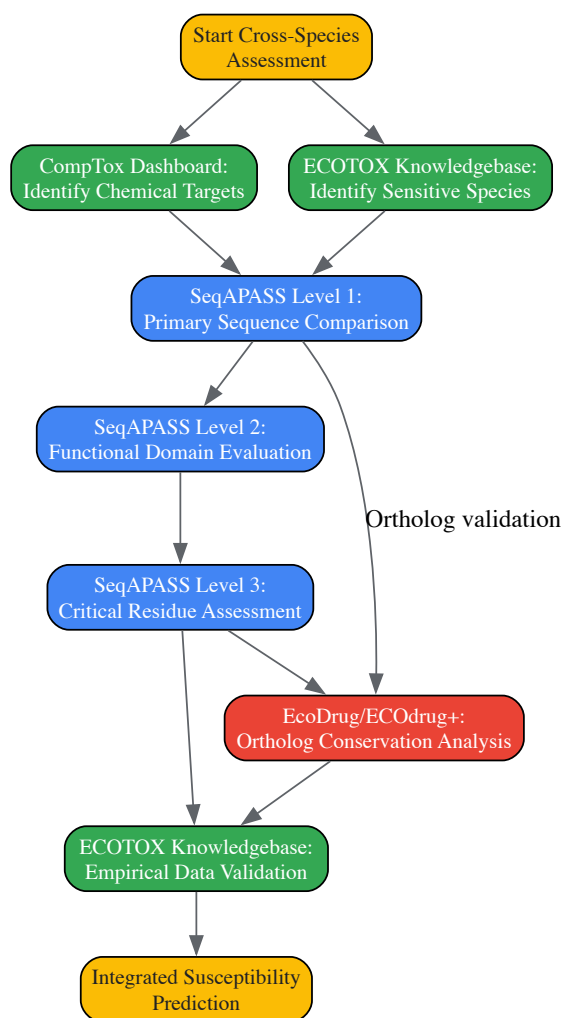
- Refine predictions by evaluating conservation of specific functional domains
- Use COBALT alignment tools to generate multiple sequence alignments
- Apply domain-specific susceptibility criteria based on known functional regions [71]

Step 4: Critical Residue Assessment (SeqAPASS Level 3)

- Input specific amino acid residues known to mediate chemical binding
- Generate multiple sequence alignments focused on critical residues
- Apply residue-specific scoring matrix to predict susceptibility [74]

Step 5: Empirical Data Validation

- Use the ECOTOX widget in SeqAPASS to compare predictions with empirical toxicity data [74]
- Execute targeted queries in ECOTOX Knowledgebase for species with predicted susceptibility
- Validate predictions against existing toxicity data for related chemicals or species



Integrated Workflow for Cross-Species Susceptibility Assessment

Pharmaceutical Environmental Risk Assessment Protocol

The interconnection between EcoDrug and ECOTOX provides a specialized workflow for **pharmaceutical environmental risk assessment**:

Step 1: Drug Target Identification

- Query EcoDrug using drug name or ATC code to identify human protein targets [76]

- Retrieve ortholog predictions across taxonomic groups using majority vote approach
- Identify potentially susceptible non-target species based on target conservation

Step 2: Taxonomic Susceptibility Profiling

- Use EcoDrug's taxonomic group view to identify clades with conserved drug targets
- Analyze taxonomic patterns of target conservation to predict vulnerable species groups
- Download ortholog prediction tables for further analysis

Step 3: Empirical Effect Concentration Retrieval

- Query ECOTOX Knowledgebase using identified pharmaceutical compound
- Retrieve effect concentrations for species with available toxicity data
- Compare empirical sensitivity with target conservation predictions

Step 4: Read-Across Extrapolation

- Apply Generalized Read-Across (GenRA) tools available through CompTox Dashboard
- Extrapolate toxicity values from tested to untested species based on target conservation
- Validate predictions against any available in vivo or in vitro data

Research Applications and Case Studies

Endocrine Disruption Pathway Conservation

Researchers have successfully applied the **SeqAPASS-ECOTOX interoperability** to evaluate the conservation of endocrine pathways across taxonomic groups. In one case study focused on **estrogen receptor binding**:

- SeqAPASS analysis demonstrated high conservation of estrogen receptor ligand-binding domains across vertebrate species
- Predictions identified potential susceptibility in fish, amphibians, and birds for human estrogenic chemicals
- ECOTOX data validation confirmed susceptibility patterns in species with available toxicity data
- The integrated approach supported chemical prioritization for EPA's Endocrine Disruptor Screening Program [\[71\]](#)

Insecticide Specificity and Pollinator Risk Assessment

The tools have been deployed to understand **insecticide selectivity** and assess potential risks to **non-target pollinators**:

- SeqAPASS analysis of tobacco budworm ecdysone receptor identified specific sequence differences in honey bees
- These differences explained the selective toxicity of insect growth regulators to pest species
- EcoDrug analysis confirmed conservation of nicotinic acetylcholine receptor targets across bee species
- Integrated assessment informed regulatory decisions on pollinator protection [\[71\]](#)

Pharmaceutical Hazard Identification for Aquatic Organisms

EcoDrug and ECOTOX have been used to screen **human pharmaceuticals** for potential hazards to aquatic organisms:

- EcoDrug analysis identified beta-blockers with conserved targets in fish species
- Ortholog predictions for adrenergic receptors showed high conservation across vertebrates
- ECOTOX data mining confirmed cardiovascular effects in fish for propranolol
- Integration enabled prioritization of pharmaceuticals for environmental risk assessment [\[72\]](#)

The Scientist's Toolkit: Essential Research Reagent Solutions

Table 3: Essential Computational Tools for Ecotoxicology Research

Tool/Resource	Function	Access Method	Key Applications
ECOTOX Knowledgebase Explore Feature	Targeted toxicity data retrieval	Web interface, API connection	Species-specific effect data retrieval, sensitivity distribution development
SeqAPASS Level 3 Heat Map	Visualization of critical residue conservation	Web interface following sequence analysis	Rapid susceptibility prediction across taxonomic groups
CompTox Batch Search	High-throughput chemical data retrieval	Web interface, API access	Chemical list characterization, QSAR input parameter generation

Tool/Resource	Function	Access Method	Key Applications
EcoDrug Majority Vote Tables	Consensus ortholog prediction	Web interface, downloadable CSV	Drug target conservation analysis, susceptible species identification
ECOTOX Data Visualization	Interactive toxicity plotting	Web interface following data queries	Data exploration, concentration-response visualization

The interoperability between ECOTOX, SeqAPASS, EcoDrug, and the CompTox Chemicals Dashboard represents a **paradigm shift** in ecotoxicological assessment, moving from purely empirical approaches to **mechanistically-driven predictions**. The integration of protein sequence conservation, drug target orthology, chemical properties, and empirical toxicity data creates a powerful framework for **cross-species extrapolation** that can prioritize chemicals for testing, reduce animal use, and improve the efficiency of ecological risk assessment [71] [72] [1].

Future developments in this field will likely focus on **enhanced structural biology integration**, with SeqAPASS Version 8's incorporation of I-TASSER protein modeling representing just the beginning [75]. The expansion of platforms like EcoDrug+ to include **environmental exposure data** and **geo-referenced concentration measurements** will further strengthen the integration of hazard and exposure assessment [73]. Additionally, the growing availability of **APIs and programming interfaces** across these tools will enable more sophisticated automated workflows and custom applications [44].

For researchers, mastering the interoperability between these resources is becoming increasingly essential for state-of-the-art ecotoxicological investigation. The protocols and workflows outlined in this technical guide provide a foundation for leveraging these integrated tools to advance the understanding of chemical impacts across the diversity of species and ecosystems.

Conclusion

The ECOTOX Knowledgebase stands as an indispensable, validated resource that has fundamentally advanced the field of ecotoxicology. By providing comprehensive, curated toxicity data through an accessible and continuously updated platform, it empowers researchers, risk assessors, and drug development professionals to make informed decisions on chemical safety and environmental protection. Its integration into regulatory processes for over three decades and its adaptability to new scientific challenges, such as assessing PFAS and pharmaceuticals in the environment, underscore its enduring value. Future directions will likely involve enhanced interoperability with bioinformatics tools, expanded support for evolutionary toxicology and read-across approaches, and growing applications in global chemical assessment frameworks, further solidifying its role as a cornerstone of evidence-based environmental science.

References

1. [Ecotoxicology \(ECOTOX\) Knowledgebase Resource Hub](https://www.epa.gov/comptox-tools/ecotoxicology-ecotox-knowledgebase-resource-hub) [https://www.epa.gov/comptox-tools/ecotoxicology-ecotox-knowledgebase-resource-hub]
2. [Assessment of Ecotoxicity - A Framework to Guide ...](https://www.ncbi.nlm.nih.gov/books/NBK253975/) - NCBI [https://www.ncbi.nlm.nih.gov/books/NBK253975/]
3. [Submission guidelines | Ecotoxicology](https://link.springer.com/journal/10646/submission-guidelines) [https://link.springer.com/journal/10646/submission-guidelines]
4. [The ECOTOXicology Knowledgebase: A Curated Database of ...](https://pmc.ncbi.nlm.nih.gov/articles/PMC9408435/) [https://pmc.ncbi.nlm.nih.gov/articles/PMC9408435/]
5. [A Curated Database of Ecologically Relevant Toxicity Tests to ...](https://academic.oup.com/etc/article/41/6/1520/7730623) [https://academic.oup.com/etc/article/41/6/1520/7730623]
6. [Ecotoxicological Studies Market Size & Outlook, 2025-2033](https://straitsresearch.com/report/ecotoxicological-studies-market) [https://straitsresearch.com/report/ecotoxicological-studies-market]
7. [EPA Ecotoxicology Knowledgebase \(ECOTOX\)](https://ntp.niehs.nih.gov/iccvamreport/2023/technology/data-resources/01-epa-ecotox) [https://ntp.niehs.nih.gov/iccvamreport/2023/technology/data-resources/01-epa-ecotox]
8. [Conducting an Ecological Risk Assessment](https://www.epa.gov/risk/conducting-ecological-risk-assessment) [https://www.epa.gov/risk/conducting-ecological-risk-assessment]
9. [Assessing the relevance of ecotoxicological studies for ...](https://pubmed.ncbi.nlm.nih.gov/27599457/) [https://pubmed.ncbi.nlm.nih.gov/27599457/]
10. [Evaluation Guidelines for Ecological Toxicity Data in the ...](https://www.epa.gov/pesticide-science-and-assessing-pesticide-risks/evaluation-guidelines-ecological-toxicity-data-open) [https://www.epa.gov/pesticide-science-and-assessing-pesticide-risks/evaluation-guidelines-ecological-toxicity-data-open]
11. [A benchmark dataset for machine learning in ecotoxicology](https://www.nature.com/articles/s41597-023-02612-2) [https://www.nature.com/articles/s41597-023-02612-2]
12. [Downloadable Computational Toxicology Data](https://www.epa.gov/comptox-tools/downloadable-computational-toxicology-data) [https://www.epa.gov/comptox-tools/downloadable-computational-toxicology-data]
13. [Advancing the Science of Behavioral Ecotoxicology and ...](https://www.setac.org/resource/behavioral-ecotoxicology-use-of-behavioral-endpoints-ecotoxicity-environmental-assessments.html) [https://www.setac.org/resource/behavioral-ecotoxicology-use-of-behavioral-endpoints-ecotoxicity-environmental-assessments.html]
14. [Ecosystem Toxicology - an overview](https://www.sciencedirect.com/topics/medicine-and-dentistry/ecosystem-toxicology) [https://www.sciencedirect.com/topics/medicine-and-dentistry/ecosystem-toxicology]
15. [Data Evaluation - Toxicity Testing](https://www.ncbi.nlm.nih.gov/books/n/nap317/ddd00052/?report=reader) [https://www.ncbi.nlm.nih.gov/books/n/nap317/ddd00052/?report=reader]
16. [RSEI Toxicity-Data and Calculations](https://www.epa.gov/rsei/rsei-toxicity-data-and-calculations) [https://www.epa.gov/rsei/rsei-toxicity-data-and-calculations]
17. [FDA Releases New Tool for Toxicity Screening of ...](https://www.fda.gov/food/hfp-constituent-updates/fda-releases-new-tool-toxicity-screening-chemicals-food) [https://www.fda.gov/food/hfp-constituent-updates/fda-releases-new-tool-toxicity-screening-chemicals-food]
18. [Presenting Quantitative Data Graphically](https://courses.lumenlearning.com/waymakermath4libarts/chapter/presenting-quantitative-data-graphically/) [https://courses.lumenlearning.com/waymakermath4libarts/chapter/presenting-quantitative-data-graphically/]
19. [The unexplored environmental impacts of pilot-scale ...](https://www.sciencedirect.com/science/article/pii/S240584402409830X) [https://www.sciencedirect.com/science/article/pii/S240584402409830X]
20. [Common issues of data science on the eco-environmental ...](https://www.sciencedirect.com/science/article/pii/S0160412025000522) [https://www.sciencedirect.com/science/article/pii/S0160412025000522]

- [21. CompTox Chemicals Dashboard \[https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard\]](https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard)
- [22. CompTox Chemicals Dashboard Resource Hub \[https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard-resource-hub\]](https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard-resource-hub)
- [23. An integrated chemical environment with tools for ... \[https://www.rti.org/publication/integrated-chemical-environment-tools-chemical-safety-testing\]](https://www.rti.org/publication/integrated-chemical-environment-tools-chemical-safety-testing)
- [24. ECOTOXr: An R package for reproducible and transparent ... \[https://www.sciencedirect.com/science/article/pii/S0045653524019751\]](https://www.sciencedirect.com/science/article/pii/S0045653524019751)
- [25. Data Interoperability - an overview \[https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/data-interopability\]](https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/data-interopability)
- [26. CompTox Chemicals Dashboard: Release Notes \[https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard-release-notes\]](https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard-release-notes)
- [27. Enhancing Data Integration, Interoperability, and Reuse to ... \[https://pmc.ncbi.nlm.nih.gov/articles/PMC9227711/\]](https://pmc.ncbi.nlm.nih.gov/articles/PMC9227711/)
- [28. Using in vitro aromatase inhibition data to predict ... \[https://pmc.ncbi.nlm.nih.gov/articles/PMC10782516/\]](https://pmc.ncbi.nlm.nih.gov/articles/PMC10782516/)
- [29. Aquatic Life Benchmarks and Ecological Risk Assessments ... \[https://www.epa.gov/pesticide-science-and-assessing-pesticide-risks/aquatic-life-benchmarks-and-ecological-risk\]](https://www.epa.gov/pesticide-science-and-assessing-pesticide-risks/aquatic-life-benchmarks-and-ecological-risk)
- [30. Retrospective stepwise prioritization of chemicals detected in ... \[https://academic.oup.com/etc/article/44/7/2048/8087440\]](https://academic.oup.com/etc/article/44/7/2048/8087440)
- [31. Procedures for Chemical Risk Evaluation Under the Toxic ... \[https://www.federalregister.gov/documents/2025/09/23/2025-18431/procedures-for-chemical-risk-evaluation-under-the-toxic-substances-control-act-tsca\]](https://www.federalregister.gov/documents/2025/09/23/2025-18431/procedures-for-chemical-risk-evaluation-under-the-toxic-substances-control-act-tsca)
- [32. How to Customize Search Filters for Academic Research \[https://www.sourcery.net/resources/how-to-customize-search-filters-for-academic-research\]](https://www.sourcery.net/resources/how-to-customize-search-filters-for-academic-research)
- [33. Top 6 Visualizations for Quantitative Data Analysis Methods \[https://chartexpo.com/blog/quantitative-data-analysis-methods\]](https://chartexpo.com/blog/quantitative-data-analysis-methods)
- [34. Visualizing Quantitative Data: Best Experience \[https://powerdrill.ai/blog/visualizing-quantitative-data-best-experience\]](https://powerdrill.ai/blog/visualizing-quantitative-data-best-experience)
- [35. Standartox: Standardizing Toxicity Data \[https://www.mdpi.com/2306-5729/5/2/46\]](https://www.mdpi.com/2306-5729/5/2/46)
- [36. 5.3 Standardized toxicity tests and endpoints \[https://fiveable.me/ecotoxicology/unit-5/standardized-toxicity-tests-endpoints/study-guide/3b3dpliOtxYz1fGP\]](https://fiveable.me/ecotoxicology/unit-5/standardized-toxicity-tests-endpoints/study-guide/3b3dpliOtxYz1fGP)
- [37. 5.6 Data Visualization \[https://web.stevenson.edu/mbranson/m4tp/version1/greenhouse-gas-math-topic-data-visualization.html\]](https://web.stevenson.edu/mbranson/m4tp/version1/greenhouse-gas-math-topic-data-visualization.html)
- [38. Text has enhanced contrast | ACT Rule | WAI \[https://www.w3.org/WAI/standards-guidelines/act/rules/0905cg/\]](https://www.w3.org/WAI/standards-guidelines/act/rules/0905cg/)
- [39. Good Practices in Omics Data Interpretation for Regulatory ... \[https://www.ecetoc.org/task-force/good-practices-in-omics-data-interpretation-for-regulatory-applications/\]](https://www.ecetoc.org/task-force/good-practices-in-omics-data-interpretation-for-regulatory-applications/)
- [40. EPA NAMs training program \[https://ntp.niehs.nih.gov/iccvarreport/2023/confidence/comm/06-epa-training\]](https://ntp.niehs.nih.gov/iccvarreport/2023/confidence/comm/06-epa-training)
- [41. ECOTOX Database \[https://norecopa.no/3r-guide/ecotox-database/\]](https://norecopa.no/3r-guide/ecotox-database/)
- [42. New Approach Methods \(NAMs\) Training \[https://www.epa.gov/chemical-research/new-approach-methods-nams-training\]](https://www.epa.gov/chemical-research/new-approach-methods-nams-training)
- [43. ECOTOX Knowledgebase and PFAS Updates \[https://waterprogramportal.org/event/epa-tools-resources-training-webinar-ecotox-knowledgebase-and-pfas-updates/\]](https://waterprogramportal.org/event/epa-tools-resources-training-webinar-ecotox-knowledgebase-and-pfas-updates/)
- [44. Computational Toxicology and Exposure Online Resources \[https://www.epa.gov/comptox-tools\]](https://www.epa.gov/comptox-tools)
- [45. NAMs Training Worksheets \[https://www.epa.gov/chemical-research/nams-training-worksheets\]](https://www.epa.gov/chemical-research/nams-training-worksheets)
- [46. NAMs Training Tool Tips \[https://www.epa.gov/chemical-research/nams-training-tool-tips\]](https://www.epa.gov/chemical-research/nams-training-tool-tips)
- [47. Development of the toxicity values database, ToxValDB \[https://www.sciencedirect.com/science/article/abs/pii/S246811325000258\]](https://www.sciencedirect.com/science/article/abs/pii/S246811325000258)
- [48. Next Generation Ecological Risk Assessment - HESI \[https://hesiglobal.org/ecorisk/\]](https://hesiglobal.org/ecorisk/)
- [49. High Time to Update Statistical Practices in Ecotoxicology \[https://www.setac.org/resource/high-time-to-update-statistical-practices-in-ecotoxicology.html\]](https://www.setac.org/resource/high-time-to-update-statistical-practices-in-ecotoxicology.html)
- [50. 14 Comparing quantitative data between individuals \[https://bookdown.org/pkaldunn/SRM-Textbook/BetweenQuantData.html\]](https://bookdown.org/pkaldunn/SRM-Textbook/BetweenQuantData.html)
- [51. Coloring in R's Blind Spot - The R Journal \[https://journal.r-project.org/articles/RJ-2023-071/\]](https://journal.r-project.org/articles/RJ-2023-071/)
- [52. Top R Color Palettes to Know for Great Data Visualization \[https://www.datanovia.com/en/blog/top-r-color-palettes-to-know-for-great-data-visualization/\]](https://www.datanovia.com/en/blog/top-r-color-palettes-to-know-for-great-data-visualization/)
- [53. Water Quality Criteria | US EPA \[https://www.epa.gov/wqc\]](https://www.epa.gov/wqc)
- [54. National Recommended Water Quality Criteria Tables \[https://www.epa.gov/wqc/national-recommended-water-quality-criteria-tables\]](https://www.epa.gov/wqc/national-recommended-water-quality-criteria-tables)
- [55. Upcoming Registration Review Actions \[https://www.epa.gov/pesticide-reevaluation/upcoming-registration-review-actions\]](https://www.epa.gov/pesticide-reevaluation/upcoming-registration-review-actions)
- [56. EPA Releases Reports as Part of Agency Efforts to ... \[https://www.lawbc.com/epa-releases-reports-as-part-of-agency-efforts-to-optimize-pesticide-registration-processes/\]](https://www.lawbc.com/epa-releases-reports-as-part-of-agency-efforts-to-optimize-pesticide-registration-processes/)
- [57. Curated mode-of-action data and effect concentrations for ... \[https://www.nature.com/articles/s41597-023-02904-7\]](https://www.nature.com/articles/s41597-023-02904-7)
- [58. CompTox Chemicals Dashboard: Latest News \[https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard-latest-news\]](https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard-latest-news)
- [59. Databases - Environmental Toxicology - Research Guides \[https://guides.library.ucdavis.edu/environmental-toxicology/databases\]](https://guides.library.ucdavis.edu/environmental-toxicology/databases)
- [60. Databases - Ecotoxicology and Models \[https://www.ecotoxmodels.org/resources/databases/\]](https://www.ecotoxmodels.org/resources/databases/)
- [61. Time to strengthen the governance of new contaminants in ... \[https://www.nature.com/articles/s41467-025-63217-4\]](https://www.nature.com/articles/s41467-025-63217-4)
- [62. 9 Site Risk Assessment \[https://pfas-1.itrcweb.org/9-site-risk-assessment/\]](https://pfas-1.itrcweb.org/9-site-risk-assessment/)
- [63. Ecotoxicological Assessment & Modeling | US EPA \[https://www.epa.gov/chemical-research/ecotoxicological-assessment-modeling\]](https://www.epa.gov/chemical-research/ecotoxicological-assessment-modeling)
- [64. Ecotox REACH 2025 Conference: Key Takeaways \[https://www.complianceandrisk.com/cr-news/ecotox-reach-2025-conference-key-takeaways-on-reach-2-0-clp-revision-and-pfas-restrictions/\]](https://www.complianceandrisk.com/cr-news/ecotox-reach-2025-conference-key-takeaways-on-reach-2-0-clp-revision-and-pfas-restrictions/)
- [65. New Approach Methodologies \[https://www.setac.org/page/sciencehts/\]](https://www.setac.org/page/sciencehts/)
- [66. International Consortium to Advance Cross-Species ... \[https://pmc.ncbi.nlm.nih.gov/articles/PMC9161440/\]](https://pmc.ncbi.nlm.nih.gov/articles/PMC9161440/)

- [67. New Approach Methodologies \(NAMs\): The Future of ...](https://emulatebio.com/new-approach-methodologies-nams-the-future-of-toxicology-and-safety-testing/) [https://emulatebio.com/new-approach-methodologies-nams-the-future-of-toxicology-and-safety-testing/]
- [68. The use of new approach methodologies for ...](https://pubmed.ncbi.nlm.nih.gov/36741274/) [https://pubmed.ncbi.nlm.nih.gov/36741274/]
- [69. Physiological variables in machine learning QSARs allow ...](https://www.sciencedirect.com/science/article/pii/S0147651323007546) [https://www.sciencedirect.com/science/article/pii/S0147651323007546]
- [70. First webinar on new approach methodologies \(NAMs\) in ...](https://www.ema.europa.eu/en/events/first-webinar-new-approach-methodologies-nams-ecotoxicology-state-science-bioaccumulation) [https://www.ema.europa.eu/en/events/first-webinar-new-approach-methodologies-nams-ecotoxicology-state-science-bioaccumulation]
- [71. Sequence Alignment to Predict Across Species ...](https://www.epa.gov/comptox-tools/sequence-alignment-predict-across-species-susceptibility-seqapass-resource-hub) [https://www.epa.gov/comptox-tools/sequence-alignment-predict-across-species-susceptibility-seqapass-resource-hub]
- [72. ECOdrug: a database connecting drugs and conservation of ...](https://pmc.ncbi.nlm.nih.gov/articles/PMC5753218/) [https://pmc.ncbi.nlm.nih.gov/articles/PMC5753218/]
- [73. EcoDrugPlus \(EcoDrug+\).](https://ecodrugplus.helsinki.fi/AboutEcodrug) [https://ecodrugplus.helsinki.fi/AboutEcodrug]
- [74. Demonstration of the Sequence Alignment to Predict ...](https://pmc.ncbi.nlm.nih.gov/articles/PMC10758989/) [https://pmc.ncbi.nlm.nih.gov/articles/PMC10758989/]
- [75. EPA Releases SeqAPASS Version 8](https://www.epa.gov/sciencematters/epa-releases-seqapass-version-8) [https://www.epa.gov/sciencematters/epa-releases-seqapass-version-8]
- [76. ECOdrug.](https://ecodrug.org/) [https://ecodrug.org/]